GALOIS MODULE STRUCTURE OF DELIGNE–LUSZTIG VARIETIES FOR $\mathbf{G} = \mathrm{GL}_3$

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Abstract. Let $\overline{Y}(w) \to \overline{X}(w)$ be the ramified Galois cover of compactified Deligne–Lusztig varieties. Our objects of study are the Lefschetz numbers for the action of the finite covering group T^{wF} on the coherent cohomology of the structure sheaf of $\overline{Y}(w)$. The full collection of Lefschetz numbers determines the virtual representation of T^{wF} on the alternating sum of these cohomology groups. In the two cases of Deligne–Lusztig surfaces of type A_2 and ${}^2A_2$, we give explicit formulas for the Lefschetz numbers for a generator of the stabilizer of an irreducible component. It follows that the equivariant Euler characteristic of the compactification $\overline{Y}(w)$ is not virtually free as a virtual module over the covering group.

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1. Introduction

Let $k = \mathbb{F}_p$ and $\bar{k} = \overline{\mathbb{F}}_p$. Let G be a connected reductive group over $\bar{k}$, let $F\colon G \to G$ be an isogeny such that some positive power of F is a Frobenius endomorphism, and let w belong to the Weyl group W of G. Associated to this datum, Deligne and Lusztig defined two varieties $X(w)$ and $Y(w)$ that fit into a finite étale map

$$Y(w) \longrightarrow X(w).$$

Furthermore, these varieties are equipped with actions of a finite group of Lie type G^F on the left and a finite group T^{wF} on the right such that, $X(w) = Y(w)/T^{wF}$ [DL76, §1]. For a reduced expression $w = s_1 \cdots s_r$, Deligne and Lusztig also constructed a Bott–Samelson-type smooth projective compactification $\overline{X}(w)$ whose boundary is a simple normal-crossings divisor [DL76, §§9.10–9.11].

Later, Bonnafé and Rouquier constructed the normalization $\overline{Y}(w)$ of $\overline{X}(w)$ in $Y(w)$. They proved that the right T^{wF}-action extends to the normalization and that its finite quotient is $\overline{X}(w)$ [BR09, Theorem 1.2]. Thus one has a commutative diagram

$$\begin{array}{ccc} Y(w) & \hookrightarrow & \overline{Y}(w) \\ \downarrow & & \downarrow \\ X(w) & \hookrightarrow & \overline{X}(w) \end{array}$$

Although $Y(w) \to X(w)$ is unramified, the compactified cover $\overline{Y}(w) \to \overline{X}(w)$ is ramified along the boundary. Furthermore, $p \nmid |T^{wF}|$ and Maschke's theorem implies that $\bar{k}[T^{wF}]$ is semisimple, and the boundary ramification is tame.

Deligne–Lusztig varieties were introduced in [DL76] to construct representations of finite groups of Lie type through their compactly supported ℓ-adic étale cohomology. They also occur in the geometry of Shimura varieties. In particular, certain generalized Deligne–Lusztig varieties for unitary groups describe the closures of strata in the supersingular loci of unitary Shimura varieties [VW11]. Their intersection theory enters the study of special cycles [KR11, KR14] and plays a role in Li and Zhang's proof of the local Kudla–Rapoport conjecture [LZ22].

In this paper, we study the equivariant Euler characteristic of the structure sheaf of the compactified Deligne–Lusztig cover $\overline{Y}(w)$, with respect to the action of its covering group. We begin by recalling a central question in "geometric Galois module theory" that motivates our study.

Let G be a finite group acting on a proper $\bar{k}$-scheme Y, and let $\mathcal{F}$ be a G-equivariant coherent sheaf. Its equivariant Euler characteristic is

$$\chi_G(Y,\mathcal{F}) := \sum_i (-1)^i [H^i(Y,\mathcal{F})] \in G_0(\bar{k}[G]) \tag{1.1}$$

When $p \nmid |G|$, Maschke's theorem identifies the Grothendieck group of finitely generated $\bar{k}[G]$-modules $G_0(\bar{k}[G])$ with the Grothendieck group of finitely generated projective $\bar{k}[G]$-modules $K_0(\bar{k}[G])$. We call this alternating class the virtual Galois module. For a p-regular element $\tau \in G$, we define its Lefschetz number $L_{\mathrm{Br}}(\tau, Y, \mathcal{F})$ to be

$$L_{\mathrm{Br}}(\tau, Y, \mathcal{F}) := \sum_i (-1)^i \,\mathrm{BrTr}\big(\tau \mid H^i(Y,\mathcal{F})\big),$$

where BrTr is the Brauer trace.

One would like to compute the class $\chi_G(Y,\mathcal{F})$ in $K_0(\bar{k}[G])$, or equivalently to determine all of its Brauer-character values. A basic question is whether this class is *virtually free*, that is, whether

$$\chi_G(Y,\mathcal{F}) \in \mathbb{Z} \cdot [\bar{k}[G]] \subseteq K_0(\bar{k}[G]).$$

Since the Brauer character of the regular representation $\bar{k}[G]$ vanishes at every non-identity element of G, if $\chi_G(Y,\mathcal{F})$ is virtually free, then $L_{\mathrm{Br}}(\tau, Y, \mathcal{F}) = 0$ for every non-identity p-regular element τ. Thus a single nonzero Lefschetz number at a non-identity element is enough to show that the equivariant Euler characteristic is not virtually free, which is what is done in this paper for some specific compactified Deligne-Lusztig covers.

The above discussed "virtual freeness question" has its origins in the classical *normal integral basis problem* which we state below. Let N/K be a tamely ramified Galois extension of number fields with Galois group G. E. Noether proved that $\mathcal{O}_N$ is a locally free $\mathcal{O}_K[G]$-module of rank one. It therefore determines a stable class

$$[\mathcal{O}_N]^{\mathrm{stab}} := [\mathcal{O}_N] - [\mathcal{O}_K[G]] \in \mathrm{Cl}(\mathcal{O}_K[G]) := K_0(\mathcal{O}_K[G])/\langle [\mathcal{O}_K[G]] \rangle.$$

If N/K admits a normal integral basis, then $\mathcal{O}_N \cong \mathcal{O}_K[G]$, and hence $[\mathcal{O}_N]^{\mathrm{stab}} = 0$. Conversely, the vanishing of $[\mathcal{O}_N]^{\mathrm{stab}}$ means that $\mathcal{O}_N$ is stably free, i.e., there is an integer $m \geq 0$ such that

$$\mathcal{O}_N \oplus \mathcal{O}_K[G]^m \cong \mathcal{O}_K[G]^{m+1}.$$

Thus, $[\mathcal{O}_N]^{\mathrm{stab}}$ is the obstruction to a stable, or virtual, normal integral basis. An additional cancellation result is required to obtain an actual normal integral basis. Fröhlich gave a character-theoretic description of the corresponding class over $\mathbb{Z}[G]$ in terms of Artin root numbers, and his conjecture was proved by Taylor [Tay81].

Over a field, a particularly close geometric result is Nakajima's theorem for finite étale covers. Let $f\colon Z \longrightarrow W$ be a finite étale G-torsor between projective varieties over a field k, and let $\mathcal{F}$ be a coherent sheaf on W. Nakajima proves that the derived complex $R\Gamma(Z, f^*\mathcal{F})$ is represented by a bounded complex $L^\bullet$ of finitely generated free $k[G]$-modules [Nak84, Theorem 1]. Consequently, we have that the equivariant Euler characteristic $\chi_G(Z, f^*\mathcal{F}) = \sum_j (-1)^j \,\mathrm{rank}_{k[G]}(L^j)[k[G]]$ is an integral multiple of the regular representation class. Furthermore, he proves that if $H^i(Z, f^*\mathcal{F}) = 0$ for every $i \neq i_0$, then $H^{i_0}(Z, f^*\mathcal{F})$ is itself a free $k[G]$-module [Nak84, Theorem 2].

In subsequent work on tamely ramified covers of varieties, Nakajima proves instead that the coherent cohomology complex can in general be represented by a bounded complex of finitely generated projective $k[G]$-modules [Nak86]. Thus, tameness guarantees that the derived cohomology is represented by a perfect $k[G]$-complex and hence defines an Euler class in $K_0(k[G])$, but it does not imply virtual freeness.

Chinburg extended the study of Galois module structure to tame covers of schemes by constructing invariants from equivariant Euler characteristics of truncated de Rham complexes [Chi94]. For projective covers over finite fields, under regularity and normal-crossings hypotheses, he proved that his invariant, after restriction of scalars to $\mathbb{F}_p$, determines and is determined by the p-adic absolute values of the associated epsilon-constants. He also expressed its image in $\mathrm{Cl}(\mathbb{Z}[G])$ as the sum of a symplectic root-number class and a ramification class, giving a geometric counterpart to Fröhlich's conjecture [Chi94, Theorems 0.1–0.2]. In subsequent joint work, Chinburg, Erez, Pappas, and Taylor expressed, under suitable hypotheses, the projective Euler characteristic of a suitable de Rham complex for a tame cover over $\mathbb{Z}$ in terms of epsilon-constants of Hasse–Weil–Artin L-functions, giving a higher-dimensional counterpart to Fröhlich's conjecture [CEPT97].

In the situation considered in this paper, the boundary ramification of $\overline{Y}(w) \to \overline{X}(w)$ is consequently the source of the non-vanishing of the reduced equivariant Euler class, and hence of the failure of virtual freeness in the sense defined above. We use the Lefschetz–Riemann–Roch theorem to answer the virtual freeness problem algebro-geometrically. We first state the Lefschetz–Riemann–Roch theorem for surfaces below.

Theorem 1.1. *(Lefschetz–Riemann–Roch) For a smooth projective surface Z and a finite-order automorphism h of order prime to p, whose fixed locus consists of isolated points and curves,*

$$L_{\mathrm{Br}}(h, Z, \mathcal{O}_Z) = \sum_{P \subset Z^h} \frac{1}{(1-[\lambda_{P,1}])(1-[\lambda_{P,2}])} + \sum_{D \subset Z^h} \left(\frac{1-g_D}{1-[\nu_D]^{-1}} - \frac{[\nu_D]^{-1} D^2}{(1-[\nu_D]^{-1})^2} \right). \tag{1.2}$$

Here, P runs over the isolated fixed points, D runs over the fixed curves, $\lambda_{P,1}, \lambda_{P,2}$ are the cotangent eigenvalues at P, ν_D is the normal eigenvalue along D, and the brackets $[\cdot]$ denote Teichmüller lifts. A more general statement of this theorem and its background is given in Section 2.6.

For Deligne–Lusztig varieties, related results on Galois module structure are known for the Drinfeld curve and its smooth projective compactification. These results concern the action of $\mathrm{SL}_2(\mathbb{F}_q)$, whereas we study the action of the torus covering group. Laurent and Köck [LK24] determine the canonical representation of $\mathrm{SL}_2(\mathbb{F}_q)$ on the space of holomorphic differentials of the smooth projective Drinfeld curve $C\colon XY^q - X^qY = Z^{q+1}$. For every prime power $q = p^r$, they obtain the decomposition

$$H^0(C, \Omega_C) \simeq \bigoplus_{j=0}^{q-2} \mathrm{Sym}^j(\bar{k}^{\,2})$$

into indecomposable modules, where $\bar{k}^2$ carries the standard representation. In particular, the canonical representation is semisimple if and only if $q = p$.

Other related results include that of Köck and Marchment [MK26], who study the action of $\mathrm{SL}_2(\mathbb{F}_q)$ on the spaces $H^0(C, \Omega_C^{\otimes m})$, $m \geq 2$, of holomorphic polydifferentials on the smooth projective Drinfeld curve $C\colon XY^q - X^qY = Z^{q+1}$, in odd characteristic p. For $q = p$, they determine the complete decomposition into indecomposable modules. For arbitrary powers q of p, they construct an explicit basis and obtain a partial module decomposition.

Our main results are stated below, following which we explain the proof methology in Section 1.2. After specializing to the (split) A_2 and (non-split) ${}^2A_2$ cases, and fixing the Coxeter element $w = s_1s_2 \in W$, we write $X, Y, \overline{X}, \overline{Y}$ for the associated Deligne–Lusztig varieties and Γ for the covering group T^{wF}.

1.1. **Main results.** This paper answers the *virtual freeness* problem discussed above in the negative for Deligne-Lusztig surfaces for $G = \mathrm{GL}_3$ of both split and non-split type. The word *split* refers to the standard $\mathbb{F}_p$-structure on $G = \mathrm{GL}_3$ given by

$$F_{\mathrm{sp}}(g) = g^{(p)}, \qquad G^{F_{\mathrm{sp}}} = \mathrm{GL}_3(\mathbb{F}_p),$$

for which the Frobenius acts trivially on the Dynkin diagram of type A_2. The *non-split*, or unitary, structure is defined by

$$F_{\mathrm{un}}(g) = J_+(g^{(p)})^{-t}J_+^{-1}, \qquad J_+ = \mathrm{antidiag}(1,1,1).$$

Here F_{un}^2 is the p^2-Frobenius, $G^{F_{\mathrm{un}}} = \mathrm{U}_3(\mathbb{F}_p)$, and the induced automorphism on the Dynkin diagram interchanges the two simple reflections. Thus the two cases have the same group GL_3 after base change to $\bar{k}$, but different Frobenius structures: the first is the untwisted type A_2 and the second is the twisted type ${}^2A_2$. We fix the Weyl group element w in both the cases to be a Coxeter element $w = s_1s_2$.

Theorem 1.2 (Split case)**.** *Choose $\mu \in \mathbb{F}_{p^3}^{\times}$ of order $p^3 - 1$, and set*

$$\tau := \mathrm{diag}(\mu, \mu^p, \mu^{p^2}), \qquad N := p^2 + p + 1, \qquad \sigma := \tau^{p-1}, \qquad \bar{\zeta} := \mu^{1-p}.$$

Then τ generates T^{wF}, while σ generates the stabilizer of each of the $(p-1)$-many irreducible components $\overline{Y}_\ell$ of the compactified cover $\overline{Y}$. Put $\widehat{\zeta} = [\bar{\zeta}] \in W(\mathbb{F}_{p^3})^{\times}$. Write $\overline{Y} = \bigsqcup_{\ell \in \mathbb{F}_p^{\times}} \overline{Y}_\ell$. For every such component,

$$\begin{aligned} L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell, \mathcal{O}_{\overline{Y}_\ell}) &= N(p+1)\left(\frac{\widehat{\zeta}(\widehat{\zeta}^{p-1} - 1)}{(1-\widehat{\zeta})^2(1-\widehat{\zeta}^p)}\right) \\ &\quad + N\frac{1 + (p-1)\widehat{\zeta}}{(1-\widehat{\zeta})^2} + N\frac{\widehat{\zeta}^{2p}}{(1-\widehat{\zeta}^p)^2}. \end{aligned} \tag{1.3}$$

And, the Lefschetz number of the entire normal compactification is given by

$$L_{\mathrm{Br}}(\sigma, \overline{Y}(w)) = (p-1)L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell(w)).$$

Furthermore, both the Lefschetz number in (1.3) and the value for the entire normal compactification are nonzero.

Theorem 1.3 (Non-split unitary case)**.** *Choose $\alpha \in \mathbb{F}_{p^2}^{\times}$ of order $p^2 - 1$, and set*

$$\tau := \mathrm{diag}(\alpha^{p-1}, \alpha, \alpha^{-p}), \qquad \bar{\xi} := \alpha^p, \qquad \widehat{\xi} := [\bar{\xi}] \in W(\mathbb{F}_{p^2})^{\times}.$$

Then, $T^{wF} = \left\{\mathrm{diag}(\zeta, a, a^{-p}) : \zeta^{p+1} = 1, \quad a^{p^2-1} = 1\right\} \simeq \mu_{p+1} \times \mu_{p^2-1}$. *The normal compactification* $\overline{Y}$ *has exactly* $p+1$ *irreducible components* $\overline{Y}_\ell$, *and* τ *generates the stabilizer* H *of each of those components. Fix one of them and denote it by* $\overline{Y}_\ell$. *Set*

$$c_0 = \frac{(p+1)(p-2)}{2}, \qquad c_u = \frac{(p+1)(3p-2)}{2}, \qquad c_v = \frac{(p+1)(3p-4)}{2}.$$

These are integers for every prime p. *For this chosen component,*

$$L_{\mathrm{Br}}(\tau, \overline{Y}_\ell) = \frac{c_u\widehat{\xi} - c_0}{(1-\widehat{\xi})^2} - \frac{(p^3+1)\widehat{\xi}}{(1-\widehat{\xi})(1-\widehat{\xi}^p)} + \frac{\widehat{\xi}^p(c_v - c_0\widehat{\xi}^p)}{(1-\widehat{\xi}^p)^2}. \tag{1.4}$$

Furthermore, both the Lefschetz number in (1.4) *and the Lefschetz number for the entire normal compactification are nonzero.*

1.2. **Proof methodology.** We fix the Deligne–Lusztig datum as discussed above restricting ourselves to the rank two case, $G = \mathrm{GL}_3$. The goal is to check whether the equivariant Euler characteristic $\chi(\overline{Y}, \mathcal{O}_{\overline{Y}})$ of the normal compactification $\overline{Y}$ is a virtual multiple of the regular representation of its finite covering group Γ. To do so, in both the cases, we explicitly show that the Lefschetz number at a specific non-identity element of the covering group is non-zero, and employ the same methods discussed below.

The group Γ acts transitively on the irreducible components of $\overline{Y}$. We therefore fix one normal component $Z := \overline{Y}_\ell$, let $\Gamma_0 \subset \Gamma$ be its setwise stabilizer, and first compute the Lefschetz number of a generator h of Γ_0 on Z. In the split case $|\Gamma_0| = p^2 + p + 1$, and in the non-split case $|\Gamma_0| = p^2 - 1$.

The singular locus of $\overline{Y}$ is a disconnected set of points. We explicitly construct the variety $\overline{Y}$ as the normalization of $\overline{X}$ in Y, by first analyzing the geometry of $\overline{X}$. We would like to point out that Bonnafe and Rouquier [BR09] construct $\overline{Y}$ as a quotient of a smooth quasi-projective scheme $\tilde{Y}$ by $\mathbb{G}_m^2$, but we do not employ those methods to construct $\overline{Y}$, and we work with smooth projective desingularizations of $\overline{Y}$ instead, for the applications considered in this paper.

At a boundary point of $\overline{X}$, we construct an auxiliary étale-local cover Y' whose normalization gives a toric model for the singularities of $\overline{Y}_\ell$. Using étale slice theorems, we verify that the resulting singularity types match with those arising in the compactifications constructed by Bonnafé and Rouquier [BR09].

Since it is convenient to apply the Lefschetz-Riemann-Roch theorem for non-singular varieties, we construct explicit Γ_0-equivariant resolution of singularities $\widetilde{Z}$ of $\overline{Y}_\ell$. We also determine the fixed curves, isolated fixed points, and their normal and tangent characters for the action of the generator σ of Γ_0. The genera or self-intersections of the strict transforms of the boundary curves are later obtained from the incidence geometry of the compactified base $\overline{X}$. The results from toric geometry that we use are collected in Section 2.7. For the equivariant minimal resolution $f\colon \widetilde{Z} \longrightarrow Z$, the quotient singularities are rational, so

$$f_*\mathcal{O}_{\widetilde{Z}} = \mathcal{O}_Z, \text{ and } \quad R^j f_*\mathcal{O}_{\widetilde{Z}} = 0 \quad (j > 0).$$

The Leray spectral sequence

$$E_2^{i,j} = H^i\big(Z, R^j f_* \mathcal{O}_{\widetilde{Z}}\big) \Longrightarrow H^{i+j}(\widetilde{Z}, \mathcal{O}_{\widetilde{Z}})$$

therefore has only its $j = 0$ row and degenerates. Because f is equivariant, its edge maps give isomorphisms of $\bar{k}[\langle h \rangle]$-modules

$$H^i(Z, \mathcal{O}_Z) \simeq H^i(\widetilde{Z}, \mathcal{O}_{\widetilde{Z}}) \qquad (i \geq 0).$$

Consequently, the resolution has the same equivariant coherent cohomology and hence the same Lefschetz numbers: $L_{\mathrm{Br}}(h, Z) = L_{\mathrm{Br}}(h, \widetilde{Z})$.

We finally insert the required data about the fixed loci into the Lefschetz–Riemann–Roch formula of Section 2.6 to compute the Lefschetz numbers. We show that these numbers are non-zero, thereby ruling out virtual freeness, first for the chosen component under Γ_0 and, hence also for the entire normal compactification $\overline{Y}$.

Remark 1.1. As the singularities of $\overline{Y}(w)$ are at worst abelian quotient singularities, we would like to point out that our methods can be used to study the geometry and the Galois module structure of varieties $\overline{Y}(w)$ in other group types and general elements $w \in W$ for which $\overline{Y}(w)$ is a surface.

1.3. **Plan of the paper.** In Section 2, we fix notation, recall Deligne–Lusztig varieties and their compactifications, and introduce virtual Galois modules and Brauer traces. We also state the Lefschetz–Riemann–Roch theorem, recall the results on normalization and resolution of toric surfaces needed below, and describe the Bonnafé–Rouquier slice construction.

In Section 3, we study the split case. We give explicit equations, determine the local normalizations and toric resolutions, and compute the fixed loci and intersection numbers. In Section 3.7, we use these calculations to prove Theorem 1.2. In Section 3.8, we establish the non-vanishing of the resulting expression and deduce the Galois-module consequence.

In Section 4, we study the unitary case and carry out similar calculations as in the previous section. In Section 4.7, we prove Theorem 1.3, and in Section 4.8, we establish the non-vanishing of the Lefschetz numbers and deduce the corresponding Galois-module obstruction.

2. Preliminaries

We collect the conventions and geometric facts used throughout the paper. Our presentation follows the original construction of Deligne and Lusztig [DL76, §§1 and 9] and the modern treatment of Bonnafe-Rouquier [BR09] and Görtz-Schröer [GS26, §1]. Recall that $k = \mathbb{F}_p$ and $\bar{k} = \overline{\mathbb{F}}_p$.

2.1. **Reductive groups, Frobenius, and Weyl-group data.** Let G be a connected reductive group over $\bar{k}$. Let $F : G \to G$ be an isogeny such that F^d is a Frobenius endomorphism for some $d \geq 1$. The fixed-point group

$$G^F = \{g \in G(\bar{k}) : F(g) = g\}$$

is a finite group of Lie type. By the Lang–Steinberg theorem, one may choose an F-stable Borel subgroup B, an F-stable maximal torus $T \subseteq B$, and the unipotent radical U of B. We use the standard conventions of [DM20, MT11]; for the Lang map and the fixed-point theory, see [Lan56, Ste68].

Let $W = N_G(T)/T$ be the Weyl group. The choice of B determines the set S of simple reflections, the length function $\ell : W \to \mathbb{Z}_{\geq 0}$, and the Bruhat order $\leq$. For $w \in W$, we choose a representative $\dot{w} \in N_G(T)$. The induced endomorphism of the torus is denoted

$$wF : T \longrightarrow T, \qquad wF(t) = \dot{w}F(t)\dot{w}^{-1},$$

and its finite group of fixed points is T^{wF}. The actions of F on the character and cocharacter lattices are adjoint for the natural pairing, and F permutes the simple roots and simple reflections; see [DM20, MT11].

2.2. **Relative position and Deligne–Lusztig varieties.** Write $\mathcal{B} = \mathrm{Bor}(G)$ for the variety of Borel subgroups of G. The choice of the Borel subgroup B gives an identification

$$G/B \xrightarrow{\sim} \mathcal{B}, \qquad gB \longmapsto gBg^{-1}.$$

For every algebraically closed extension $\bar{k}/k$, the Bruhat decomposition gives a disjoint union

$$\mathcal{B}(\bar{k}) \times \mathcal{B}(\bar{k}) = \coprod_{w \in W} \mathcal{O}(w)(\bar{k}),$$

where the diagonal G-orbit indexed by w is

$$\mathcal{O}(w) = \{(g_1B, g_2B) : g_1^{-1}g_2 \in B\dot{w}B\}.$$

Consequently there is a well-defined relative-position map on geometric points

$$\mathrm{inv}\colon \mathcal{B}(\bar{k}) \times \mathcal{B}(\bar{k}) \longrightarrow W, \qquad \mathrm{inv}(g_1B, g_2B) = w \iff g_1^{-1}g_2 \in B\dot{w}B. \tag{2.1}$$

It is independent of the representatives g_1, g_2. Scheme-theoretically, $\mathcal{O}(w)$ is the fiber over w of the quotient-stack map

$$\mathcal{B} \times \mathcal{B} \longrightarrow [G\backslash(\mathcal{B} \times \mathcal{B})] \simeq [B\backslash G/B].$$

Thus two Borel subgroups $\mathcal{B}_1, \mathcal{B}_2$ are said to be *in relative position* w, or equivalently that $\mathcal{B}_2$ has relative position w with respect to $\mathcal{B}_1$, if

$$\mathrm{inv}(\mathcal{B}_1, \mathcal{B}_2) = w.$$

Following the notation of Bonnafé–Rouquier [BR09, §1.1], we may also write this ordered relation as

$$\mathcal{B}_1 \xrightarrow{w} \mathcal{B}_2.$$

The order is significant: $\mathcal{B}_1 \xrightarrow{w} \mathcal{B}_2$ if and only if $\mathcal{B}_2 \xrightarrow{w^{-1}} \mathcal{B}_1$. Scheme-theoretically, the first relation means precisely that the ordered pair lies in the locally closed subscheme $\mathcal{O}(w)$. Bruhat closure gives

$$\overline{\mathcal{O}(w)} = \bigcup_{v \leq w} \mathcal{O}(v). \tag{2.2}$$

If $w = w_1w_2$ and $\ell(w) = \ell(w_1) + \ell(w_2)$, the intermediate Borel subgroup in a chain of relative positions w_1, w_2 is unique. Equivalently,

$$\mathcal{O}(w_1) \times_{\mathcal{B}} \mathcal{O}(w_2) \xrightarrow{\sim} \mathcal{O}(w).$$

These facts are recalled in [DL76, §1] and [GS26, §1.3].

2.2.1. *Relative position in type A.* We use only root data of type A in this paper. For $G = \mathrm{GL}_n$, with B upper triangular and T diagonal, one has $W \simeq S_n$. In this case, we identify $\mathcal{B}$ with the full flag variety Fl; its $\bar{k}$-points are flags

$$\mathcal{F} = (0 = V_0 \subset V_1 \subset \cdots \subset V_n = K^n), \qquad \dim V_i = i.$$

For $w \in S_n$, put

$$r_w(i,j) := \#\{\nu : 1 \le \nu \le j,\ w(\nu) \le i\}.$$

If $\mathcal{F} = (V_i)$ and $\mathcal{F}' = (V'_j)$, then

$$\mathrm{inv}(\mathcal{F}, \mathcal{F}') = w \quad \Longleftrightarrow \quad \dim(V_i \cap V'_j) = r_w(i,j) \quad \text{for all } i,j. \tag{2.3}$$

Moreover, the closure condition is

$$\mathrm{inv}(\mathcal{F}, \mathcal{F}') \le w \quad \Longleftrightarrow \quad \dim(V_i \cap V'_j) \ge r_w(i,j) \quad \text{for all } i,j. \tag{2.4}$$

Thus, under this GL_n-specific identification, relative position records all incidence dimensions between the two flags, while the Bruhat order records their possible degenerations; see [GS26, §§1.3–1.4].

There are two Frobenius structures of type A_{n-1} relevant here. In the split case,

$$F_{\mathrm{sp}}(g) = g^{(p)}, \qquad G^{F_{\mathrm{sp}}} = \mathrm{GL}_n(\mathbb{F}_p),$$

and F_{sp} acts trivially on the Dynkin diagram. In the non-split case, put

$$J_+ = \mathrm{antidiag}(1, \ldots, 1), \qquad F_{\mathrm{un}}(g) = J_+ (g^{(p)})^{-t} J_+^{-1}.$$

Then F_{un}^2 is the p^2-Frobenius, $G^{F_{\mathrm{un}}} = \mathrm{U}_n(\mathbb{F}_p)$, and the induced diagram involution is $s_i \longmapsto s_{n-i}$. These are the untwisted type A_{n-1} and twisted type ${}^2A_{n-1}$, respectively. For $n = 3$, they give exactly the split and non-split type-A_2 surfaces treated in this paper.

2.2.2. *Deligne–Lusztig strata.* The graph morphism of F is

$$\Gamma_F \colon \mathcal{B} \longrightarrow \mathcal{B} \times \mathcal{B}, \qquad \mathcal{C} \longmapsto (\mathcal{C}, F(\mathcal{C})).$$

The Deligne–Lusztig variety attached to w is the inverse image of the relative-position stratum $\mathcal{O}(w)$:

$$\begin{aligned} X(w) &= \{gB \in \mathcal{B} : g^{-1}F(g) \in B\dot{w}B\} \\ &= \{\mathcal{C} \in \mathcal{B} : \mathrm{inv}(\mathcal{C}, F(\mathcal{C})) = w\} \\ &= \{\mathcal{C} \in \mathcal{B} : \mathcal{C} \xrightarrow{w} F(\mathcal{C})\}. \end{aligned} \tag{2.5}$$

Equivalently, it is the transverse intersection of $\mathcal{O}(w)$ with the graph of F. Since every ordered pair of Borel subgroups has a unique relative position, the variety of Borel subgroups is partitioned into Deligne–Lusztig strata

$$\mathcal{B} = \coprod_{w \in W} X(w).$$

The closed stratum $X(1) = \mathcal{B}^F$ is the finite scheme of F-stable Borel subgroups, whereas the stratum for the longest element is open. Each $X(w)$ is stable under G^F, because simultaneous left translation by an F-fixed element does not change relative position. In particular, the Coxeter element $w = s_1 s_2$ produces the surfaces studied in this paper.

Proposition 2.1 (Basic geometry of $X(w)$)**.** *The scheme $X(w)$ is smooth and equidimensional of dimension $\ell(w)$, and G^F acts on it by left translation. Its schematic closure in $\mathcal{B}$ is normal and Cohen–Macaulay, and its underlying stratification is*

$$\overline{X(w)}^{\text{naive}} = \bigcup_{v \le w} X(v).$$

Proof. This follows by comparing the Lang morphism with the Bruhat cell $B\dot{w}B/B$; see [GS26, Lemma 1.10], whose proof uses [Lan56, Ste68]. □

The covering variety. Let $U = R_u(B)$, and choose a representative $\dot{w} \in N_G(T)$ of w. The corresponding Deligne–Lusztig covering variety is

$$Y(w) = \{gU \in G/U : g^{-1}F(g) \in U\dot{w}U\}. \tag{2.6}$$

Write

$$wF(t) = \dot{w}F(t)\dot{w}^{-1} \qquad (t \in T),$$

and let T^{wF} be the finite group of fixed points of this twisted Frobenius wF. Left multiplication by G^F preserves $Y(w)$. There is also a right action

$$(gU) \cdot t = gtU \qquad (t \in T^{wF})$$

because the condition $wF(t) = t$ gives $(gt)^{-1}F(gt) \in U\dot{w}U$. The left and right actions commute.

The quotient map $G/U \to G/B$ restricts to

$$\pi_w : Y(w) \longrightarrow X(w), \qquad Y(w)/T^{wF} \xrightarrow{\sim} X(w). \tag{2.7}$$

Indeed, changing a lift of a point of $X(w)$ from G/B to G/U amounts to right multiplication by an element of $T = B/U$; the Frobenius condition restricts the possible changes exactly to T^{wF}. Thus every geometric fiber of π_w is a simply transitive T^{wF}-set. In particular, π_w is a finite étale right T^{wF}-torsor [DL76, §1].

2.3. **The smooth compactification and its boundary.** Fix a reduced expression $w = s_1 \cdots s_r$. Let $\mathcal{B} = \mathrm{Bor}(G)$. Following Deligne–Lusztig [DL76, §§9.1–9.2], consider the Bott–Samelson–Demazure–Hansen variety

$$\overline{\mathcal{O}}(w) = \left\{(\mathcal{B}_0, \dots, \mathcal{B}_r) : \ \mathrm{inv}(\mathcal{B}_{i-1}, \mathcal{B}_i) \in \{1, s_i\} \quad (1 \le i \le r) \ \right\}. \tag{2.8}$$

Here each $\mathcal{B}_i$ is a Borel subgroup. The condition in (2.8) means that $(\mathcal{B}_{i-1}, \mathcal{B}_i)$ lies in the closure $\overline{\mathcal{O}(s_i)} = \mathcal{O}(1) \cup \mathcal{O}(s_i)$ of the Bruhat orbit corresponding to s_i. Projection to $\mathcal{B}_0$ exhibits $\overline{\mathcal{O}}(w)$ as an iterated $\mathbb{P}^1$-bundle over $\mathcal{B}$. In particular, it is smooth and projective. Its endpoint morphism is

$$\Gamma_w \colon \overline{\mathcal{O}}(w) \longrightarrow \mathcal{B} \times \mathcal{B}, \qquad (\mathcal{B}_0, \dots, \mathcal{B}_r) \longmapsto (\mathcal{B}_0, \mathcal{B}_r).$$

Let $\Gamma_F \subset \mathcal{B} \times \mathcal{B}$ be the graph of F. The compactification used by Deligne and Lusztig is the fiber product

$$\overline{X}(w) = \overline{\mathcal{O}}(w) \times_{\mathcal{B}\times\mathcal{B}} \Gamma_F. \tag{2.9}$$

Equivalently, it is the closed subvariety of (2.8) defined by $\mathcal{B}_r = F(\mathcal{B}_0)$.

For a subset $I \subseteq \{1, \dots, r\}$, put

$$x_{I,i} = \begin{cases} 1, & i \in I, \\ s_i, & i \notin I, \end{cases}$$

and define the locally closed stratum

$$X_I = \left\{(\mathcal{B}_0, \dots, \mathcal{B}_r) \in \overline{X}(w) : \mathrm{inv}(\mathcal{B}_{i-1}, \mathcal{B}_i) = x_{I,i} \text{ for every } i\right\}.$$

One then has the disjoint stratification

$$\overline{X}(w) = \bigsqcup_{I \subseteq \{1,\dots,r\}} X_I, \qquad X(w) = X_\varnothing. \tag{2.10}$$

For $1 \le i \le r$, put

$$D_i = \{\mathrm{inv}(\mathcal{B}_{i-1}, \mathcal{B}_i) = 1\} = \bigsqcup_{\substack{I \subseteq \{1,\dots,r\} \\ i \in I}} X_I. \tag{2.11}$$

The divisor D_i is the boundary condition obtained by replacing the i-th reflection in the chosen reduced expression for w by the identity. It need not be connected after imposing $\mathcal{B}_r = F(\mathcal{B}_0)$.

Proposition 2.2 (Smooth compactification)**.** *The variety $\overline{X}(w)$ is smooth, projective, and of pure dimension $r = \ell(w)$, and $X(w)$ is a dense open subvariety. Its boundary*

$$\partial\overline{X}(w) = \overline{X}(w) \setminus X(w) = D_1 \cup \cdots \cup D_r$$

is a simple normal-crossings divisor. More precisely, for $I \subseteq \{1, \dots, r\}$, every nonempty intersection

$$D_I := \bigcap_{i \in I} D_i$$

is smooth of codimension $|I|$ in $\overline{X}(w)$.

Projection to the first Borel subgroup defines a proper morphism

$$\rho_w \colon \overline{X}(w) \longrightarrow \mathcal{B}, \qquad (\mathcal{B}_0, \dots, \mathcal{B}_r) \longmapsto \mathcal{B}_0.$$

It restricts to the usual identification of the open stratum with $X(w)$, and its image is the closure of $X(w)$ in $\mathcal{B}$. Hence it gives a proper birational morphism from $\overline{X}(w)$ to that naive closure.

Proof. The boundary divisors of $\overline{\mathcal{O}}(w)$ and all of their intersections are smooth. Deligne and Lusztig prove that Γ_F is transverse to Γ_w and to the restriction of Γ_w to every such intersection [DL76, Lemma 9.11]. Taking the fiber product with Γ_F therefore preserves smoothness and the normal-crossings boundary. The dimension is r, and projectivity follows from the closed embedding $\overline{X}(w) \hookrightarrow \mathcal{B}^{r+1}$. On the locus where every relative position is s_i, the chain is uniquely determined by its first Borel subgroup; this identifies the open stratum with $X(w)$. The final assertion follows by properness and density of this open stratum. See also the formulation in [BR09, §1.2]. □

For the length-two reduced expression $w = s_1 s_2$, the deepest stratum is

$$X_{\{1,2\}} = D_1 \cap D_2 = \{(\mathcal{B}, \mathcal{B}, \mathcal{B}) : F(\mathcal{B}) = \mathcal{B}\}.$$

Thus its points are the F-fixed Borel subgroups. Let $\alpha_i^\vee$ be the co-root corresponding to s_i, and put

$$\beta_i = s_1 \cdots s_{i-1}(\alpha_i^\vee) \in Y(T).$$

[DL76, Lemma 9.13] or equivalently [BR09, Corollary 1.4] says that the right T^{wF}-torsor over the open stratum has, along D_i, the same ramification as the pullback by $\beta_i\colon \mathbb{G}_m \to T$ of the Lang covering

$$L_{wF}\colon T \longrightarrow T, \qquad t \longmapsto t^{-1}wF(t),$$

when that pullback is extended across 0. This determines the tame local monodromy along each boundary component.

2.4. **A description in terms of flags.** For $G = \mathrm{GL}_3$, recall that the variety $\mathcal{B}$ of Borel subgroups is the full flag variety, and its rational point is a complete flag $(L \subset H)$, where L is a line and H is a plane in k^3. With the upper-triangular Borel convention and $S = \{s_1, s_2\}$, the Bruhat conditions for two flags $(L \subset H)$ and $(L' \subset H')$ are

w	1	s_1	s_2	s_1s_2	s_2s_1
$\mathrm{inv}((L,H),(L',H')) \le w$	$L = L',\ H = H'$	$H = H'$	$L = L'$	$L' \subset H$	$L \subset H'$

See [GS26, Lemma 1.5]. In particular, for $w = s_1s_2$, we compare the two flags (L, H) and $(F(L), F(H))$ and see that the naive compactification is cut out by the condition $F(L) \subseteq H$.

2.5. **Galois modules and Brauer traces.** In our setting, the open Deligne–Lusztig cover $Y(w) \longrightarrow X(w)$ is a finite étale right T^{wF}-torsor. The right T^{wF}-action extends to the normal compactification $\overline{Y}(w)$, and the resulting finite morphism $\overline{Y}(w) \longrightarrow \overline{X}(w)$ is the quotient by T^{wF} is ramified along the boundary. Set

$$\Gamma = T^{wF}, \qquad Z = \overline{Y}(w).$$

The order of Γ, and hence that of every component stabilizer, is prime to p. Maschke's theorem therefore implies that $\bar{k}[\Gamma]$ is semisimple and identifies $G_0(\bar{k}[\Gamma])$ with $K_0(\bar{k}[\Gamma])$. For a Γ-equivariant coherent sheaf $\mathcal{E}$ on Z, we may thus write

$$\chi_\Gamma(Z, \mathcal{E}) = \sum_i (-1)^i [H^i(Z, \mathcal{E})] \in K_0(\bar{k}[\Gamma])$$

This is the equivariant Euler characteristic introduced in (1.1), and it agrees with the projective Euler characteristic of $R\Gamma(Z, \mathcal{E})$. The class is *virtually free* if

$$\chi_\Gamma(Z, \mathcal{E}) \in \mathbb{Z} \cdot [\bar{k}[\Gamma]]$$

Since Z is normal, its irreducible components are disjoint. If Γ acts transitively on these components, Z_0 is one component, and Γ_0 is its stabilizer, then

$$H^i(Z, \mathcal{O}_Z) \simeq \mathrm{Ind}_{\Gamma_0}^{\Gamma} H^i(Z_0, \mathcal{O}_{Z_0})$$

for every i. Consequently,

$$\chi_\Gamma(Z, \mathcal{O}_Z) = \mathrm{Ind}_{\Gamma_0}^{\Gamma} \chi_{\Gamma_0}(Z_0, \mathcal{O}_{Z_0}) \tag{2.12}$$

Note that as Γ is given by the subgroup of rational points of a torus, every element of Γ is p-regular, and Brauer characters determine the classes in $K_0(\bar{k}[\Gamma])$. Let $[\,\cdot\,]\colon \bar{k}^\times \longrightarrow W(\bar{k})^\times$ denote the Teichmüller lift, where $W(\bar{k})$ is the ring of p-typical Witt vectors. If $\tau \in \Gamma$ acts on a finite-dimensional $\bar{k}$-representation V with eigenvalues $\lambda_1, \ldots, \lambda_m$, its Brauer trace is

$$\mathrm{BrTr}(\tau \mid V) = \sum_{j=1}^{m} [\lambda_j]$$

The eigenvalues are roots of unity of order prime to p, and their Teichmüller lifts are the corresponding roots of unity in $W(\bar{k})$.

Recall that the Brauer–Lefschetz number is defined by

$$\begin{aligned} L_{\mathrm{Br}}(\tau, Z, \mathcal{E}) &:= \sum_i (-1)^i \operatorname{BrTr}\big(\tau \mid H^i(Z, \mathcal{E})\big), \\ L_{\mathrm{Br}}(\tau, Z) &:= L_{\mathrm{Br}}(\tau, Z, \mathcal{O}_Z). \end{aligned} \tag{2.13}$$

Thus the function $\tau \mapsto L_{\mathrm{Br}}(\tau, Z, \mathcal{E})$ is precisely the Brauer character of $\chi_\Gamma(Z, \mathcal{E})$, and its values determine this virtual module.

If $\chi_\Gamma(Z, \mathcal{E}) = m[\bar{k}[\Gamma]]$ for some $m \in \mathbb{Z}$, its value at every nonidentity element is zero, because the regular representation has character zero away from the identity. Consequently,

$$\tau \neq 1 \quad \text{and} \quad L_{\mathrm{Br}}(\tau, Z, \mathcal{E}) \neq 0 \quad \Longrightarrow \quad \chi_\Gamma(Z, \mathcal{E}) \notin \mathbb{Z} \cdot [\bar{k}[\Gamma]]. \tag{2.14}$$

Remark 2.1. In our main results in Theorems 1.2 and 1.3, the Brauer–Lefschetz expression has the form $R(\widehat{\omega})$, where $R(T) \in \mathbb{Q}(T)$, M is prime to p, and $\widehat{\omega}$ is the Teichmüller lift of a primitive M-th root of unity in $\bar{k}$. Write $R(T) = P(T)/Q(T)$, with $P(T), Q(T) \in \mathbb{Q}[T]$, and assume that $Q(\widehat{\omega}) \neq 0$. Set $\zeta_M = e^{2\pi i/M}$. The assignment $\zeta_M \longmapsto \widehat{\omega}$ defines a field embedding $\mathbb{Q}(\zeta_M) \hookrightarrow \operatorname{Frac} W(\bar{k})$. It follows that $Q(\zeta_M) \neq 0$ and $R(\widehat{\omega}) \neq 0 \iff R(\zeta_M) \neq 0$. It therefore suffices to show $R(\zeta_M) \neq 0$ in $\mathbb{C}$.

2.6. **The Lefschetz–Riemann–Roch Theorem.** Lefschetz–Riemann–Roch expresses an alternating trace on coherent cohomology as a sum of contributions supported on the fixed locus. Donovan established the algebraic formula for periodic automorphisms of smooth projective varieties over an algebraically closed field [Don69, Theorem 5.4 and Corollary 5.5]. Baum, Fulton, and Quart subsequently developed a Lefschetz–Riemann–Roch transformation for singular varieties [BFQ79]. Thomason later formulated the theorem and its coherent trace formula in equivariant algebraic K-theory [Tho86]. We now state the theorem for smooth projective varieties.

Let Z be a smooth projective variety over $\bar{k}$, let G be a finite group acting on Z, and let $\mathcal{E}$ be a G-equivariant vector bundle. Fix an element $h \in G$ whose order is prime to p. The fixed subscheme Z^h is then smooth. Write its decomposition into connected components as $Z^h = \coprod_F F$.

Theorem 2.3 (Lefschetz–Riemann–Roch)**.** *With the preceding notation,*

$$L_{\mathrm{Br}}(h, Z, \mathcal{E}) = \sum_{F \subset Z^h} \int_F \frac{\operatorname{ch}_h(\mathcal{E}|_F)\, \operatorname{Td}(T_F)}{\operatorname{ch}_h\Big(\lambda_{-1}(N_{F/Z}^\vee)\Big)}, \tag{2.15}$$

where for a connected component $F \subset Z^h$, its normal bundle is

$$N_{F/Z} = T_Z|_F / T_F,$$

and its dual $N_{F/Z}^\vee$ is the conormal bundle. The class in the denominator is

$$\lambda_{-1}(N_{F/Z}^\vee) := \sum_{j=0}^{\operatorname{codim}(F,Z)} (-1)^j [\Lambda^j N_{F/Z}^\vee]. \tag{2.16}$$

Put $c = \operatorname{codim}(F, Z)$. For $x \in F$, let $\rho_1, \dots, \rho_c$ be the eigenvalues of h on the normal space

$$N_{F/Z,x} = T_x Z / T_x F.$$

The eigenvalues on the conormal space $N^\vee_{F/Z,x}$ are then $\rho_1^{-1}, \dots, \rho_c^{-1}$, and

$$\mathrm{BrTr}\Big(h \mid \lambda_{-1}(N^\vee_{F/Z,x})\Big) = \prod_{i=1}^{c} (1 - [\rho_i]^{-1}).$$

We use the specialization of the theorem to $\mathcal{E} = \mathcal{O}_Z$ when Z is a surface in which case its fixed locus is a disjoint union of isolated points P and smooth curves D. If λ_1, λ_2 are the eigenvalues of h on the conormal space $T^\vee_P Z = \mathfrak{m}_P/\mathfrak{m}_P^2$, then the contribution of P is

$$a_{\mathrm{Br}}(P) = \frac{1}{(1 - [\lambda_1])(1 - [\lambda_2])}. \tag{2.17}$$

If D has genus g_D, self-intersection D^2, and normal eigenvalue ν, then its contribution is

$$b_{\mathrm{Br}}(D) = \frac{1 - g_D}{1 - [\nu]^{-1}} - \frac{[\nu]^{-1} D^2}{(1 - [\nu]^{-1})^2} = \frac{[\nu](1 - g_D)}{[\nu] - 1} - \frac{[\nu] D^2}{([\nu] - 1)^2}. \tag{2.18}$$

Consequently,

$$L_{\mathrm{Br}}(h, Z) = \sum_P a_{\mathrm{Br}}(P) + \sum_D b_{\mathrm{Br}}(D).$$

Here $1 - g_D = \chi(D, \mathcal{O}_D)$, while D^2 is the degree of the normal line bundle $N_{D/Z}$. Thus these two terms are precisely the degree-one parts of the Todd class and the inverse conormal factor in (2.15). All denominators are units because the displayed eigenvalues are nontrivial roots of unity of order prime to p.

In this paper, the above theorem is applied to the equivariant minimal resolution $\widetilde{Y}'$ of a component of the normal compactification. The required input is therefore concrete: the isolated fixed points and their local eigenvalues, the fixed curves and their normal eigenvalues, and the genus and self-intersections of those curves.

2.7. **Toric geometry preliminaries.** We recall the results from toric geometry used to normalize and resolve the local surface models in both the split and the non-split cases. Our conventions follow [CLS11, Chapters 1–4 and §10.2].

Let $N \simeq \mathbb{Z}^2$ be a lattice, let $M = \mathrm{Hom}(N, \mathbb{Z})$ be its dual, and put $N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}$. A strongly convex rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ determines the normal affine toric surface

$$U_\sigma = \operatorname{Spec} k[S_\sigma], \qquad S_\sigma := \sigma^\vee \cap M. \tag{2.19}$$

Conversely, suppose that we have a semigroup algebra $k[S]$, where $S \subset M$ is finitely generated and spans M as a group. If $C = \mathrm{Cone}(S) \subset M_{\mathbb{R}}$, then its normalization is

$$\operatorname{Spec} k[C \cap M]. \tag{2.20}$$

Normalization can also be described in terms of semigroups, see [CLS11, Theorem 1.3.5]. Let S be a finitely generated submonoid of a lattice, put $M_S = \mathbb{Z}S$, and consider $V = \operatorname{Spec} \bar{k}[S]$. The saturation of S in M_S is

$$S^{\mathrm{sat}} := \{m \in M_S : nm \in S \text{ for some } n \in \mathbb{Z}_{>0}\} = \mathrm{Cone}(S) \cap M_S.$$

The integral closure of $\bar{k}[S]$ in its field of fractions is $\bar{k}[S^{\mathrm{sat}}]$. Hence the inclusion $S \subseteq S^{\mathrm{sat}}$ induces the normalization map

$$\operatorname{Spec} \bar{k}[S^{\mathrm{sat}}] \longrightarrow \operatorname{Spec} \bar{k}[S],$$

which is finite, birational, and equivariant for the common dense torus $\operatorname{Spec} \bar{k}[M_S]$. In particular, V is normal if and only if $S = S^{\mathrm{sat}}$; see [CLS11, Theorem 1.3.5 and Proposition 1.3.8]. Thus, to normalize an affine toric model, one replaces its defining semigroup by all lattice points of its cone in the lattice generated by that semigroup.

The orbit–cone correspondence identifies the torus orbits of U_σ with the faces of σ: a face τ corresponds to an orbit of dimension $2 - \dim \tau$. In particular, the zero face corresponds to the dense torus, the rays correspond to one-dimensional orbits, and σ itself corresponds to the unique closed orbit, a point denoted by o_σ.

Let u_0, u_{r+1} be the primitive generators of the two rays of σ. The surface U_σ is smooth if and only if these generators form a basis of N, or equivalently,

$$|\det(u_0, u_{r+1})| = 1, \tag{2.21}$$

where the determinant is computed with respect to a lattice basis of N. More generally, choose primitive lattice vectors $u_1, \dots, u_r$ in the interior of σ, ordered from u_0 to u_{r+1}, and let Σ be the fan whose maximal cones are

$$\sigma_i = \operatorname{Cone}(u_{i-1}, u_i), \qquad 1 \le i \le r+1.$$

The subdivision induces a proper birational toric morphism

$$\pi_\Sigma \colon X_\Sigma \longrightarrow U_\sigma,$$

which is an isomorphism away from o_σ. If every consecutive pair satisfies

$$|\det(u_{i-1}, u_i)| = 1,$$

then X_Σ is smooth, and π_Σ is a resolution of singularities. Each maximal cone σ_i gives an affine chart $U_{\sigma_i} \simeq \mathbb{A}^2$.

For the singularities occurring in this paper, choose a basis e_1, e_2 of N in which

$$\sigma = \operatorname{Cone}(e_2, de_1 - ae_2), \qquad 0 < a < d, \qquad \gcd(a, d) = 1. \tag{2.22}$$

The negative Hirzebruch–Jung continued fraction

$$\frac{d}{a} = [[b_1, \dots, b_r]] := b_1 - \cfrac{1}{b_2 - \cfrac{1}{\ddots - \cfrac{1}{b_r}}}, \qquad b_i \ge 2, \tag{2.23}$$

determines a regular subdivision as follows. Starting with $u_0 = e_2$ and $u_1 = e_1$, define the remaining vectors by

$$u_{i-1} + u_{i+1} = b_i u_i \qquad (1 \le i \le r). \tag{2.24}$$

Then $u_{r+1} = de_1 - ae_2$, the vectors $u_1, \dots, u_r$ are primitive and lie in the interior of σ, and every consecutive pair forms a basis of N. Thus they give a smooth toric resolution as above; see [CLS11, Theorems 10.2.3 and 10.2.5].

Write $\rho_i = \mathbb{R}_{\ge 0} u_i$. The exceptional divisors

$$E_i = V(\rho_i) \subset X_\Sigma, \qquad 1 \le i \le r,$$

are isomorphic to $\mathbb{P}^1$ and form a chain. Consecutive curves meet transversely at one point, and their intersection numbers are

$$E_i^2 = -b_i, \qquad E_i \cdot E_{i+1} = 1, \qquad E_i \cdot E_j = 0 \quad (|i-j| > 1); \tag{2.25}$$

see [CLS11, Theorem 10.4.4]. Since $b_i \geq 2$, no exceptional curve has self-intersection -1, so this is the minimal resolution.

The same fan describes the torus action on the resolution. On the chart associated with $\sigma_i = \mathrm{Cone}(u_{i-1}, u_i)$, let $m_i, n_i \in M$ be the basis dual to u_{i-1}, u_i. The corresponding characters

$$x_i = \chi^{m_i}, \qquad y_i = \chi^{n_i}$$

are coordinates on $U_{\sigma_i} \simeq \mathbb{A}^2$. For a torus element t, denote translation by t by r_t. Its pullback on character functions is

$$r_t^*(\chi^m) = \chi^m(t)\chi^m.$$

Thus, on this chart,

$$r_t^* x_i = \chi^{m_i}(t) x_i, \qquad r_t^* y_i = \chi^{n_i}(t) y_i.$$

In particular, the fixed subscheme is defined by

$$\big(\chi^{m_i}(t) - 1\big)x_i = 0, \qquad \big(\chi^{n_i}(t) - 1\big)y_i = 0.$$

These equations determine the fixed points and fixed curves directly from the character values.

The fan also determines the orders of characters along the torus-invariant divisors. For $m \in M$, the character χ^m, regarded as a rational function on X_Σ, has divisor

$$\mathrm{div}_{X_\Sigma}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho, \tag{2.26}$$

where $\Sigma(1)$ is the set of rays of Σ, u_ρ is the primitive generator of ρ, and $D_\rho = V(\rho)$ is the corresponding torus-invariant prime divisor; see [CLS11, Propositions 4.1.1–4.1.2]. The coefficient $\langle m, u_\rho \rangle$ is the order of vanishing, or the negative of the order of pole, of χ^m along D_ρ. We use these orders in the pullback and ramification calculations below.

The exceptional curves E_i are complete, and their self-intersections are determined by the local fan. By contrast, the divisors associated with the two extremal rays are noncomplete in X_Σ. They are the strict transforms of the two torus-invariant boundary divisors of U_σ, and describe only local portions of the corresponding curves on the global surface. Their global self-intersections therefore cannot be recovered from this local fan alone. We compute those numbers using the global boundary divisors and the projection formula in Sections 3.6 and 4.6.

2.8. **The Bonnafé–Rouquier normal compactification.** We first recall the main theorems and the precise ingredients from [BR09, §2.4] that enter the local calculations.

Notation and boundary strata.

Let $\mathcal{B}$ be the variety of Borel subgroups of G, and write $\mathrm{pos}(B, B') \in W$ for the relative position of two Borel subgroups. The compactification $\overline{X}(w)$ of $X(w)$ has

the following description:

$$\overline{X}(w) = \left\{ (B_0, \dots, B_r) \in \mathcal{B}^{r+1} \;\middle|\; \begin{array}{l} B_r = F(B_0), \\ \mathrm{pos}(B_{i-1}, B_i) \in \{1, s_i\} \quad (1 \le i \le r) \end{array} \right\}.$$

The open locus on which all relative positions are s_i is identified with $X(w)$ by projection to B_0. Allowing relative position 1, equivalently $B_{i-1} = B_i$, adds the boundary.

For words $x = (x_1, \dots, x_r)$ and $w = (s_1, \dots, s_r)$, we say $x \preccurlyeq w$ if $x_i \preccurlyeq s_i$ in the Bruhat order. For $x \preccurlyeq w$, we define $I_x = \{i : x_i = 1\} \subseteq \{1, \dots, r\}$. For $I \subseteq \{1, \dots, r\}$, put

$$x_i = \begin{cases} 1, & i \in I, \\ s_i, & i \notin I, \end{cases} \qquad x_I = x_1 \cdots x_r,$$

and define the locally closed stratum

$$X_I = \left\{ (B_0, \dots, B_r) \in \overline{X}(w) \;\middle|\; \mathrm{pos}(B_{i-1}, B_i) = x_i \quad (1 \le i \le r) \right\}.$$

Thus

$$\overline{X}(w) = \coprod_{I \subseteq \{1, \dots, r\}} X_I, \qquad X_\varnothing \simeq X(w),$$

and the nonempty subsets I index the boundary strata.

For type A_{n-1}, the simple reflection s_i is the transposition $(i\ i+1)$. In the rank-two case considered in this paper,

$$w = s_1 s_2 = (12)(23) = (123)$$

Here every subexpression is reduced after its identity terms are omitted, and

$$X_\varnothing \simeq X(s_1 s_2), \qquad X_{\{1\}} \simeq X(s_2), \qquad X_{\{2\}} \simeq X(s_1), \qquad X_{\{1,2\}} \simeq X(1).$$

2.8.1. *The compactification $\overline{Y}(w)$ of Bonnafé–Rouquier.* Let G be a connected reductive group over $\bar{k}$, and let $F : G \to G$ be an isogeny some positive power of which is a Frobenius endomorphism. Fix an F-stable Borel subgroup B, an F-stable maximal torus $T \subseteq B$, and write $U = R_u(B)$ for the unipotent radical. Let

$$W = N_G(T)/T$$

be the Weyl group. The choice of B determines a set Δ of simple roots and the corresponding set $S = \{s_\alpha : \alpha \in \Delta\}$ of simple reflections. For $\alpha \in \Delta$, write $\alpha^\vee \in Y(T)$ for the associated coroot, where

$$Y(T) = \mathrm{Hom}(\mathbb{G}_m, T)$$

is the cocharacter lattice, and fix a representative $\dot{s}_\alpha \in N_G(T)$ of s_α. The endomorphism induced by F on W, T, and $Y(T)$ is again denoted by F. Thus, for $v \in W$, the notation vF denotes the corresponding twisted endomorphism; on T, it is

$$t \longmapsto \dot{v} F(t) \dot{v}^{-1},$$

where $\dot{v} \in N_G(T)$ represents v. In particular, T^{vF} denotes its finite fixed-point group.

Fix the reduced expression $w = s_1 \cdots s_r$. For $I \subseteq \{1, \dots, r\}$, let $x_i = 1$ for $i \in I$ and $x_i = s_i$ for $i \notin I$, and put

$$Y_{w,I} = \sum_{i \in I} \mathbb{Z} s_1 \cdots s_{i-1}(\alpha_i^\vee) \subseteq Y(T).$$

Choose $d \geq 1$ and a power q of p such that $(vF)^d(t) = t^q$ for every $v \in W$ and $t \in T$, and fix a primitive $(q-1)$-st root ϵ. The norm map used below is

$$N_w : Y(T) \longrightarrow T^{wF}, \qquad N_w(\lambda) = \prod_{j=0}^{d-1} (wF)^j\big(\lambda(\epsilon)\big).$$

It is surjective and induces

$$Y(T)/(wF-1)Y(T) \xrightarrow{\sim} T^{wF}.$$

For $x_I = x_1 \cdots x_r$, the notation N_{x_I} has the analogous meaning.

Theorem 2.4 (Bonnafé–Rouquier, Theorem 1.2)**.** *Let $\overline{Y}(w)$ be the normalization of the compactification $\overline{X}(w)$ in $Y(w)$, and let $\overline{\pi}_w$ be the resulting finite morphism. Then the following hold.*

(a) *The variety $\overline{Y}(w)$ is projective, normal, and rationally smooth, and*

$$\operatorname{Sing} \overline{Y}(w) \subseteq \overline{\pi}_w^{-1} \left(\bigcup_{\substack{I \subseteq \{1,\dots,r\} \\ |I| \geq 2}} X_I \right).$$

(b) *The action of T^{wF} on $Y(w)$ extends to $\overline{Y}(w)$, and*

$$\overline{Y}(w)/T^{wF} \xrightarrow{\sim} \overline{X}(w).$$

(c) *For $I \subseteq \{1,\dots,r\}$, the stabilizer in T^{wF} of every point of $\overline{\pi}_w^{-1}(X_I)$ is $N_w(Y_{w,I})$.*

(d) *For $I \subseteq \{1,\dots,r\}$, there is a canonical morphism*

$$i_I : Y_I \longrightarrow \overline{\pi}_w^{-1}(X_I)$$

over X_I, and it induces an isomorphism

$$Y_I/N_{x_I}(Y_{w,I}) \xrightarrow{\sim} \overline{\pi}_w^{-1}(X_I).$$

Part (c) identifies the inertia group along each boundary stratum. The finite stabilizer arising from the auxiliary torus quotient is described in Proposition 2.6; it is this group that acts on the normal slice.

2.8.2. *The auxiliary construction and its stabilizers.* For every i, Bonnafé–Rouquier choose the unique primitive cocharacter $\lambda_i \in Y(T)$ and unique positive integer m_i satisfying

$$\lambda_i - wF(\lambda_i) = m_i s_1 \cdots s_{i-1}(\alpha_i^\vee), \qquad Y(T)/\mathbb{Z}\lambda_i \text{ is torsion-free.} \tag{2.27}$$

The defining equality implies $m_i \mid (q-1)$; in particular, $p \nmid m_i$ [BR09, Remark 2.4].

Let $\varphi_{\alpha_i} : G_{\alpha_i} U \to \mathbb{A}^1$ be the regular function used in the construction; its zero locus is $B \cap G_{\alpha_i} U$, and it takes the value one on $U\dot{s}_i U$. The auxiliary variety consists of the tuples

$$(g_1 U, \dots, g_{r+1} U; \xi_1, \dots, \xi_r)$$

such that, for every i,

$$g_i^{-1} g_{i+1} \in G_{\alpha_i} U, \qquad \varphi_{\alpha_i}(g_i^{-1} g_{i+1}) = \xi_i^{m_i}, \qquad g_{r+1} U = F(g_1) U. \tag{2.28}$$

We denote it by $\widetilde{Y}_{\mathrm{BR}}(w)$. This Bonnafé–Rouquier auxiliary variety is distinct from both the toric surfaces used below and the minimal resolutions of the normal components $\overline{Y}_\ell(w)$.

Proposition 2.5 (Bonnafé–Rouquier, (2.5), Proposition 2.6 and §2.5)**.** *The variety $\widetilde{Y}_{\mathrm{BR}}(w)$ is smooth and pure of dimension $2r$. It carries a right action of $T^{wF} \times (\mathbb{G}_m)^r$, and the natural morphism $\widetilde{\pi}_w : \widetilde{Y}_{\mathrm{BR}}(w) \to \overline{X}(w)$ induces*

$$\widetilde{Y}_{\mathrm{BR}}(w)/\big(T^{wF} \times (\mathbb{G}_m)^r\big) \xrightarrow{\sim} \overline{X}(w).$$

Moreover,

$$\overline{Y}(w) = \widetilde{Y}_{\mathrm{BR}}(w)/(\mathbb{G}_m)^r,$$

and the locus on which every ξ_i is nonzero is the open part corresponding to $Y(w)$.

For $z = (z_1, \dots, z_r) \in (\mathbb{G}_m)^r$, set

$$\Lambda_1(z) = \lambda_1(z_1) \cdots \lambda_r(z_r), \qquad \Lambda_{i+1}(z) = s_i\big(\Lambda_i(z)\big)\alpha_i^\vee(z_i^{m_i}).$$

The auxiliary torus acts by

$$(g_1U, \dots, g_{r+1}U; \xi_1, \dots, \xi_r) * z = (g_1\Lambda_1(z)U, \dots, g_{r+1}\Lambda_{r+1}(z)U; z_1\xi_1, \dots, z_r\xi_r).$$

For $I \subseteq \{1, \dots, r\}$, define

$$\widetilde{Y}_I(w) = \{(g; \xi) \in \widetilde{Y}_{\mathrm{BR}}(w) : \xi_i = 0 \iff i \in I\}$$

and

$$H_I = \{z \in (\mathbb{G}_m)^r : \Lambda_1(z) = \cdots = \Lambda_{r+1}(z) = 1,\ z_i = 1 \text{ for } i \notin I\}.$$

Proposition 2.6 (Bonnafé–Rouquier, Proposition 2.7)**.** *For every $I \subseteq \{1, \dots, r\}$, the stabilizer in $(\mathbb{G}_m)^r$ of any point of $\widetilde{Y}_I(w)$ is exactly H_I. The group H_I is finite and is contained in $H_{\{1,\dots,r\}}$. If $|I| \leq 1$, then $H_I = 1$. Furthermore, for every $I \subseteq \{1, \dots, r\}$,*

$$\widetilde{\pi}_w^{-1}(X_I) = \widetilde{Y}_I(w).$$

For each $I \subseteq \{1, \dots, r\}$, the construction in the proof further gives a $(\mathbb{G}_m)^r$-equivariant isomorphism

$$Y_I \times \big((\mathbb{G}_m)^r/H_I\big) \xrightarrow{\sim} \widetilde{Y}_I(w).$$

This identifies the tangent directions along a boundary stratum with the Deligne–Lusztig stratum directions together with the auxiliary-torus orbit directions.

Thus only strata with at least two vanishing boundary coordinates can produce singularities in the quotient, exactly as asserted in Theorem 2.4(a).

2.8.3. *The étale-slice in rank two.* We first recall the form of Luna's étale slice theorem that will be used. If a smooth group G_0 acts on a smooth variety V, the tangent space to the orbit $G_0 \cdot v$ accounts for the directions along which the quotient removes the G_0-action. The remaining first-order directions form the *normal representation*

$$N_v := T_vV/T_v(G_0 \cdot v)$$

of the stabilizer $(G_0)_v$. Luna's theorem asserts that, when this stabilizer is linearly reductive, the quotient is étale-locally modeled on the quotient of N_v by $(G_0)_v$.

Theorem 2.7 (Luna's étale slice theorem, quotient form)**.** *Let V be a quasi-separated algebraic space locally of finite type over the algebraically closed field k, and let a smooth affine group scheme G_0 act on V. Fix $v \in V(k)$, put $H = (G_0)_v$, and suppose that H is linearly reductive. Then there are an affine H-scheme S, an H-fixed point $s \in S(k)$, and an H-equivariant unramified morphism*

$$(S, s) \longrightarrow (V, v)$$

such that the induced morphism from the contracted product

$$G_0 \times^H S \longrightarrow V$$

is étale.

If, in addition, V admits a good quotient $\pi\colon V \to V/\!/G_0$ and the orbit $G_0 \cdot v$ is closed in V, the slice can be chosen so that $S/\!/H \to V/\!/G_0$ is étale and

$$G_0 \times^H S \simeq V \times_{V/\!/G_0} (S/\!/H).$$

Suppose now that V admits a good quotient, that $G_0 \cdot v$ is closed, and that V is smooth at v. The étale morphisms $V/\!/G_0 \leftarrow S/\!/H \to N_v/\!/H$ identify the completed local ring of $V/\!/G_0$ at $\bar{v} = \pi(v)$ with that of $N_v/\!/H$ at 0. If moreover H is finite of order prime to $p = \operatorname{char} k$, then H is linearly reductive, and completing $k[N_v]^H$ at the origin gives $k[[N_v^\vee]]^H$. Hence there is a non-canonical isomorphism

$$\widehat{\mathcal{O}}_{V/\!/G_0,\bar{v}} \simeq k[[N_v^\vee]]^H. \tag{2.29}$$

In particular, the étale-local singularity type of $V/\!/G_0$ at $\bar{v}$ is determined by the representation of H on N_v.

For more details, see [AHR20, Theorem 4.5 and Remark 4.6].

We now specialize this statement to $G = \mathrm{GL}_3$, $r = 2$, and the reduced expression $w = s_1 s_2$. Let

$$Y(T) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3, \qquad \alpha_1^\vee = e_1 - e_2, \qquad \alpha_2^\vee = e_2 - e_3,$$

where s_1 interchanges e_1, e_2 and s_2 interchanges e_2, e_3. Set

$$V = \widetilde{Y}_{\mathrm{BR}}(w), \qquad G_0 = (\mathbb{G}_m)^2.$$

By Proposition 2.5, V is smooth of dimension four and its geometric quotient by G_0 is $\overline{Y}(w)$. On the invariant affine neighborhoods occurring in the slice theorem this is a good quotient, so its quotient clause applies.

Choose a point $\widetilde{P}$ of the deepest boundary stratum $\xi_1 = \xi_2 = 0$. Then the stabilizer $H_{\{1,2\}}$ is given by

$$K_{12} = \left\{(z_1, z_2) \in (\mathbb{G}_m)^2 : z_1^{m_1} = z_2^{m_2} = 1,\ \lambda_1(z_1)\lambda_2(z_2) = 1\right\}. \tag{2.30}$$

We write K_{12} to distinguish this auxiliary slice stabilizer from the component stabilizers inside T^{wF}. By Proposition 2.6, the stabilizers on the other strata are trivial. Since $m_i \mid (q-1)$, the finite group K_{12} has order prime to p, so Theorem 2.7 applies.

The G_0-orbit through $\widetilde{P}$ has dimension two, and hence the normal representation

$$N_{\widetilde{P}} = T_{\widetilde{P}}V / T_{\widetilde{P}}(G_0 \cdot \widetilde{P})$$

has dimension two. The two boundary parameters are transverse to the deepest stratum. Indeed, for $I = \{1, 2\}$, the factor $Y_{\{1,2\}}$ in the preceding stratum description is zero-dimensional, so the tangent space to $\widetilde{Y}_{\{1,2\}}(w)$ at $\widetilde{P}$ is the tangent space to the G_0-orbit. The morphism $(\xi_1, \xi_2) : V \to \mathbb{A}^2$ is smooth at $\widetilde{P}$: in the proof of [BR09, Proposition 2.3], the successive fibres have coordinates (x, ξ) on $\mathbb{A}^2 \setminus \{0\}$, and fixing the ξ_i gives smooth fibres. The graph-transversality lemma [BR09, Lemma 2.5] then gives smooth fibres of dimension two on V. Thus the classes of ξ_1, ξ_2 form a basis of $N_{\widetilde{P}}^\vee$. From the action in Proposition 2.5, one reads

$$(z_1, z_2)\colon (\xi_1, \xi_2) \longmapsto (z_1\xi_1, z_2\xi_2). \tag{2.31}$$

Consequently the two characters of K_{12} obtained by restricting the coordinate characters z_1, z_2 are exactly the two slice weights.

Suppose $K_{12} \simeq \mu_d$ and a chosen generator is embedded as

$$\eta \longmapsto (\eta^a, \eta^b).$$

Then, for P the image of $\widetilde{P}$ in $\overline{Y}(w)$, (2.29) becomes

$$\widehat{\mathcal{O}}_{\overline{Y}(w),P} \simeq \bar{k}[[u,v]]^{\mu_d}, \qquad \eta(u,v) = (\eta^a u, \eta^b v).$$

When a is invertible modulo d, replacing the generator by $\eta^{a^{-1}}$ puts the action in the standard cyclic quotient form

$$\frac{1}{d}(1,c), \qquad c \equiv a^{-1}b \pmod{d}.$$

This description is evaluated in Section 3.3 for the split Frobenius and in Section 4.4 for the unitary Frobenius. In each case the cocharacters λ_i determine K_{12}, while (2.31) determines its two weights.

2.9. **Equivariant comparison with the normal compactification.**

Proposition 2.8 (Cohomology under resolution of rational surface singularities)**.** *Let X be a normal projective surface with rational singularities, and let*

$$f\colon Y \longrightarrow X$$

be a projective resolution of singularities. Then the natural pullback maps are isomorphisms

$$H^i(X, \mathcal{O}_X) \xrightarrow{\sim} H^i(Y, \mathcal{O}_Y) \qquad (i \geq 0). \tag{2.32}$$

If a finite group A acts on X, the action lifts to Y, and f is A-equivariant, then these are isomorphisms of $\bar{k}[A]$-modules.

Proof. Because X is normal and f is proper and birational, Zariski's Main Theorem gives

$$f_*\mathcal{O}_Y = \mathcal{O}_X.$$

The assumption that X has rational singularities gives

$$R^j f_*\mathcal{O}_Y = 0 \qquad (j > 0).$$

Thus the Leray spectral sequence

$$E_2^{i,j} = H^i\big(X, R^j f_*\mathcal{O}_Y\big) \Longrightarrow H^{i+j}(Y, \mathcal{O}_Y)$$

has only its $j = 0$ row. It therefore degenerates and its edge maps give

$$H^i(Y, \mathcal{O}_Y) \cong H^i(X, f_*\mathcal{O}_Y) = H^i(X, \mathcal{O}_X)$$

for every $i \geq 0$. When f is A-equivariant, the derived pushforward, the Leray spectral sequence, and its edge maps are all A-equivariant. The resulting isomorphisms therefore respect the $\bar{k}[A]$-module structures. □

Lemma 2.9 (Rational singularities)**.** *Let X be a normal integral variety over a perfect field K. Suppose that there is a family of étale morphisms $u_\alpha\colon U_\alpha \to X$ whose images cover X, such that each U_α is a tame quotient in the sense of* [CR11, Definition 4.2.6]*, i.e. U_α is integral and normal and admits a finite surjective morphism from a smooth integral K-scheme whose degree is invertible in K. Then every*

resolution $g\colon \widetilde{X} \to X$, that is, every proper birational morphism with $\widetilde{X}$ smooth and integral, satisfies

$$g_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X, \qquad R^i g_*\mathcal{O}_{\widetilde{X}} = 0 \quad (i > 0).$$

In particular, X has rational singularities in the cohomological sense used in Proposition 2.8.

Proof. Since X is integral and g is proper and birational, there is a dense open subset $V \subset X$ such that $g^{-1}(V) \to V$ is an isomorphism. Let $u\colon U \to X$ be one of the morphisms u_α and form the cartesian square

$$\begin{array}{ccc} \widetilde{U} := \widetilde{X} \times_X U & \xrightarrow{u'} & \widetilde{X} \\ {\scriptstyle g_U}\downarrow & & \downarrow{\scriptstyle g} \\ U & \xrightarrow[u]{} & X. \end{array}$$

The morphism g_U is proper by base change, and $\widetilde{U}$ is smooth over K because u' is étale and $\widetilde{X}$ is smooth over K.

We claim that $\widetilde{U}$ is integral and that g_U is birational. Since u is open and X is irreducible, the nonempty open set $u(U)$ meets V; hence $U_V := u^{-1}(V)$ is a nonempty, hence dense and irreducible, open subset of U, and g_U restricts to an isomorphism $\widetilde{U}_V := g_U^{-1}(U_V) \xrightarrow{\sim} U_V$. As $\widetilde{U}$ is smooth, it is the disjoint union of finitely many connected components, each of which is open and integral. For such a component W, the restriction $u'|_W$ is étale, so $u'(W)$ is a nonempty open subset of the irreducible scheme $\widetilde{X}$ and therefore meets the dense open subset $g^{-1}(V)$. Thus W meets $u'^{-1}(g^{-1}(V)) = \widetilde{U}_V$. Since $\widetilde{U}_V \simeq U_V$ is irreducible and the components are pairwise disjoint and open, $\widetilde{U}$ has exactly one component. Hence $\widetilde{U}$ is integral, $\widetilde{U}_V$ is dense in $\widetilde{U}$, and g_U is birational.

Consequently $\widetilde{U}$ is a tame quotient (it is smooth, hence normal, and integral; take the identity, of degree 1), and U is a tame quotient by hypothesis. By [CR11, Corollary 4.3.12], applied to the proper birational morphism g_U,

$$R(g_U)_*\mathcal{O}_{\widetilde{U}} \simeq \mathcal{O}_U.$$

Since u is flat and g is proper, flat base change gives

$$u^* R^i g_*\mathcal{O}_{\widetilde{X}} \simeq R^i(g_U)_*\mathcal{O}_{\widetilde{U}},$$

which vanishes for $i > 0$.

Now fix $i > 0$ and put $\mathcal{F} := R^i g_*\mathcal{O}_{\widetilde{X}}$, a coherent $\mathcal{O}_X$-module. Given $x \in X$, choose α and $y \in U_\alpha$ with $u_\alpha(y) = x$. The local homomorphism $\mathcal{O}_{X,x} \to \mathcal{O}_{U_\alpha,y}$ is flat, hence faithfully flat, and $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{U_\alpha,y} = (u_\alpha^*\mathcal{F})_y = 0$. Therefore $\mathcal{F}_x = 0$ for every $x \in X$, i.e. $R^i g_*\mathcal{O}_{\widetilde{X}} = 0$ for $i > 0$.

Finally, $g_*\mathcal{O}_{\widetilde{X}}$ is a coherent $\mathcal{O}_X$-algebra, since g is proper, and it is contained in the function field $K(\widetilde{X}) = K(X)$, since g is birational and $\widetilde{X}$ is integral. Its local sections are therefore rational functions that are integral over $\mathcal{O}_X$, and normality of X gives $g_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$. □

Let $f\colon \widetilde{Z} \to Z$ denote either of the minimal resolutions

$$\begin{aligned} \widetilde{\overline{Y}}_\ell(w) &\longrightarrow \overline{Y}_\ell(w) && \text{in the split case,} \\ \widetilde{\overline{Y}}_{\delta_0}(w) &\longrightarrow \overline{Y}_{\delta_0}(w) && \text{in the non-split case, with } \delta_0 \in \mu_{p+1} \text{ fixed.} \end{aligned}$$

Recall that Z is an integral proper surface over $\bar{k}$. By the quotient clause of Theorem 2.7, every singular point $z \in Z$ has an étale neighborhood

$$u_z \colon U_z = S_z /\!/ \mu_d \longrightarrow Z,$$

where S_z is affine and

$$d = N = p^2+p+1 \quad \text{in the split case,} \qquad d = N_{\mathrm{un}} = p^2-1 \quad \text{in the non-split case.}$$

After replacing S_z by a μ_d-stable open neighborhood of the point over z contained in its smooth locus, and then by the connected component through that point (which is μ_d-stable because the point is μ_d-fixed), we may assume that S_z is smooth, connected and affine, hence integral. Since $p \nmid d$, the group scheme μ_d is the constant group $\mathbb{Z}/d$ over $\bar{k}$. The quotient morphism $S_z \to U_z$ is therefore finite and surjective, U_z is integral and normal, and the degree of $S_z \to U_z$ equals the order of the image of μ_d in $\mathrm{Aut}(S_z)$, which divides d. Since $N \equiv 1$ and $N_{\mathrm{un}} \equiv -1 \pmod{p}$, this degree is invertible in $\bar{k}$, so U_z is a tame quotient in the sense of [CR11, Definition 4.2.6]. The smooth locus $Z^{\mathrm{sm}} \subset Z$ is a tame quotient via the identity, and the open immersion $Z^{\mathrm{sm}} \hookrightarrow Z$ is étale. Hence the morphisms u_z, together with $Z^{\mathrm{sm}} \hookrightarrow Z$, form an étale cover of Z by tame quotients. In particular Z is normal, because normality is local in the étale topology. Lemma 2.9 therefore shows that Z has rational singularities:

$$f_*\mathcal{O}_{\widetilde{Z}} = \mathcal{O}_Z, \qquad R^j f_*\mathcal{O}_{\widetilde{Z}} = 0 \quad (j > 0). \tag{2.33}$$

Let h be the element under consideration, namely $h = \sigma$ in the split case and $h = \tau$ in the non-split case; it acts on Z by $\bar{k}$-automorphisms. Since Z is a normal surface, its minimal resolution is unique up to unique isomorphism. Hence the automorphism h of Z lifts uniquely to an automorphism of $\widetilde{Z}$ with $f \circ h = h \circ f$, so f is $\langle h \rangle$-equivariant. Proposition 2.8, applied to the equivariant resolution f and combined with (2.33), gives canonical isomorphisms of $\bar{k}[\langle h \rangle]$-modules

$$f^* \colon H^i(Z, \mathcal{O}_Z) \xrightarrow{\sim} H^i(\widetilde{Z}, \mathcal{O}_{\widetilde{Z}}) \qquad (i \geq 0). \tag{2.34}$$

(Explicitly, (2.33) and the Leray spectral sequence show that f^* is an isomorphism, and it commutes with the action of h because $f \circ h = h \circ f$.) Since Z and $\widetilde{Z}$ are proper surfaces over $\bar{k}$, these modules are finite-dimensional and vanish for $i > 2$. Since the order of h is prime to p, their Brauer characters are defined at h, and taking alternating Brauer traces of (2.34) gives

$$\boxed{L_{\mathrm{Br}}(h, Z) = L_{\mathrm{Br}}(h, \widetilde{Z}).} \tag{2.35}$$

Thus

$$L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell(w)) = L_{\mathrm{Br}}(\sigma, \widetilde{\overline{Y}}_\ell(w))$$

in the split case, and

$$L_{\mathrm{Br}}(\tau, \overline{Y}_{\delta_0}(w)) = L_{\mathrm{Br}}(\tau, \widetilde{\overline{Y}}_{\delta_0}(w))$$

in the non-split case. These equalities will be applied to the fixed-locus calculations in Sections 3.7 and 4.7, respectively, to obtain the formulas for the normal compactifications.

3. Split Case

3.1. **Split setup and main result.** Recall that $k = \mathbb{F}_p$ and $\bar{k} = \overline{\mathbb{F}}_p$. We take $G = \mathrm{GL}_3$ over $\bar{k}$, with split Frobenius F, upper triangular Borel subgroup B, and diagonal torus

$$T = \{\mathrm{diag}(t_1, t_2, t_3) : t_i \in \bar{k}^\times\} \simeq (\mathbb{G}_m)^3.$$

Its Weyl group is S_3. All geometric fixed-locus and coherent-cohomology calculations are understood after base change to $\bar{k}$. For the Coxeter element $w = (123)$, choose the lift that cyclically permutes the diagonal entries. Then

$$wF(t_1, t_2, t_3) = (t_3^p, t_1^p, t_2^p), \qquad (wF)^3(t_1, t_2, t_3) = (t_1^{p^3}, t_2^{p^3}, t_3^{p^3}). \tag{3.1}$$

Consequently,

$$T^{wF} = \{\mathrm{diag}(a, a^p, a^{p^2}) : a \in \mathbb{F}_{p^3}^\times\} \simeq \mathbb{F}_{p^3}^\times,$$

so the finite group acting on the right is cyclic of order $p^3 - 1$. Equivalently, $Y(w) \to X(w)$ is a right torsor under this cyclic group. Fix a primitive $(p^3 - 1)$-st root $\mu \in \mathbb{F}_{p^3}^\times$ and let

$$\tau := \mathrm{diag}(\mu, \mu^p, \mu^{p^2})$$

be the corresponding generator.

By Proposition 3.6, both $Y(w)$ and its normal compactification $\overline{Y}(w)$ are disjoint unions of $p-1$ irreducible components. The generator τ permutes them transitively, whereas its $(p-1)$-st power

$$\sigma := \tau^{p-1}$$

preserves every component and has order $N = p^2 + p + 1$. If μ is the primitive $(p^3 - 1)$-st root used above, then $\bar{\zeta} := \mu^{1-p}$ is a primitive N-th root. Thus the Teichmüller parameter relevant to the action on a chosen component is

$$\widehat{\zeta} = [\bar{\zeta}] = [\mu^{1-p}] \in W(\mathbb{F}_{p^3})^\times.$$

The Brauer–Lefschetz calculation for σ is stated in Theorem 1.2 and proved in Section 3.7.

3.2. **Explicit description of the varieties.** Let V be a three-dimensional vector space over $k = \mathbb{F}_p$, where p is prime. Let Ω_V denote the Drinfeld half-space, the complement of all k-rational lines in $\mathbb{P}(V) = \mathbb{P}^2$.

Proposition 3.1. *The variety Ω_V is isomorphic to the Deligne–Lusztig variety $X := X(w)$ corresponding to the Coxeter element $w = (123) \in S_3$ for $G = \mathrm{GL}_3$.*

Proof. See [DL76, §2.1]. □

To describe this embedding explicitly, fix homogeneous coordinates x_0, x_1, x_2 on $\mathbb{P}(V) \simeq \mathbb{P}^2$ and dual homogeneous coordinates y_0, y_1, y_2 on $\mathbb{P}(V^*) \simeq \mathbb{P}^2$. We recall the definition of Moore determinant.

Definition 3.2 (**The Moore determinant**)**.** For indeterminates $w_1, \ldots, w_n$, define the Moore determinant $\Delta_p(w_1, \ldots, w_n)$ by

$$\begin{vmatrix} w_1 & w_2 & \cdots & w_n \\ w_1^p & w_2^p & \cdots & w_n^p \\ \vdots & \vdots & \ddots & \vdots \\ w_1^{p^{n-1}} & w_2^{p^{n-1}} & \cdots & w_n^{p^{n-1}} \end{vmatrix}$$

viewed as an element of $\mathbb{F}_p[w_1, \dots, w_n]$.

Lemma 3.3. *The Moore determinant is the product of all normalized linear forms over $\mathbb{F}_p$, where the last nonzero coefficient is normalized to one; explicitly,*

$$\Delta_p(w_1, \dots, w_n) = \prod_{i=1}^{n} \prod_{(a_1,\dots,a_{i-1}) \in \mathbb{F}_p^{i-1}} (a_1 w_1 + \cdots + a_{i-1} w_{i-1} + w_i)\,.$$

This identity holds in $\mathbb{F}_p[w_1, \dots, w_n]$.

Proof. See [Gos96, Corollary 1.3.7]. □

Definition 3.4. Let $\Pi \in k[X, Y, Z]$ be the product of the linear forms indexed by the nonzero vectors of k^3:

$$\Pi(X, Y, Z) := \prod_{(a_0,a_1,a_2) \in k^3 \setminus \{0\}} (a_0 X + a_1 Y + a_2 Z). \tag{3.2}$$

For each point of $\mathbb{P}^2(k)$, choose the unique representative whose last nonzero coordinate is one, and define

$$\Theta(X, Y, Z) := \prod_{[a_0:a_1:a_2] \in \mathbb{P}^2(k)} (a_0 X + a_1 Y + a_2 Z), \tag{3.3}$$

using these representatives. With this normalization, the Moore-product formula gives the identity

$$\Theta(X, Y, Z) = \Delta_p(X, Y, Z). \tag{3.4}$$

Remark 3.1. The chosen representatives, whose last nonzero coordinate is one, make Θ a specified polynomial. If arbitrary representatives are used, then Θ changes by a scalar in $k^\times$; hence Θ^{p-1}, $D_+(\Theta)$, and the localization at Θ do not change, although the labels of the components Y_ℓ may be permuted.

With the stated normalization, group the factors of Π by their projective classes. If L is the normalized linear form for one class, then

$$\prod_{c \in k^\times} cL = \left(\prod_{c \in k^\times} c \right) L^{p-1} = -L^{p-1}$$

in $k[X, Y, Z]$. Since $|\mathbb{P}^2(k)| = p^2 + p + 1$ is odd, it follows that

$$\Pi = -\Theta^{p-1} = (-1)^p \Theta^{p-1} \qquad \text{in } k[X, Y, Z]. \tag{3.5}$$

The preceding description gives the following affine presentation.

Lemma 3.5. *Put*

$$t = \frac{x_1}{x_0}, \qquad v = \frac{x_2}{x_0}, \qquad \theta := \Theta(1, t, v).$$

Then $X = D_+(\Theta)$ lies in the chart $x_0 \neq 0$, and

$$X \simeq \operatorname{Spec} k[t, v, \theta^{-1}].$$

We next describe the affine cover Y. By [DL76, §2.2], it is defined in $\mathbb{A}^3_k$, with coordinates X_0, X_1, X_2, by

$$\Delta_p(X_0, X_1, X_2)^{p-1} = 1.$$

Since $X_0 \neq 0$ on this hypersurface, set

$$(t, v, z) = (X_1/X_0, X_2/X_0, 1/X_0), \qquad (X_0, X_1, X_2) = (1/z, t/z, v/z).$$

Homogeneity gives $\Delta_p(X_0, X_1, X_2) = X_0^N \theta(t, v)$, and hence

$$z^{p^3-1} = \theta(t, v)^{p-1}. \tag{3.6}$$

The right action of τ multiplies the first column by μ, so $\tau^* z = \mu^{-1} z$ and $\sigma^* z = \mu^{1-p} z = \bar{\zeta} z$. Because $\mathbb{F}_p^\times$ is the set of roots of $W^{p-1} - 1$,

$$W^{p-1} - 1 = \prod_{\ell \in \mathbb{F}_p^\times} (W - \ell).$$

Writing $N = p^2 + p + 1$, the defining polynomial factors exactly as

$$z^{p^3-1} - \theta^{p-1} = \prod_{\ell \in \mathbb{F}_p^\times} (z^N - \ell\theta). \tag{3.7}$$

Proposition 3.6. *For $G = \mathrm{GL}_3$, the group T^{wF} acting on the right is cyclic of order $p^3 - 1$. The affine scheme*

$$Y := \operatorname{Spec} \frac{k[t, v, \theta^{-1}, z]}{(z^{p^3-1} - \theta^{p-1})}$$

is a finite étale right T^{wF}-torsor over X, and it is the disjoint union of the $p - 1$ integral schemes Y_ℓ:

$$Y = \bigsqcup_{\ell \in \mathbb{F}_p^\times} Y_\ell, \tag{3.8}$$

where

$$Y_\ell := \operatorname{Spec} \frac{k[t, v, \theta^{-1}, z]}{(z^N - \ell\theta)}. \tag{3.9}$$

Proof. The factors in (3.7) are pairwise comaximal over $k[t, v, \theta^{-1}]$: for $\ell \neq \ell'$, their difference is $(\ell' - \ell)\theta$, which is a unit. The Chinese remainder theorem therefore gives the displayed disjoint union.

It remains to prove that each factor is integral. Let $K = k(t, v)$, fix $\ell \in \mathbb{F}_p^\times$, and choose any nonconstant linear factor L of θ. Its discrete valuation satisfies

$$\nu_L(\ell\theta) = 1.$$

If $\alpha^N = \ell\theta$ and $\widetilde{\nu}$ is an extension of ν_L to $K(\alpha)$, with ramification index e, then

$$N\widetilde{\nu}(\alpha) = \widetilde{\nu}(\ell\theta) = e.$$

Hence $N \mid e$. Since

$$e \leq [K(\alpha) : K] \leq N,$$

all three quantities equal N. Thus $z^N - \ell\theta$ is irreducible over K, and Gauss's lemma shows that its quotient over the localization $k[t, v, \theta^{-1}]$ is a domain. The same valuation argument over $\bar{k}(t, v)$ proves that each Y_ℓ is geometrically integral. Finally, $p \nmid N$ and z is invertible on this quotient, so its derivative Nz^{N-1} is

invertible. Each $Y_\ell \to X$ is therefore finite étale of degree N, and their disjoint union has degree $(p-1)N = p^3 - 1$. □

These presentations fit into the following commutative diagram.

$$\begin{array}{ccc}
Y = \mathrm{Spec}\Big(\frac{k[t,v,\theta^{-1},z]}{(z^{p^3-1}-\theta^{p-1})}\Big) = \bigsqcup_{\ell\in\mathbb{F}_p^\times} Y_\ell & \hookrightarrow & \overline{Y} = \bigsqcup_{\ell\in\mathbb{F}_p^\times} \overline{Y}_\ell \\
\downarrow{\scriptstyle \mathbb{Z}/(p^3-1)\mathbb{Z}} & & \downarrow{\scriptstyle \mathbb{Z}/(p^3-1)\mathbb{Z}} \\
X = \mathrm{Spec}(k[t,v,\theta^{-1}]) & \hookrightarrow & \overline{X} = \mathrm{V}(f_0,f_1) = \mathrm{Bl}_Z\mathbb{P}^2 \subset \mathbb{P}^2\times\mathbb{P}^2 \\
\downarrow & & \downarrow \\
\mathbb{P}^2_{[1;t;v]} & \xrightarrow{\quad\cong\quad} & \mathbb{P}^2
\end{array}$$

Remark 3.2. The schemes Y and $\overline{Y}$ are disjoint unions of $p-1$ irreducible components. The generator τ of T^{wF} cyclically permutes them: it sends Y_ℓ to $Y_{\mu^{-N}\ell}$, and μ^{-N} generates $\mathbb{F}_p^\times$. For $p > 2$, its alternating Brauer trace on the cohomology of the whole disjoint union is zero. For $p = 2$, there is one component and $\tau = \sigma$. We therefore fix a component $\overline{Y}_\ell$ and compute $L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell)$ for its stabilizer generator $\sigma = \tau^{p-1}$.

We use Langer's description [Lan19] of the smooth compactification $\overline{X}$ constructed in [DL76, §9.10]. Let x_0, x_1, x_2 and y_0, y_1, y_2 be the homogeneous coordinates on $\mathbb{P}(V) = \mathbb{P}^2$ and $\mathbb{P}(V^*) = \mathbb{P}^2$, respectively.

Proposition 3.7. *The surface $\overline{X}$ is isomorphic to the complete intersection*

$$\overline{X} = \mathrm{V}(f_0, f_1) \subset \mathbb{P}(V)\times\mathbb{P}(V^*) = \mathbb{P}^2\times\mathbb{P}^2,$$

where

$$f_0 = x_0y_0 + x_1y_1 + x_2y_2, \qquad f_1 = x_0^p y_0 + x_1^p y_1 + x_2^p y_2.$$

Proof. See [Lan19, §5, Proposition 5.2]. □

This is the wonderful compactification of the hyperplane arrangement in the sense of De Concini–Procesi [DCP95]. It is the blow-up of $\mathbb{P}^2$ at all k-rational points [Lan19, §5].

Recall that $V = \mathbb{F}_p^3$. Let π_1 and π_2 be the two projections from $\overline{X} \subset \mathbb{P}^2 \times \mathbb{P}^2$. The morphism $\pi_1 \colon \overline{X} \to \mathbb{P}^2$ is the blow-up at the finite set $\mathbb{P}^2(\mathbb{F}_p)$.

We denote by $\mathcal{L}(V)$ the set of all proper k-linear subspaces of $\mathbb{P}(V)$. Every k-linear subspace $L \in \mathcal{L}(V)$ is of the form $\mathbb{P}(V/W)$ for some k-linear subspace $W \subset V$. For $L = \mathbb{P}(V/W) \subseteq \mathbb{P}(V)$ we set

$$L^\perp = \mathbb{P}(W^*) \subseteq \mathbb{P}(V^*).$$

If η_L denotes the generic point of L, then we set

$$E_L := \overline{\pi_1^{-1}(\eta_L)} \subset \overline{X}.$$

We use the following description from [Lan19, §5, Lemma 5.1].

Proposition 3.8. *For any $L \in \mathcal{L}(V)$ we have*

$$E_L = L \times_k L^\perp \quad \text{and} \quad \pi_1^{-1}(L) = \bigcup_{\substack{L' \in \mathcal{L}(V) \\ L' \subset L}} E_{L'}.$$

In particular, each E_L is a smooth curve over k, and $\pi_1^{-1}(L)$ is of pure dimension 1.

Therefore, $\pi_1^{-1}(X) = \overline{X} \setminus \bigcup_{L \in \mathcal{L}(V)} E_L$.

Proposition 3.9 (The split boundary and its incidence count)**.** *Put $N = p^2+p+1$. The boundary $\overline{X} \setminus X$ has $2N$ irreducible components:*

$$\overline{X} \setminus X = \bigcup_{P \in \mathbb{P}^2(\mathbb{F}_p)} E_P \ \cup \bigcup_{L \in (\mathbb{P}^2)^\vee(\mathbb{F}_p)} \widetilde{L}.$$

Here E_P is the exceptional curve over the rational point P, and $\widetilde{L}$ is the strict transform of the rational line L. Their self-intersections are

$$E_P^2 = -1, \qquad \widetilde{L}^2 = 1 - (p+1) = -p.$$

Moreover, E_P and $\widetilde{L}$ meet transversely exactly when $P \in L$. There are $N(p+1)$ such incidences.

Proof. There are N rational points and, by projective duality, N rational lines in $\mathbb{P}^2_{\mathbb{F}_p}$. Blowing up a point produces an exceptional curve of self-intersection -1. A rational line has self-intersection 1 in $\mathbb{P}^2$ and contains exactly $p+1$ rational points. Blowing up those points therefore changes its self-intersection to $1-(p+1) = -p$. The final assertion follows from the fact that every one of the N points lies on $(p+1)$-many rational lines. □

3.3. **The Bonnafé–Rouquier slice in the split case.** We now evaluate the slice mechanism of Section 2.8 for the split Frobenius. On $Y(T)$, the map F is multiplication by p. Put

$$N = p^2 + p + 1.$$

The solutions of (2.27) are

$$\lambda_1 = (p+1, -1, -p), \qquad \lambda_2 = (1, p, -p-1), \qquad m_1 = m_2 = N. \tag{3.10}$$

Indeed, since

$$w(e_1) = e_2, \qquad w(e_2) = e_3, \qquad w(e_3) = e_1,$$

we have

$$\lambda_1 - pw(\lambda_1) = N(e_1 - e_2)$$

and

$$\lambda_2 - pw(\lambda_2) = N(e_1 - e_3) = Ns_1(\alpha_2^\vee).$$

Both λ_i are primitive, so the torsion-free conditions in (2.27) are satisfied.

For $(z_1, z_2) \in (\mathbb{G}_m)^2$,

$$\lambda_1(z_1)\lambda_2(z_2) = \mathrm{diag}\Big(z_1^{p+1}z_2, z_1^{-1}z_2^{p}, z_1^{-p}z_2^{-p-1}\Big).$$

Substitution in (2.30) gives

$$K_{12} = \Big\{(\zeta, \zeta^{-(p+1)}) : \zeta^N = 1\Big\} \simeq \mu_N. \tag{3.11}$$

Consequently, the action on the normal slice is

$$\zeta(x, y) = (\zeta x, \zeta^{-(p+1)}y) = (\zeta x, \zeta^{p^2}y),$$

because $-(p+1) \equiv p^2 \pmod N$. Luna's theorem therefore gives the cyclic quotient singularity

$$\boxed{\frac{1}{p^2+p+1}(1, p^2)}. \tag{3.12}$$

This agrees with the normalized toric cone computed below. That cone is

$$\mathrm{Cone}\big((1,0),(p+1,N)\big).$$

With $e_2 = (1,0)$ and $e_1 = (1,1)$, its second primitive ray is

$$(p+1, N) = Ne_1 - p^2e_2.$$

Thus its cyclic quotient parameters are also (N, p^2). Equivalently, since $p^3 \equiv 1 \pmod N$, the quotient may be written as $\frac{1}{N}(1,p)$; this reverses the exceptional chain. Finally,

$$\frac{N}{p^2} = [[\underbrace{2, \ldots, 2}_{p-1 \text{ terms}}, p+2]],$$

which is the continued fraction appearing in the split resolution.

3.4. **Local study of the normalization $\overline{Y}_\ell$.** The global surface $\overline{Y}_\ell(w)$ is the normalization of $\overline{X}(w)$ in the function field of a fixed component $Y_\ell(w)$. We now compute its completed local rings over the zero-dimensional boundary strata. These local calculations are used to determine the type of singularities that occur in $\overline{Y}_\ell(w)$. Consider the affine chart

$$U_{10} = \{x_1 \neq 0,\ y_0 \neq 0\} \subset \mathbb{P}^2 \times \mathbb{P}^2.$$

The six charts $U_{ij} = \{x_i \neq 0,\ y_j \neq 0\}$, with $i \neq j$, cover $\overline{X}$. On U_{10}, put

$$t = \frac{x_0}{x_1}, \qquad v = \frac{x_2}{x_1}, \qquad r = \frac{y_1}{y_0}, \qquad w = \frac{y_2}{y_0}.$$

The equations of $\overline{X}$ become

$$t + r + vw = 0, \qquad t^p + r + v^p w = 0.$$

Eliminating r gives

$$R = k[U_{10} \cap \overline{X}] = \frac{k[t, v, w]}{(t - t^p + vw - v^p w)}. \tag{3.13}$$

The Kummer presentation of Proposition 3.6 is written in the chart $x_0 \neq 0$: there $\theta = \Theta(1, x_1/x_0, x_2/x_0)$, and the Kummer generator is $z = 1/X_0$. Near the boundary points studied below we work instead in the chart $x_1 \neq 0$, so both

the defining polynomial and the Kummer generator have to be changed. With $t = x_0/x_1$ and $v = x_2/x_1$ as above, put

$$\theta_{10} := \Theta(t, 1, v). \tag{3.14}$$

This is the chart representative of Θ for $x_1 \neq 0$; the expression $\Theta(1, t, v)$ belongs to the chart $x_0 \neq 0$. Since Θ is homogeneous of degree $N = p^2 + p + 1$,

$$\Theta(x_0, x_1, x_2) = x_0^N\, \Theta\Big(1, \frac{x_1}{x_0}, \frac{x_2}{x_0}\Big) = x_1^N\, \Theta\Big(\frac{x_0}{x_1}, 1, \frac{x_2}{x_1}\Big),$$

and hence, on the overlap of the two charts,

$$\theta = \Big(\frac{x_1}{x_0}\Big)^N \theta_{10} = t^{-N}\theta_{10}. \tag{3.15}$$

Accordingly, we replace the Kummer generator $z = 1/X_0$ by

$$z_{10} := zt = \frac{1}{X_1}. \tag{3.16}$$

The linear forms x_0 and x_1 are factors of Θ, so t is a unit on X and z_{10} is a unit on Y_ℓ. By (3.15), the defining equation $z^N = \ell\theta$ of Y_ℓ becomes

$$z_{10}^N = \ell\, \theta_{10}. \tag{3.17}$$

Equivalently, (3.17) is obtained by repeating the proof of Proposition 3.6 with X_1 in place of X_0, using $\Delta_p(X_0, X_1, X_2) = X_1^N\, \theta_{10}(X_0/X_1, X_2/X_1)$. Since t is pulled back from $\overline{X}$, it is fixed by σ, and therefore

$$\sigma^* z_{10} = \bar{\zeta}\, z_{10},$$

exactly as for z. All local calculations below use the generator z_{10} and the equation (3.17). We proceed to construct the normalization of the local Kummer extension

$$\frac{R[z_{10}]}{(z_{10}^N - \ell\theta_{10})}, \qquad \ell \in \mathbb{F}_p^\times.$$

Lemma 3.10. *Let*

$$P = ([0:1:0], [1:0:0]) \in \overline{X}(k).$$

In the chart U_{10}, the point P is the origin $t = v = r = w = 0$. The elements

$$u := w, \qquad v := \frac{x_2}{x_1}$$

form a regular system of parameters at P, and there is a unit $\varepsilon \in \widehat{\mathcal{O}}_{\overline{X},P}^\times$ such that

$$\theta_{10} = \varepsilon v^{p+1} u. \tag{3.18}$$

More explicitly, if

$$E(t, v) := \prod_{a,b \in \mathbb{F}_p} (1 + at + bv)$$

and

$$\beta := -\frac{1 - v^{p-1}}{1 - t^{p-1}},$$

then $t = vw\beta$ in the completed local ring and one may take

$$\varepsilon = E(t, v)(1 - v^{p-1})(1 + w^{p-1}\beta^p). \tag{3.19}$$

The same factorization, up to multiplication of ε by a nonzero scalar, holds at every k-rational point of $\overline{X}$.

Proof. Write

$$f = t - t^p + vw - v^p w.$$

At the origin, $\partial f/\partial t = 1$. Thus the formal implicit function theorem applied to (3.13) gives

$$\widehat{\mathcal{O}}_{\overline{X},P} \simeq k[[v, w]],$$

so $u = w$ and $v = x_2/x_1$ are indeed a regular system of parameters.

We next factor the defining polynomial in the correct chart. From the definition

$$\theta_{10} = \prod_{[a_0:a_1:a_2]\in\mathbb{P}^2(\mathbb{F}_p)} (a_0 t + a_1 + a_2 v),$$

the factors that vanish at P are precisely those with $a_1 = 0$. They are

$$t \quad \text{and} \quad at + v \qquad (a \in \mathbb{F}_p).$$

Their product is $-v(t^p - tv^{p-1})$. The factors with $a_1 \neq 0$ have product $-E(t, v)$: renormalizing them to $a_1 = 1$ contributes $(\prod_{c\in\mathbb{F}_p^\times} c)^p = -1$. The polynomial $E(t, v)$ is a unit because $E(0, 0) = 1$. Using

$$\prod_{c\in\mathbb{F}_p} (T + cQ) = T^p - TQ^{p-1},$$

we obtain the exact identity

$$\theta_{10} = E(t, v)v(t^p - tv^{p-1}). \tag{3.20}$$

The equation $f = 0$ gives

$$t^p - t = vw(1 - v^{p-1}),$$

and hence

$$t^p - tv^{p-1} = (1 - v^{p-1})(t + vw). \tag{3.21}$$

It also gives

$$t(1 - t^{p-1}) = -vw(1 - v^{p-1}).$$

Both parenthesized factors are units at P, so

$$t = vw\beta, \qquad \beta = -\frac{1 - v^{p-1}}{1 - t^{p-1}} \in \widehat{\mathcal{O}}^\times_{\overline{X},P}.$$

Finally, using $t + vw = t^p + v^p w$, we find

$$t + vw = v^p w(1 + w^{p-1}\beta^p). \tag{3.22}$$

Substituting (3.21) and (3.22) into (3.20) yields

$$\theta_{10} = E(t, v)(1 - v^{p-1})(1 + w^{p-1}\beta^p)v^{p+1}w = \varepsilon v^{p+1}u.$$

Each of the three factors defining ε has nonzero constant term (in fact, constant term 1), so ε is a unit. This proves the claimed factorization at P.

The group $\mathrm{PGL}_3(\mathbb{F}_p)$ acts transitively on the $\mathbb{F}_p$-rational flags parametrized by $\overline{X}(\mathbb{F}_p)$. It permutes the linear factors of Θ, changing their product only by a nonzero scalar. Transporting the calculation above therefore proves the final assertion. □

For the later toric notation, set $s := u = w$ and retain the coordinate $v = x_2/x_1$. If $N = p^2 + p + 1$, then $p \nmid N$ and $\varepsilon \equiv 1 \pmod{(u, v)}$. Hensel's lemma therefore gives a unique $\eta \in 1 + (u, v)$ with $\eta^N = \varepsilon$. Replacing the Kummer coordinate by z/η removes the unit. The remaining nonzero component scalar can be normalized to one up to an isomorphism of complete local k-algebras by rescaling a boundary parameter. Thus, we get the following equation:

$$z^N = v^{p+1}s. \tag{3.23}$$

3.4.1. *Local calculations.* The wonderful compactification $\overline{X}(w)$ is the blow-up of $\mathbb{P}^2$ at all its $\mathbb{F}_p$-rational points [Lan19]. Its boundary has one exceptional component E_P for every rational point P and one component $\widetilde{L}$, the strict transform, for every rational line L. An $\mathbb{F}_p$-rational point of $\overline{X}(w)$ is a zero-dimensional boundary stratum corresponding to an incidence $P \in L$: the exceptional curve over P parametrizes tangent directions at P, and its $\mathbb{F}_p$-points are precisely the $p+1$ rational lines through P. Thus, in an étale neighborhood of such a point, the boundary is a pair of transverse branches

$$E_P = (v = 0), \qquad \widetilde{L} = (s = 0),$$

and the completed cover has the Kummer equation

$$z^N = v^{p+1}s.$$

The following calculation explains why the normal eigenvalue is constant within each of the two boundary families.

Lemma 3.11 (Normal character along a ramified divisor)**.** *Let $K \subset L$ be the degree-N extension of function fields induced by a chosen component of the cover. Let D be a prime divisor on the base, let t be a uniformizer at its generic point, and suppose that in L*

$$z^N = \varepsilon t^a, \qquad \sigma(z) = \bar{\zeta} z,$$

where ε is a unit, σ fixes K, $\bar{\zeta}$ has order N, and $\gcd(a, N) = 1$. Then there is a unique prime divisor D' above D, and it is totally ramified. If $ab \equiv 1 \pmod{N}$, then σ acts on a local parameter of D', and hence on $I_{D'}/I_{D'}^2$, by $\bar{\zeta}^{\,b}$. In the convention of Theorem 2.3, the normal eigenvalue is therefore

$$\nu_{D'} = \bar{\zeta}^{\,-b}.$$

The same eigenvalue occurs along the strict transform of D' under any equivariant resolution that is an isomorphism at its generic point.

Proof. Let v' extend the valuation of D, and let e be its ramification index. From $Nv'(z) = ae$ and $\gcd(a, N) = 1$, one obtains $N \mid e$. Since $e \le [L : K] = N$, we have $e = N$; the fundamental inequality then implies that D' is unique and has residue degree one.

Choose $c \in \mathbb{Z}$ such that $ab + Nc = 1$. Then $\pi = z^b t^c$ has valuation one, so it is a local parameter at D', and

$$\sigma(\pi) = \bar{\zeta}^{\,b}\pi.$$

Reducing modulo π^2 gives the claimed character. An equivariant resolution that is an isomorphism at the generic point preserves this discrete valuation ring and its character. □

In the present cover, every rational line occurs with multiplicity one in the boundary divisor, whereas the exceptional divisor over a rational point occurs with multiplicity $p+1$, because exactly $p+1$ rational lines pass through that point. Consequently, at the generic points of the two boundary types one has

$$a_{\widetilde{L}} = 1, \qquad a_{E_P} = p+1.$$

Lemma 3.11 therefore gives

$$\nu_{\widetilde{L}'} = \bar{\zeta}^{-1}, \qquad \nu_{\widetilde{E}'_P} = \bar{\zeta}^{p},$$

because

$$(-p)(p+1) + N = 1.$$

This proves the constancy across all curves in either family. It also shows that the same Kummer multiplicities occur at every incidence. Along an individual fixed curve, the eigenvalue is constant for the intrinsic reason that σ acts over the identity on its normal line bundle, hence by an invertible regular function and therefore is constant on a connected projective integral curve.

3.5. **Toric normalization, resolution, and fixed locus.**

3.5.1. *Interpretation of the local model as a toric surface.* We use the semigroup, normalization, and resolution conventions collected in Section 2.7. Put $N = p^2 + p + 1$.

Introduce the affine algebraic surface

$$Y_{\mathscr{A}} := \operatorname{Spec} \frac{k[v,s,z]}{(z^N - v^{p+1}s)}.$$

Its completion at the origin is the scalar-one algebra A'_P of the preceding subsection. Let

$$\nu_\sigma \colon U_\sigma := Y_{\mathscr{A}}^{\mathrm{norm}} \longrightarrow Y_{\mathscr{A}}$$

be its normalization, and let $o_\sigma \in U_\sigma$ be the point above the origin. Compatibility of normalization with completion give

$$\widehat{\mathcal{O}}_{\overline{Y},\overline{P}} \xrightarrow{\sim} \widehat{\mathcal{O}}_{U_\sigma,o_\sigma}. \tag{3.24}$$

We first observe that we have an isomorphism of k-algebras

$$\frac{k[v,s,z]}{(z^N - v^{p+1}s)} \simeq k[X^pY, XY, XY^{p^2}] \tag{3.25}$$

given by $z \mapsto XY, v \mapsto X^pY$ and $s \mapsto XY^{p^2}$. The map is surjective. Both rings are two-dimensional domains, so its kernel is a prime ideal of height zero and hence is zero.

Let $M \simeq \mathbb{Z}^2$ be the character lattice of the dense torus, put $N_{\mathrm{lat}} = \mathrm{Hom}(M,\mathbb{Z})$, and write $\chi^{(a,b)} = t_1^a t_2^b$. Taking $t_1 = XY = z$ and $t_2 = X^pY = v$, the three monomials become

$$v = \chi^{(0,1)} = t_2, \qquad z = \chi^{(1,0)} = t_1, \qquad s = \chi^{(N,-p-1)} = t_1^N t_2^{-p-1}. \tag{3.26}$$

Consequently,

$$\mathscr{A} = \{(0,1), (N,-p-1), (1,0)\} \subset M \simeq \mathbb{Z}^2,$$

and the corresponding affine toric variety is

$$Y_{\mathscr{A}} = \operatorname{Spec} k[S],$$

where S is the semigroup generated by $\mathscr{A}$. This is the algebraic local model whose completion gives A'_P; it is not being identified with a global surface over $\overline{X}(w)$.

By (2.20), the toric normalization introduced above is

$$U_\sigma = \operatorname{Spec} k[S_\sigma],$$

where S_σ is the saturation of S and σ is the dual cone of the cone generated by $\mathscr{A}$. Its completed local ring at the torus-fixed point o_σ is identified with the completed local ring of $\overline{Y}$ at $\overline{P}$ by (3.24).

Lemma 3.12. *For the cone $\sigma = \mathrm{Cone}((1,0),(p+1,N))$, the Hilbert basis $\mathscr{H}$ is*

$$\mathscr{H} = \{u_0 := (1,0),\ u_1 := (1,1),\ u_2 := (1,2),\ \dots\ ,\ u_p := (1,p),\ u_{p+1} := (p+1,N)\}$$

Proof. The consecutive determinants are

$$\det(u_i, u_{i+1}) = 1 \quad (0 \le i < p), \qquad \det(u_p, u_{p+1}) = N - p(p+1) = 1.$$

Thus every cone spanned by two consecutive vectors is smooth, and the union of these cones is σ. Every lattice point of σ therefore lies in one of these smooth cones and is a nonnegative integral combination of its two generators. The vectors $u_0, \dots, u_p$, whose first coordinate is 1, are indecomposable in the semigroup $\sigma \cap N_{\mathrm{lat}}$; the final vector u_{p+1} is the primitive generator of the other extremal ray. Hence the displayed set is precisely the Hilbert basis. □

Take $e_1 = (1,1), e_2 = (1,0)$. Then $(p+1,N) = Ne_1 - p^2e_2$, so the cone is in Cox normal form $\sigma = \mathrm{Cone}(e_2, Ne_1 - p^2e_2)$ with numerical parameters $d = N$ and $a = p^2$. As recalled in Section 2.7, the Hirzebruch–Jung expansion of d/a determines the minimal regular subdivision and hence the minimal toric resolution $\phi : X_\Sigma \to U_\sigma$.

Proposition 3.13. *The Hirzebruch–Jung continued fraction is*

$$\frac{N}{p^2} = [[\underbrace{2, 2, \dots, 2}_{p-1 \text{ terms}}, p+2]].$$

Proof. The negative Euclidean algorithm begins with

$$b_1 = \left\lceil \frac{p^2+p+1}{p^2} \right\rceil = 2, \qquad r_1 = 2p^2 - N = p^2 - p - 1.$$

Put $r_0 = p^2$. Inductively,

$$r_j = p^2 - j(p+1) \quad (0 \le j \le p-1).$$

Indeed, for $1 \le j \le p-2$,

$$\frac{r_{j-1}}{r_j} = 1 + \frac{p+1}{r_j} \in (1,2),$$

so the next quotient is two and

$$r_{j+1} = 2r_j - r_{j-1} = p^2 - (j+1)(p+1),$$

as required. Finally,

$$r_{p-1} = p^2 - (p-1)(p+1) = 1, \qquad r_{p-2} = p+2.$$

Thus the first $p-1$ quotients are two and the last quotient is $p+2$. □

3.5.2. *The minimal resolution ϕ.* For the Hilbert-basis vectors $u_0, \dots, u_{p+1}$ above, put

$$\sigma_i := \mathrm{Cone}(u_{i-1}, u_i) \qquad (1 \le i \le p+1),$$

and let Σ consist of these cones and their faces. The determinant calculation in the preceding lemma shows that each σ_i is regular. Since $|\Sigma| = \sigma$, the subdivision defines a toric resolution

$$\phi\colon X_\Sigma \longrightarrow U_\sigma.$$

This morphism is projective; see [CLS11, §10.2]. The exceptional curves D_i, corresponding to the rays generated by u_i for $1 \le i \le p$, form a chain of smooth rational curves. The relations

$$u_{i-1} + u_{i+1} = 2u_i \quad (1 \le i < p), \qquad u_{p-1} + u_{p+1} = (p+2)u_p$$

give

$$D_i^2 = -2 \quad (1 \le i < p), \qquad D_p^2 = -(p+2).$$

Thus ϕ is the minimal resolution associated with the Hirzebruch–Jung expansion above, as recalled in Section 2.7. The surface X_Σ is covered by the $p+1$ affine charts U_{σ_i}, with

$$U_{\sigma_i} \cap U_{\sigma_j} = U_{\sigma_i \cap \sigma_j};$$

see [CLS11, Ch. 3, (3.1.3)].

Let $\rho_\ell\colon \widetilde{\overline{Y}}_\ell(w) \longrightarrow \overline{Y}_\ell(w)$ be the minimal resolution of the normal projective surface $\overline{Y}_\ell(w)$. Its existence follows from resolution theory for normal surfaces. Write

$$\overline{Y} := \overline{Y}_\ell(w), \qquad \widetilde{\overline{Y}} := \widetilde{\overline{Y}}_\ell(w),$$

and put

$$E_{\overline{P}} := \rho_\ell^{-1}(\overline{P}), \qquad E_\Sigma := \phi^{-1}(o_\sigma).$$

We compare the two resolutions on an étale neighborhood. Before the coordinate changes used in (3.23), the local Kummer equation is

$$z_0^N = \ell \varepsilon v^{p+1} u, \qquad N = p^2 + p + 1.$$

The formula in (3.19) shows that ε belongs to $\mathcal{O}^\times_{\overline{X},P}$: all its denominators are units at P. Thus this equation holds on an affine open neighborhood of P. Since $p \nmid N$, adjoining a root η of $\eta^N = \varepsilon$ gives a finite étale cover of this neighborhood. Choose a point P_V above P and an affine open neighborhood V of P_V in this cover. On V, put

$$z := z_0/\eta, \qquad s := \ell u.$$

The Kummer equation becomes

$$z^N = v^{p+1} s.$$

Since v, u are regular parameters at P, after shrinking V the morphism

$$\gamma\colon V \longrightarrow \mathbb{A}^2_k, \qquad \gamma = (v, s),$$

is étale. We may also assume that P_V is the only point of V mapping to the origin.

Normalization commutes with étale base change; see the Stacks Project, Tag 03GV. Consequently, the preceding change of coordinates induces an isomorphism

$$W := \overline{Y} \times_{\overline{X}} V \xrightarrow{\sim} U_\sigma \times_{\mathbb{A}^2_k} V.$$

In particular, the projections

$$a\colon W \longrightarrow \overline{Y}, \qquad b\colon W \longrightarrow U_\sigma$$

are étale. The scheme W is affine, since it is finite over V. Let P_W be its unique point above P_V. Then

$$a(P_W) = \overline{P}, \qquad b(P_W) = o_\sigma, \qquad \kappa(P_W) = k.$$

Since b is étale and $b^{-1}(o_\sigma)$ has only the point P_W, the surface W is regular away from P_W.

The two base changes

$$\widetilde{\overline{Y}} \times_{\overline{Y}} W \longrightarrow W, \qquad X_\Sigma \times_{U_\sigma} W \longrightarrow W$$

are resolutions of W. Over P_W, étale base change preserves the exceptional curves and their self-intersections. Hence both resolutions are minimal. Uniqueness of the minimal resolution gives an isomorphism over W,

$$\widetilde{\overline{Y}} \times_{\overline{Y}} W \xrightarrow{\sim} X_\Sigma \times_{U_\sigma} W. \tag{3.27}$$

Put $\widetilde{W} := \widetilde{\overline{Y}} \times_{\overline{Y}} W$. We obtain the commutative diagram

$$\begin{array}{ccccc}
\widetilde{\overline{Y}} & \longleftarrow & \widetilde{W} & \longrightarrow & X_\Sigma \\
\downarrow{\scriptstyle \rho_\ell} & & \downarrow & & \downarrow{\scriptstyle \phi} \\
\overline{Y} & \xleftarrow{a} & W & \xrightarrow{b} & U_\sigma \\
\downarrow{\scriptstyle \overline{\pi}_\ell} & & \downarrow & & \downarrow \\
\overline{X} & \longleftarrow & V & \xrightarrow[\gamma]{} & \mathbb{A}^2_k,
\end{array}$$

in which all four squares are cartesian.

The fibre of $\widetilde{W} \to W$ over P_W identifies with both $E_{\overline{P}}$ and E_Σ. Thus the exceptional curves above $\overline{P}$ have the same intersection matrix as the curves $D_1, \ldots, D_p$. Moreover, the affine open subsets

$$U_{\sigma_i} \times_{U_\sigma} W, \qquad 1 \le i \le p+1,$$

cover $\widetilde{W}$, with transition maps obtained by base change from those of X_Σ.

Let $G_\ell = \langle \sigma \rangle$ be the component stabilizer. It acts trivially on $\overline{X}$, and on the Kummer coordinate it acts by

$$\sigma(z_0) = \bar{\zeta} z_0,$$

where $\bar{\zeta}$ is a primitive N-th root of unity. Give V the trivial G_ℓ-action. Since η is a function on V, the change of coordinates above gives

$$\sigma(z) = \bar{\zeta} z, \qquad \sigma(v) = v, \qquad \sigma(s) = s.$$

The morphisms a and b are therefore G_ℓ-equivariant. Uniqueness of the minimal resolution lifts the actions on $\overline{Y}$ and U_σ to their resolutions and makes (3.27) equivariant. The lift on X_Σ is the restriction of its torus action. Hence the toric charts compute the stabilizer characters on the tangent and normal spaces and the fixed locus near $E_{\overline{P}}$. The lower squares in the diagram also identify the two boundary branches and preserve their ramification indices.

The number and orbits of the global boundary curves, their genera, the self-intersections of their strict transforms, and the total number of incidences require the global geometry of $\overline{X}$. Their intersection numbers are computed from pull-back divisors and the projection formula. These numbers, together with the local characters computed below, enter the Lefschetz–Riemann–Roch formula.

3.5.3. *Affine charts of the resolution.* We now write down these affine charts, compute the action of the component stabilizer $G_\ell = \langle\sigma\rangle \simeq \mathbb{Z}/N\mathbb{Z}$, and analyze its fixed locus.

The dual monoids must be computed cone by cone. For the first cone the primitive dual generators are $(0,1)$ and $(1,-1)$. For $2 \le i \le p$, the dual generators of $\sigma_i = \mathrm{Cone}(u_{i-1}, u_i)$ are

$$(i,-1), \qquad (1-i,1),$$

and for the last cone they are $(-p,1)$ and $(N,-p-1)$. Each displayed pair is the basis of M dual to the two primitive ray generators of the corresponding unimodular cone. Consequently, the charts for the local toric resolution X_Σ are as follows:

$$\begin{aligned}
U_{\sigma_1} &= \mathrm{Spec}\big(k[t_2, t_1 t_2^{-1}]\big) && \rightsquigarrow \quad \text{affine coordinates} \quad a_{1,1}, a_{1,2}\\
U_{\sigma_2} &= \mathrm{Spec}\big(k[t_1^2 t_2^{-1}, t_1^{-1} t_2]\big) && \rightsquigarrow \quad \text{affine coordinates} \quad a_{2,1}, a_{2,2}\\
U_{\sigma_3} &= \mathrm{Spec}\big(k[t_1^3 t_2^{-1}, t_1^{-2} t_2]\big) && \rightsquigarrow \quad \text{affine coordinates} \quad a_{3,1}, a_{3,2}\\
&\vdots && \vdots\\
U_{\sigma_i} &= \mathrm{Spec}\big(k[t_1^i t_2^{-1}, t_1^{1-i} t_2]\big) && \rightsquigarrow \quad \text{affine coordinates} \quad a_{i,1}, a_{i,2}\\
&\vdots && \vdots\\
U_{\sigma_{p+1}} &= \mathrm{Spec}\big(k[t_1^{-p} t_2, t_1^N t_2^{-p-1}]\big) && \rightsquigarrow \quad \text{affine coordinates} \quad a_{p+1,1},\ a_{p+1,2}
\end{aligned}$$

3.5.4. *Action of the component stabilizer and fixed locus.* The subgroup $\langle\sigma\rangle \le T^{wF}$ that stabilizes a component is cyclic of order N. On the local model

$$\mathrm{Spec}\, k[v,s,z]/(z^N - v^{p+1}s)$$

its generator acts by

$$\sigma(z) = \bar{\zeta} z, \qquad \sigma(v) = v, \qquad \sigma(s) = s,$$

where $\bar{\zeta} = \mu^{1-p}$ is a primitive N-th root. Under (3.25), this becomes

$$\sigma(t_1) = \bar{\zeta} t_1, \qquad \sigma(t_2) = t_2.$$

The action extends to $\overline{Y}_\ell(w)$ and lifts to its minimal resolution $\widetilde{\overline{Y}}_\ell(w)$. On the local toric charts it has the following weights:

$$\begin{aligned}
U_{\sigma_1}\colon &\quad (a_{1,1}, a_{1,2}) \longmapsto (a_{1,1}, \bar{\zeta} a_{1,2}),\\
U_{\sigma_i}\colon &\quad (a_{i,1}, a_{i,2}) \longmapsto (\bar{\zeta}^{\,i} a_{i,1}, \bar{\zeta}^{\,1-i} a_{i,2}) \quad (2 \le i \le p),\\
U_{\sigma_{p+1}}\colon &\quad (a_{p+1,1}, a_{p+1,2}) \longmapsto (\bar{\zeta}^{-p} a_{p+1,1}, a_{p+1,2}).
\end{aligned}$$

The endpoint charts also identify the two global boundary types. On U_{σ_1},

$$a_{1,1} = v, \qquad a_{1,2} = z/v, \qquad s = a_{1,1}^{p^2} a_{1,2}^N,$$

so $a_{1,2} = 0$ is the strict lift of the line-type branch $s = 0$, and its normal character is $\bar{\zeta}^{-1}$. On the last chart,

$$a_{p+1,1} = v/z^p, \qquad a_{p+1,2} = s, \qquad v = a_{p+1,2}^p a_{p+1,1}^N,$$

so $a_{p+1,1} = 0$ is the strict lift of the exceptional-type branch $v = 0$, and its normal character is $\bar{\zeta}^p$. In this local resolution there are exactly $p-1$ isolated fixed points

$$Q_{i-1} = \{a_{i,1} = a_{i,2} = 0\}, \qquad 2 \le i \le p, \tag{3.28}$$

and two fixed boundary branches

$$\{a_{1,2} = 0\}, \qquad \{a_{p+1,1} = 0\}. \tag{3.29}$$

Remark 3.3. In the local Galois model $Y_{\mathscr{A}} = \operatorname{Spec} k[v, s, z]/(z^N - v^{p+1}s)$, the stabilizer fixes $\{v = 0\}$ and $\{s = 0\}$ and acts on z through $\bar{\zeta}$. These are the two boundary branches through one incidence point. As the incidence point varies, the first branches are the local branches of the strict transforms of the N curves E_P, and the second branches are those of the strict transforms of the N curves $\widetilde{L}$. Thus the global fixed locus contains N curves $\widetilde{E}'_P$ and N curves $\widetilde{L}'_L$.

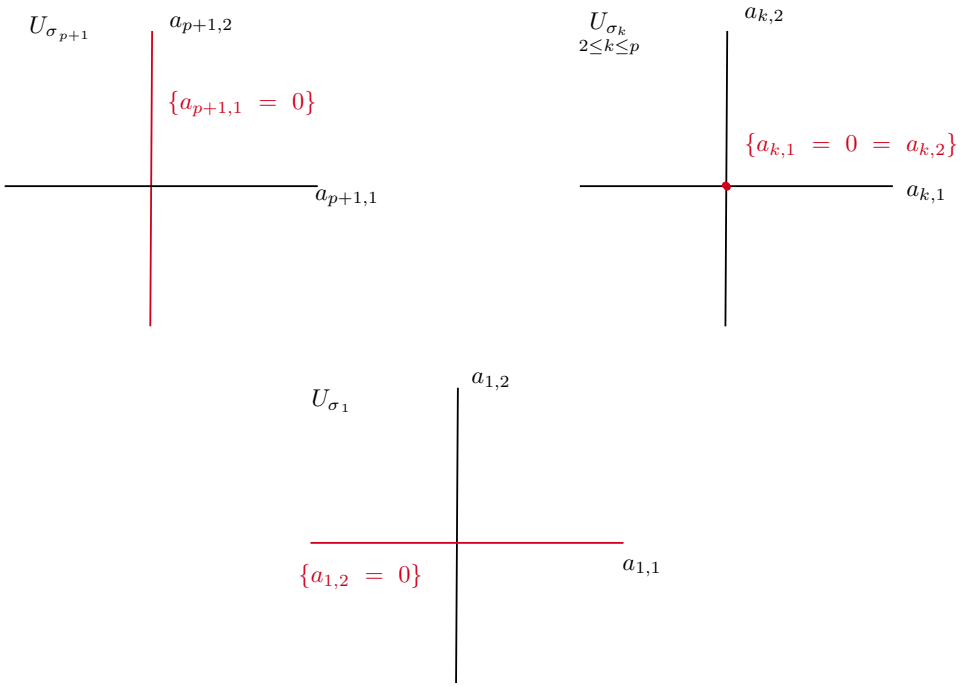


FIGURE 1. The fixed loci in the $(p+1)$ affine charts of the local toric resolution X_Σ

3.6. **Intersection number calculations.**

3.6.1. *Local computations on the minimal resolution.* The fan already contains the intersection data needed below. By the wall relations and intersection formulas (2.24)–(2.25), the following computation is entirely local. Let $D_i = V(\rho_i)$, where $\rho_i = \mathrm{Cone}(u_i)$. For the Hirzebruch–Jung coefficients

$$(b_1, \dots, b_p) = (\underbrace{2, \dots, 2}_{p-1 \text{ terms}}, p+2),$$

the wall relations $u_{i-1} + u_{i+1} = b_i u_i$ give

$$D_i \cdot D_j = \begin{cases} -b_i, & i = j, \\ 1, & |i-j| = 1, \\ 0, & |i-j| > 1, \end{cases} \qquad 1 \le i, j \le p; \tag{3.30}$$

see [CLS11, Theorem 10.2.5]. Thus the exceptional curves $E_i := D_i$, $1 \le i \le p$, form a negative-definite chain with

$$E_i^2 = -2 \quad (1 \le i < p), \qquad E_p^2 = -p - 2. \tag{3.31}$$

The exceptional locus is $\phi^{-1}(o_\sigma) = E_1 \cup \cdots \cup E_p$. The boundary divisors D_0 and D_{p+1} are noncomplete in the local toric surface; their global counterparts on $\widetilde{\overline{Y}}_\ell(w)$ are complete curves.

For any character χ^m, its torus-invariant divisor is

$$\mathrm{div}(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, u_\rho \rangle D_\rho. \tag{3.32}$$

This is the only general toric divisor formula used below; see [CLS11, Proposition 4.1.1].

By Remark 3.3, choose one curve of each global type and denote the representatives by $\widetilde{L}'$ and $\widetilde{E}'$. Locally over an incidence point of $\overline{X}$, they are the prime divisors D_0 and D_{p+1}, respectively. The characters $\chi^{(N,-p-1)} = s$ and $\chi^{(0,1)} = v$ give the total transforms of the two base boundary divisors. The same local description holds for every curve of the corresponding type.

Equation (3.32) gives

$$\mathrm{div}(\chi^{(N,-p-1)}) = N D_0 + \sum_{i=1}^{p} \big(N - (p+1)i\big) E_i, \tag{3.33}$$

$$\mathrm{div}(\chi^{(0,1)}) = N D_{p+1} + \sum_{i=1}^{p} i E_i. \tag{3.34}$$

3.6.2. *Global intersection number calculations.* Over each of the $N(p+1)$ boundary crossing points, equivalently each incident pair (P, L) with $P \in L$, there are $p-1$ isolated fixed points. Hence there are $N(p+1)(p-1) = N(p^2-1)$ isolated fixed points globally. In addition, there are N fixed curves $\widetilde{E}'_P$, one above each exceptional boundary component, and N fixed curves $\widetilde{L}'_L$, one above each line-type boundary component. All are fixed pointwise by the component stabilizer $\langle \sigma \rangle \simeq \mathbb{Z}/N\mathbb{Z}$.

Let $L \subset \mathbb{P}^2$ be a k-rational line, and let $P \in L(k)$. Write $\widetilde{L}$ for the strict transform of L in $\overline{X}$ and $\widetilde{E}_P$ for the exceptional curve above P. These curves meet

transversely in one k-rational point, as shown in Figure 2. When P is understood, we abbreviate $\widetilde{E}_P$ to $\widetilde{E}$.

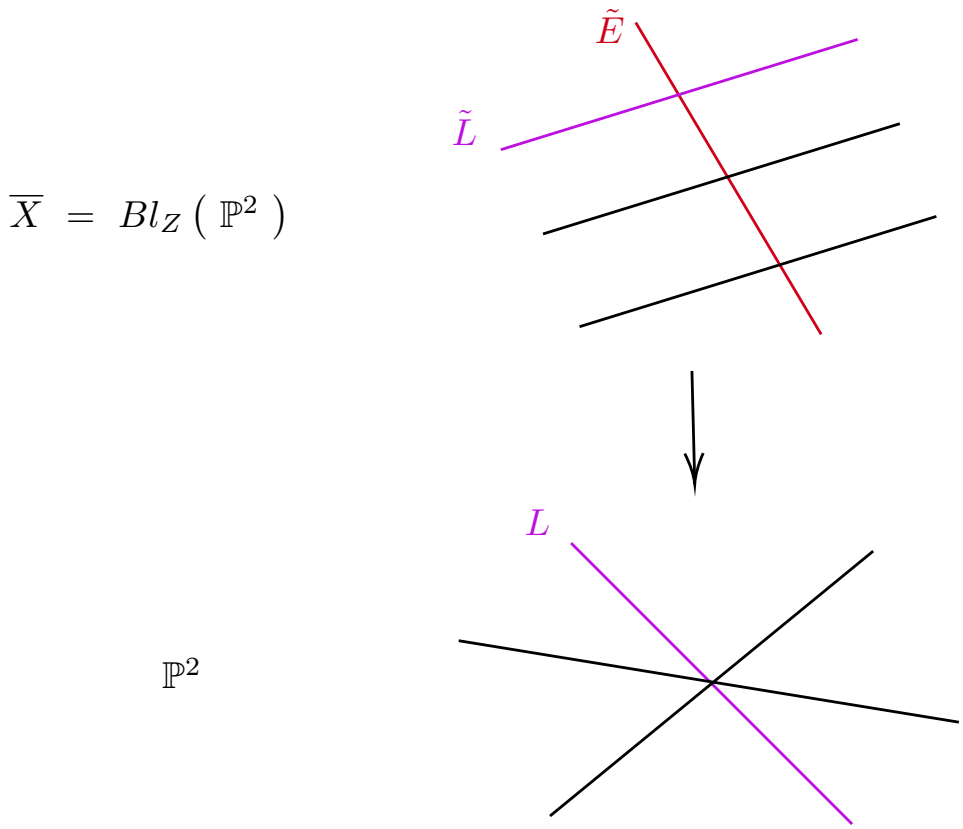


FIGURE 2. The divisors $\widetilde{L}$ and $\widetilde{E}_P$ for $p = 2$

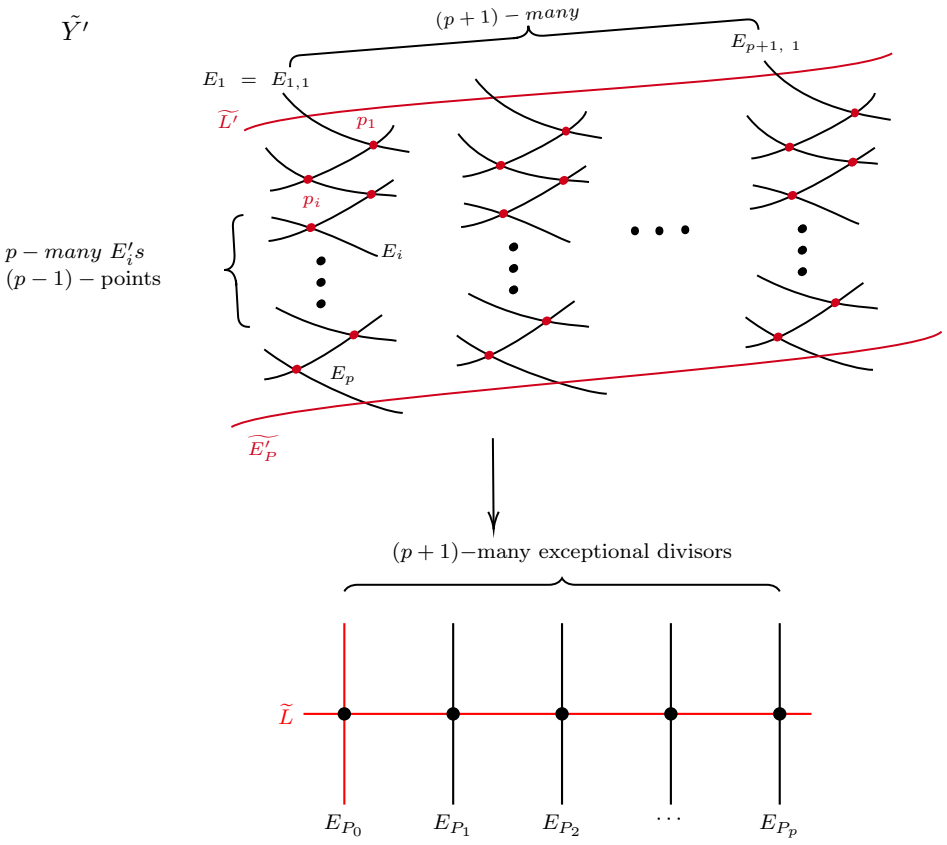
Let $\widetilde{\pi}_\ell : \widetilde{\overline{Y}}_\ell(w) \longrightarrow \overline{X}(w)$ be the generically finite morphism of degree N obtained by composing the minimal resolution ρ_ℓ with the finite morphism $\overline{\pi}_\ell : \overline{Y}_\ell(w) \to \overline{X}(w)$. Let $\widetilde{E}'$ and $\widetilde{L}'$ denote arbitrary representatives of the two types of strict transforms in $\widetilde{\overline{Y}}_\ell(w)$. Both are complete curves isomorphic to $\mathbb{P}^1$. Every base boundary component contains exactly $p+1$ incidence points, so the calculation from (3.33) is identical for every representative and gives

$$\widetilde{\pi}_\ell^*(\widetilde{L}) = N\widetilde{L}' + \sum_{j=1}^{p+1}\sum_{i=1}^{p}\big(N-(p+1)i\big)E_{i,j}, \tag{3.35}$$

$$\widetilde{\pi}_\ell^*(\widetilde{E}) = N\widetilde{E}' + \sum_{j=1}^{p+1}\sum_{i=1}^{p} iE_{i,j}. \tag{3.36}$$

In each formula, j indexes the $p+1$ crossing points on the chosen base curve; the two formulas generally involve different exceptional chains.

The following two figures illustrate the exceptional locus and the fixed locus of h on the smooth equivariant resolution $\widetilde{Y} = \bigsqcup_\ell \widetilde{Z_\ell} \to \overline{Y} = \bigsqcup_\ell \widetilde{Y}_\ell$ described above, with the fixed locus shown in red.



$$\overline{X}(w) = \mathrm{Bl}_{\mathbb{P}^2(\mathbb{F}_p)}(\mathbb{P}^2)$$

The following calculation gives the two boundary self-intersections used in the Brauer–Lefschetz formula.

Lemma 3.14 (Self-intersections of the lifted boundary curves)**.** *The strict transforms on $\widetilde{\overline{Y}}_\ell(w)$ satisfy*

$$(\widetilde{L}')^2 = -p, \qquad (\widetilde{E}')^2 = -1.$$

Proof. For coefficients $a_1, \dots, a_p$, put

$$Q_j(a) = \sum_{i=1}^{p} a_i E_{i,j}.$$

Using

$$E_{i,j}^2 = -2 \quad (i < p), \qquad E_{p,j}^2 = -p-2, \qquad E_{i,j} \cdot E_{i+1,j} = 1,$$

one obtains, by direct expansion,

$$Q_j(a)^2 = -a_1^2 - \sum_{i=1}^{p-1}(a_i - a_{i+1})^2 - (p+1)a_p^2. \tag{3.37}$$

The chains over distinct crossing points are disjoint. For the line-type divisor, set

$$c_i = N - (p+1)i.$$

Then $c_1 = p^2$, $c_p = 1$, and $c_i - c_{i+1} = p+1$. Hence

$$Q_j(c)^2 = -p^4 - (p-1)(p+1)^2 - (p+1) = -p^2 N.$$

Moreover,

$$\widetilde{L}' \cdot Q_j(c) = c_1 = p^2.$$

The projection formula and

$$\widetilde{\pi}_\ell^* \widetilde{L} = N\widetilde{L}' + \sum_{j=1}^{p+1} Q_j(c), \qquad \widetilde{L}^2 = -p,$$

give

$$-pN = N^2(\widetilde{L}')^2 + Np^2(p+1),$$

and therefore

$$(\widetilde{L}')^2 = -p. \tag{3.38}$$

For the exceptional-type divisor, set $e_i = i$. Equation (3.37) gives

$$Q_j(e)^2 = -1 - (p-1) - (p+1)p^2 = -pN.$$

Since $\widetilde{E}'$ meets only $E_{p,j}$, we have $\widetilde{E}' \cdot Q_j(e) = p$.

Now using $\widetilde{\pi}_\ell^* \widetilde{E} = N\widetilde{E}' + \sum_{j=1}^{p+1} Q_j(e)$, and $\widetilde{E}^2 = -1$ the projection formula yields $-N = N^2(\widetilde{E}')^2 + Np(p+1)$. Thus we obtain

$$(\widetilde{E}')^2 = -1. \tag{3.39}$$

□

3.7. **Brauer–Lefschetz calculation.** Work over $\bar{k}$, and let $N = p^2+p+1$. Recall that the group T^{wF} acting on the right is generated by an element τ of order $(p-1)N$, while

$$\sigma = \tau^{p-1}$$

generates the stabilizer of a chosen component and has exact order N. The local covering coordinate z transforms as

$$\sigma(z) = \bar{\zeta} z, \qquad \operatorname{ord}(\bar{\zeta}) = N.$$

Since $N \mid p^3 - 1$, we may take $\bar{\zeta} \in \mathbb{F}_{p^3}^\times$. Put

$$\widehat{\zeta} = [\bar{\zeta}] \in W(\mathbb{F}_{p^3})^\times.$$

Then $\widehat{\zeta}^N = 1$, its reduction modulo p is $\bar{\zeta}$, and

$$[\bar{\zeta}^m] = \widehat{\zeta}^m \qquad (m \in \mathbb{Z}).$$

Because $0 < |m| \le p < N$, each $1 - \widehat{\zeta}^m$ occurring below is a unit in $W(\mathbb{F}_{p^3})$.

3.7.1. *Fixed-locus data.* The inputs to Lefschetz–Riemann–Roch for one split component are summarized below. For an isolated point Q_{i-1} indexed by $2 \le i \le p$, the last column lists the two cotangent eigenvalues; for a curve it lists the normal eigenvalue.

fixed stratum	number	genus	self-intersection	eigenvalue(s)
Q_{i-1}	$N(p+1)$ for each i	–	–	(ζ^i, ζ^{1-i})
$\widetilde{L}'$	N	0	$-p$	$\bar{\zeta}^{-1}$
$\widetilde{E}'$	N	0	-1	$\bar{\zeta}^{p}$

3.7.2. *Isolated fixed points.* For each relevant base point, the toric resolution has the isolated fixed points

$$Q_{i-1} \in \operatorname{Spec} k[t_1^i t_2^{-1}, t_1^{1-i} t_2], \qquad 2 \le i \le p.$$

In the local coordinates $a_{i,1}, a_{i,2}$, the action of σ is

$$a_{i,1} \longmapsto \bar{\zeta}^{\,i} a_{i,1}, \qquad a_{i,2} \longmapsto \bar{\zeta}^{\,1-i} a_{i,2}.$$

Thus the lifted cotangent eigenvalues are $\widehat{\zeta}^i$ and $\widehat{\zeta}^{1-i}$, and Theorem 2.3 gives

$$a_{\mathrm{Br}}(Q_{i-1}) = \frac{1}{(1-\widehat{\zeta}^i)(1-\widehat{\zeta}^{1-i})}. \tag{3.40}$$

The contribution of the $p-1$ points over one base point is therefore

$$S_{\mathrm{iso}} = \sum_{i=2}^{p} \frac{1}{(1-\widehat{\zeta}^i)(1-\widehat{\zeta}^{1-i})}.$$

For every summand,

$$\frac{1}{(1-\widehat{\zeta}^i)(1-\widehat{\zeta}^{1-i})} = -\frac{1}{1-\widehat{\zeta}} \left(\frac{1}{1-\widehat{\zeta}^{i-1}} - \frac{1}{1-\widehat{\zeta}^i} \right).$$

The sum now telescopes, giving

$$\boxed{S_{\mathrm{iso}} = \frac{\widehat{\zeta}(\widehat{\zeta}^{p-1}-1)}{(1-\widehat{\zeta})^2(1-\widehat{\zeta}^p)}}. \tag{3.41}$$

3.7.3. *Fixed curves.* There are N fixed curves $\widetilde{E}'_P$ of exceptional type and N fixed curves $\widetilde{L}'_L$ of line type. Choose one representative of each type and abbreviate it to $\widetilde{E}'$ and $\widetilde{L}'$. Both are isomorphic to $\mathbb{P}^1$, and the intersection calculations above give

$$(\widetilde{E}')^2 = -1, \qquad (\widetilde{L}')^2 = -p.$$

Along $\widetilde{L}'$, the normal eigenvalue is $\bar{\zeta}^{-1}$. Consequently, (2.18) gives

$$\begin{aligned} b_{\mathrm{Br}}(\widetilde{L}') &= \frac{1}{1-\widehat{\zeta}} - \frac{\widehat{\zeta}(-p)}{(1-\widehat{\zeta})^2} \\ &= \boxed{\frac{1+(p-1)\widehat{\zeta}}{(1-\widehat{\zeta})^2}}. \end{aligned} \tag{3.42}$$

Along $\widetilde{E}'$, the normal eigenvalue is $\bar{\zeta}^{p}$. Hence

$$b_{\mathrm{Br}}(\widetilde{E}') = \frac{1}{1-\widehat{\zeta}^{-p}} - \frac{\widehat{\zeta}^{-p}(-1)}{(1-\widehat{\zeta}^{-p})^2}$$

$$= \boxed{\frac{\widehat{\zeta}^{2p}}{(1-\widehat{\zeta}^{p})^{2}}}. \tag{3.43}$$

The first term in each line is essential: since the curves are rational, $1-g_D = 1$, not 0. The uniformity of these two normal eigenvalues was proved at the generic points of all boundary components in Lemma 3.11. In particular, it does not depend on choosing projective transformations that lift to $\overline{Y}_\ell(w)$.

3.7.4. *Proof of the split main theorem.* There are $N(p+1)$ boundary crossing points, and the fixed locus over each one contributes the quantity in (3.41). There are also N curves of each boundary type. Adding all fixed-curve contributions therefore gives

$$\begin{aligned} L_{\mathrm{Br}}(\sigma, \widetilde{\overline{Y}}_\ell(w)) = N(p+1)\left(\frac{\widehat{\zeta}(\widehat{\zeta}^{p-1}-1)}{(1-\widehat{\zeta})^{2}(1-\widehat{\zeta}^{p})}\right) \\ + N\frac{1+(p-1)\widehat{\zeta}}{(1-\widehat{\zeta})^{2}} + N\frac{\widehat{\zeta}^{2p}}{(1-\widehat{\zeta}^{p})^{2}}. \end{aligned} \tag{3.44}$$

This computes the right-hand side of (1.3) on the smooth resolution. The passage to the normal component follows from (2.35). Since σ preserves all $p-1$ components and τ identifies their Brauer characters, the whole-space value is $p-1$ times the component value. This proves the trace formulas in Theorem 1.2.

3.8. **Nonvanishing and Galois-module consequence.**

Proposition 3.15. *The split Brauer–Lefschetz numbers*

$$L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell(w)) \quad \textit{and} \quad L_{\mathrm{Br}}(\sigma, \overline{Y}(w))$$

are nonzero for every prime p.

Proof. Let $R_p(T) \in \mathbb{Q}(T)$ denote the rational function on the right-hand side of (1.3). For a primitive N-th root ω, neither ω nor ω^p is one, because $p \nmid N$. Hence every denominator in $R_p(\omega)$ is nonzero. Lemma 2.1, with $M = N$ and $\widehat{\omega} = \widehat{\zeta}$, shows that it is enough to prove nonvanishing after the complex specialization $\omega = e^{2\pi i/N}$.

Put $A = \pi/N$ and $x = e^{2iA}$. The elementary identities

$$\operatorname{Im}\frac{1}{(1-x^{i})(1-x^{1-i})} = \frac{1}{4}\big(\cot(iA) - \cot((i-1)A)\big)$$

and

$$\operatorname{Im}\frac{1+(p-1)x}{(1-x)^{2}} = \frac{1}{2}\cot A, \qquad \operatorname{Im}\frac{x^{2p}}{(1-x^{p})^{2}} = -\frac{1}{2}\cot(pA)$$

give

$$\frac{1}{N}\operatorname{Im} L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell(w)) = \frac{p-1}{4}\big(\cot(pA) - \cot A\big) < 0.$$

Indeed, $0 < A < pA < \pi/2$, and cot is strictly decreasing there. Hence the value for the irreducible component, and hence also for the entire variety is, nonzero. □

Corollary 3.16 (Split Galois-module consequence)**.** *Put $\Gamma_{\mathrm{sp}} = \langle\sigma\rangle$. Then*

$$\chi_{\Gamma_{\mathrm{sp}}}\big(\overline{Y}_\ell(w), \mathcal{O}_{\overline{Y}_\ell(w)}\big)$$

is not virtually free over $\bar{k}[\Gamma_{\mathrm{sp}}]$. Moreover, $\chi_{T^{wF}}\big(\overline{Y}(w), \mathcal{O}_{\overline{Y}(w)}\big)$ is not virtually free over $\bar{k}[T^{wF}]$.

4. Non-Split Case

4.1. **Unitary notation and setup.** Let $G = \mathrm{GL}_3$ over $\bar{k}$. We equip G with the unitary Frobenius

$$F(g) = J(g^{(p)})^{-t}J^{-1}, \qquad J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \tag{4.1}$$

where $g^{(p)}$ is obtained by raising each entry to the p-th power and $(g^{(p)})^{-t} = ((g^{(p)})^{-1})^t$. The corresponding finite group of Lie type is $G^F = \mathrm{U}_3(\mathbb{F}_p)$. The choice of the matrix J is explained in Section 4.2.

Let B be the upper triangular Borel subgroup, $U = R_u(B)$ its upper unitriangular radical, and

$$T = \{\mathrm{diag}(t_1, t_2, t_3) : t_i \in \bar{k}^\times\} \simeq (\mathbb{G}_m)^3$$

the diagonal torus. On T,

$$F\big(\mathrm{diag}(t_1, t_2, t_3)\big) = \mathrm{diag}(t_3^{-p}, t_2^{-p}, t_1^{-p}).$$

The Weyl group $W = N_G(T)/T$ is S_3. Writing $\chi_i(\mathrm{diag}(t_1, t_2, t_3)) = t_i$, take

$$\alpha_1 = \chi_1 - \chi_2, \qquad \alpha_2 = \chi_2 - \chi_3.$$

Then $F(\alpha_1) = p\alpha_2$ and $F(\alpha_2) = p\alpha_1$. If e_i denotes the cocharacter in the i-th diagonal entry, the corresponding simple coroots are

$$\alpha_1^\vee = e_1 - e_2, \qquad \alpha_2^\vee = e_2 - e_3,$$

and

$$F(e_1) = -pe_3, \qquad F(e_2) = -pe_2, \qquad F(e_3) = -pe_1.$$

4.1.1. *The Frobenius and the order of the acting group.* Fix the Coxeter element $w = s_1 s_2 = (123)$. The action on the cocharacter lattice gives

$$|T^{wF}| = |\det(I - wF \mid Y(T))| = (p+1)(p^2-1).$$

The explicit decomposition of this finite abelian group and the stabilizer of each irreducible component are determined in Section 4.2.3 and Proposition 4.3.

4.2. **Explicit description of the Deligne–Lusztig varieties.**

4.2.1. *The Hermitian form.* Let $V = \mathbb{F}_{p^2}^3$, equipped with its standard basis, and put $V_{\bar{k}} = V \otimes_{\mathbb{F}_{p^2}} \bar{k}$. Vectors are written as columns. For a matrix g with entries in $\bar{k}$ we denote by $g^{(p)}$ the matrix obtained by raising each entry of g to the p-th power, and we write $g^{-t} = (g^{-1})^t$. Put

$$J_+ = \mathrm{antidiag}(1, 1, 1), \qquad J_- = \mathrm{antidiag}(1, -1, 1).$$

For $J \in \{J_+, J_-\}$ we have $J = J^t = J^{-1}$, and we consider the Hermitian form on V (extended to $V_{\bar{k}}$ by the same formula) and the isogeny of $\mathrm{GL}_{3,\bar{k}}$ given by

$$\langle v, v' \rangle_J = v^{(p)t} J v', \qquad F_J(g) = J\,(g^{(p)})^{-t} J^{-1}.$$

Görtz and Schröer [GS26] describe the unitary Frobenius and its action on the flag variety using J_+; in this paper we work with J_- instead. For $z = (z_0, z_1, z_2)^t \in V_{\bar{k}}$ put

$$\mathcal{H}_\pm(z) := \langle z, z \rangle_{J_\pm} = z_0 z_2^p \pm z_1^{p+1} + z_2 z_0^p,$$

so that $\mathcal{H}_\pm = 0$ is the Hermitian curve in $\mathbb{P}(V_{\bar{k}})$ attached to $J_\pm$.

The two conventions are equivalent. Since the norm map $\mathbb{F}_{p^2}^{\times} \to \mathbb{F}_p^{\times}$ is surjective, there is $\varepsilon \in \mathbb{F}_{p^2}^{\times}$ with $\varepsilon^{p+1} = -1$; for $p = 2$ we take $\varepsilon = 1$, and then $J_+ = J_-$. With $D = \operatorname{diag}(1, \varepsilon, 1)$ we have

$$D^{(p)t} J_+ D = J_-. \tag{4.2}$$

Hence D is an isometry from $(V, \langle \cdot, \cdot \rangle_{J_-})$ onto $(V, \langle \cdot, \cdot \rangle_{J_+})$; in particular, $\mathcal{H}_+(Dz) = \mathcal{H}_-(z)$. Moreover, (4.2) implies $F_{J_+}(DgD^{-1}) = D\, F_{J_-}(g)\, D^{-1}$, and therefore

$$(Dg)^{-1} F_{J_+}(Dg) = g^{-1} F_{J_-}(g) \cdot D^{-1} F_{J_+}(D) \in g^{-1} F_{J_-}(g)\, T,$$

where $T \subset \mathrm{GL}_{3,\bar{k}}$ denotes the diagonal torus. Consequently, left multiplication by D identifies the Deligne–Lusztig varieties attached to F_{J_-}, together with their compactifications, with those attached to F_{J_+}. This allows us to apply the results of [GS26] in our setting.

Following [GS26, (7.2)] and [Lan19], we regard the flag variety as an incidence subvariety

$$\mathrm{Flag}(V_{\bar{k}}) \hookrightarrow \mathbb{P}^2_{\bar{k}} \times \mathbb{P}^2_{\bar{k}}, \qquad (L \subset P) \longmapsto ([x], [y]),$$

where $L = \bar{k}x$ and $P = \{v \in V_{\bar{k}} : y^t v = 0\}$ for nonzero column vectors $x = (x_0, x_1, x_2)^t$ and $y = (y_0, y_1, y_2)^t$. Thus $[x]$ is the point of $\mathbb{P}(V_{\bar{k}})$ defined by L, and $[y]$ is the coefficient vector of a linear equation for P; in other words, we identify the dual projective plane with $\mathbb{P}^2_{\bar{k}}$ by means of the standard basis. The image is the incidence divisor

$$x^t y = x_0 y_0 + x_1 y_1 + x_2 y_2 = 0.$$

The isometry D sends a flag $L \subset P$ to $DL \subset DP$. Since DP is the kernel of the covector $y^t D^{-1}$, in coordinates this reads

$$([x], [y]) \longmapsto ([Dx], [D^{-t} y]). \tag{4.3}$$

This map preserves the incidence equation and satisfies $\mathcal{H}_+(Dx) = \mathcal{H}_-(x)$ and $\mathcal{H}_+(D^{-t}y) = \mathcal{H}_-(y)$; thus it carries the Hermitian equations for J_- to those for J_+.

From now on we fix $J = J_-$ and write $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_J$, $F = F_J$ and $\mathcal{H} = \mathcal{H}_-$; explicitly,

$$F(g) = J(g^{(p)})^{-t} J, \qquad \mathcal{H}(z) = z_0 z_2^p - z_1^{p+1} + z_2 z_0^p.$$

Let $B \subset \mathrm{GL}_{3,\bar{k}}$ be the Borel subgroup of upper triangular matrices and U its unipotent radical; the isogeny F preserves B, U and T. Let $w = s_1 s_2$, which corresponds to the 3-cycle $(1\,2\,3) \in S_3$, and choose the representative

$$\dot{w} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The Deligne–Lusztig variety and its covering variety are then

$$X(w) = \{gB : g^{-1} F(g) \in B \dot{w} B\}, \qquad Y(w) = \{gU : g^{-1} F(g) \in U \dot{w} U\},$$

and the covering map $Y(w) \to X(w)$ is $gU \mapsto gB$.

4.2.2. *The compactification $\overline{X}(w)$ and its boundary.* For a subspace $W \subseteq V_{\bar{k}}$ put $W^\perp = \{v' \in V_{\bar{k}} : \langle v, v' \rangle = 0 \text{ for all } v \in W\}$. By [GS26, Lemma 7.1], the Frobenius acts on flags by

$$F(L \subset P) = (P^\perp \subset L^\perp).$$

Two flags $(L \subset P)$ and $(L' \subset P')$ are in relative position $w = s_1 s_2$ if and only if $L \not\subset P'$ and $L' \subset P$ [GS26, §7.3]. Hence $X(w) = \{(L \subset P) : P^\perp \subset P,\ L \not\subset L^\perp\}$, and the smooth compactification attached to the reduced expression $s_1 s_2$ is the closure

$$\overline{X}(w) = \{(L \subset P) : P^\perp \subset P\}. \tag{4.4}$$

It is the union of the four Deligne–Lusztig strata $X(1)$, $X(s_1)$, $X(s_2)$ and $X(w)$. Its boundary has two irreducible components,

$$B_u := \overline{X}(s_1) = \{(L \subset P) : P = L^\perp\}, \tag{4.5}$$

$$B_v := \overline{X}(s_2) = \{(L \subset P) : L = P^\perp\}, \tag{4.6}$$

which meet transversally along $X(1) = B_u \cap B_v$. The latter consists of the $p^3 + 1$ flags $L \subset L^\perp$ with L an $\mathbb{F}_{p^2}$-rational isotropic line. Thus

$$X(w) = \overline{X}(w) \setminus (B_u \cup B_v). \tag{4.7}$$

These statements follow from the relative-position description and [GS26, §7.3, especially Proposition 7.2]; the transversality can also be read off from the local computation (4.16) below.

We now pass to the coordinates $([x], [y])$. Since $\langle x, v \rangle = (Jx^{(p)})^t v$, the plane $L^\perp$ has coefficient vector $Jx^{(p)}$. As v' runs through P, the vector $v'^{(p)}$ runs through the kernel of $y^{(p)t}$; hence $P^\perp = \bar{k}\, Jy^{(p)}$. Since J is symmetric, $\mathcal{H}(y) = y^t J y^{(p)}$, so that $P^\perp \subset P$ if and only if $\mathcal{H}(y) = 0$. We recover [GS26, Proposition 7.2]:

$$\overline{X}(w) = \left\{ ([x], [y]) : \begin{array}{l} x_0 y_0 + x_1 y_1 + x_2 y_2 = 0, \\ y_0 y_2^p - y_1^{p+1} + y_2 y_0^p = 0 \end{array} \right\}, \tag{4.8}$$

and the boundary curves are

$$B_u = \{([x], [y]) \in \overline{X}(w) : [y] = [Jx^{(p)}]\}, \tag{4.9}$$

$$B_v = \{([x], [y]) \in \overline{X}(w) : [x] = [Jy^{(p)}]\}. \tag{4.10}$$

Moreover, $L \subset L^\perp$ if and only if $\mathcal{H}(x) = 0$, so

$$X(w) = \overline{X}(w) \cap \{\mathcal{H}(x) \neq 0\}, \qquad B_u \cup B_v = \overline{X}(w) \cap \{\mathcal{H}(x) = 0\}. \tag{4.11}$$

Let

$$C_x = \{\mathcal{H}(x) = 0\} \subset \mathbb{P}^2_x, \qquad C_y = \{\mathcal{H}(y) = 0\} \subset \mathbb{P}^2_y$$

be the two copies of the Hermitian curve; they are smooth plane curves of degree $p + 1$, hence of genus $p(p-1)/2$. The second projection exhibits $\overline{X}(w)$ as a $\mathbb{P}^1$-bundle over C_y, the fibre over $[y]$ being the line $\{[x] : x^t y = 0\}$. By (4.10), B_v is the graph of the morphism $C_y \to \mathbb{P}^2_x$, $[y] \mapsto [Jy^{(p)}]$, so the second projection restricts to an isomorphism

$$B_v \xrightarrow{\sim} C_y,$$

and B_v is a section of the ruling. By (4.9) and the identity $x^t J x^{(p)} = \mathcal{H}(x)$, the first projection identifies B_u with C_x, and under this identification the morphism $B_u \to C_y$ factors as

$$C_x \xrightarrow{F_{C_x/k}} C_x^{(p)} \xrightarrow[\sim]{J} C_y,$$

where $F_{C_x/k}$ is the relative Frobenius and the second arrow is the linear isomorphism induced by J. Consequently, $B_u \to C_y$ is finite and radicial of degree p, whereas $B_v \to C_y$ has degree one: B_v is a section and B_u is a purely inseparable p-section of the ruling. In particular, both boundary components are Hermitian curves of genus $p(p-1)/2$.

4.2.3. *The finite group T^{wF}.* For $t = \mathrm{diag}(t_1, t_2, t_3) \in T$ we have

$$F(t) = \mathrm{diag}(t_3^{-p}, t_2^{-p}, t_1^{-p}).$$

Put $F_w = \mathrm{Ad}(\dot{w}) \circ F$ and $T^{wF} = \{t \in T : F_w(t) = t\}$. Since $\dot{w}\,\mathrm{diag}(a_1, a_2, a_3)\dot{w}^{-1} = \mathrm{diag}(a_3, a_1, a_2)$, we get $F_w(t) = \mathrm{diag}(t_1^{-p}, t_3^{-p}, t_2^{-p})$. Hence $t \in T^{wF}$ if and only if $t_1 = t_1^{-p}$, $t_2 = t_3^{-p}$ and $t_3 = t_2^{-p}$, that is,

$$t_1^{p+1} = 1, \qquad t_2^{p^2-1} = 1, \qquad t_3 = t_2^{-p}.$$

Therefore

$$T^{wF} = \left\{ \mathrm{diag}(\zeta, a, a^{-p}) : \zeta^{p+1} = 1,\ a^{p^2-1} = 1 \right\} \cong \mu_{p+1} \times \mu_{p^2-1}.$$

The group T^{wF} acts on $Y(w)$ from the right by $gU \cdot t = gtU$; this is well defined because t normalizes U and $t^{-1}\dot{w}F(t) = \dot{w}$. The action is free, and $Y(w) \to X(w)$ is a T^{wF}-torsor.

4.2.4. *Coordinates on $Y(w)$.* We now describe the T^{wF}-torsor $Y(w) \to X(w)$ explicitly. Let $g \in \mathrm{GL}_3(\bar{k})$ have columns v_1, v_2, v_3, and put

$$x := v_1, \qquad y := v_1 \times v_2, \qquad \delta := \det(g),$$

where $\times$ denotes the usual cross product on $\bar{k}^3$. Right multiplication by an element of U replaces v_2 by $v_2 + c_1 v_1$ and v_3 by $v_3 + c_2 v_1 + c_3 v_2$ for some $c_1, c_2, c_3 \in \bar{k}$, which changes none of x, y and δ. Hence the entries of x and y, and δ, are regular functions on $\mathrm{GL}_{3,\bar{k}}/U$.

By Cramer's rule, $g^{-t} = \delta^{-1}(v_2 \times v_3,\ v_3 \times v_1,\ v_1 \times v_2)$. Applying this to $g^{(p)}$, and noting that right multiplication by J moves the third column to the first position, we find that the first column of $F(g)$ is

$$F(g)_1 = \delta^{-p}\, J y^{(p)}. \tag{4.12}$$

Proposition 4.1. *Let $Z^\circ \subset \mathbb{A}^3 \times \mathbb{A}^3 \times \mathbb{G}_m$ be the locally closed subscheme of triples (x, y, δ) with $y \neq 0$ and*

$$x \times J y^{(p)} = \delta^p y, \qquad \mathcal{H}(x) = 1, \qquad \delta^{p+1} = 1. \tag{4.13}$$

Then $gU \mapsto (x, y, \delta)$ defines an isomorphism $Y(w) \xrightarrow{\sim} Z^\circ$.

Proof. Put $A = g^{-1}F(g)$. Multiplying out $u\dot{w}u'$ with $u, u' \in U$ shows that

$$U\dot{w}U = \left\{ A = \begin{pmatrix} * & * & * \\ 1 & * & * \\ 0 & 1 & * \end{pmatrix} : \det A = 1 \right\}.$$

We translate these conditions on A into conditions on (x, y, δ). First, $\det F(g) = \det(g^{(p)})^{-1} = \delta^{-p}$, so $\det A = \delta^{-(p+1)}$, and $\det A = 1$ if and only if $\delta^{p+1} = 1$. Second, $A_{21} = 1$ and $A_{31} = 0$ mean that $F(g)_1 \in v_2 + \bar{k}v_1$. Since $v_1 \times v_2$ and $v_1 \times v_3$ are linearly independent, this is equivalent to $x \times F(g)_1 = y$, that is, by (4.12), to $x \times Jy^{(p)} = \delta^p y$. Third, $A^{-1} = F(g)^{-1}g = Jg^{(p)t}Jg$; since the last row of J is $(1, 0, 0)$, the $(3, 1)$-entry of A^{-1} is $v_1^{(p)t}Jv_1 = \mathcal{H}(x)$. On the other hand, the cofactor formula gives $(A^{-1})_{31} = (A_{21}A_{32} - A_{22}A_{31})/\det A$. Hence, once the first two conditions hold, $A_{32} = \mathcal{H}(x)$, and $A_{32} = 1$ if and only if $\mathcal{H}(x) = 1$.

It follows that $gU \in Y(w)$ if and only if (x, y, δ) satisfies (4.13). Since v_1 and v_2 are linearly independent, $y \neq 0$, so we obtain a morphism $Y(w) \to Z^\circ$. Conversely, let $(x, y, \delta) \in Z^\circ$. Taking the dot product of the first equation with x gives $\delta^p\, x^t y = 0$, so $x^t y = 0$; moreover $x \neq 0$ because $\mathcal{H}(x) = 1$. Hence there is a vector v_2 with $x \times v_2 = y$, unique modulo $\bar{k}x$, and, as $y \neq 0$, a vector v_3 with $y^t v_3 = \delta$, unique modulo $\bar{k}x + \bar{k}v_2$. The matrix $g = (x, v_2, v_3)$ has determinant $y^t v_3 = \delta \neq 0$, its class gU does not depend on the choices, and $gU \in Y(w)$ by what we have just shown. Locally on Z° the choices can be made by regular functions: for instance, $v_2 = x_0^{-1}(0, y_2, -y_1)^t$ on $\{x_0 \neq 0\}$, and $v_3 = (\delta/y_0, 0, 0)^t$ on $\{y_0 \neq 0\}$, and similarly on the other standard opens. This defines a morphism $Z^\circ \to Y(w)$ inverse to the one above. □

Explicitly, since $Jy^{(p)} = (y_2^p, -y_1^p, y_0^p)^t$, the equations (4.13) read

$$\begin{aligned} x_1 y_0^p + x_2 y_1^p &= \delta^p y_0, \\ x_2 y_2^p - x_0 y_0^p &= \delta^p y_1, \\ -x_0 y_1^p - x_1 y_2^p &= \delta^p y_2, \\ x_0 x_2^p - x_1^{p+1} + x_2 x_0^p &= 1, \\ \delta^{p+1} &= 1. \end{aligned}$$

Lemma 4.2 (Coordinate form of the projection)**.** *Under the identification of Proposition 4.1, the projection $Y(w) \to X(w)$ is given by*

$$(x, y, \delta) \longmapsto ([x], [y]).$$

Proof. Let $g = (v_1, v_2, v_3)$. The point gB is the flag $L = \bar{k}v_1 \subset P = \bar{k}v_1 + \bar{k}v_2$, and $L = \bar{k}x$. The vector $y = v_1 \times v_2$ is nonzero and satisfies $y^t v_1 = y^t v_2 = 0$. Hence $P \subseteq \{v : y^t v = 0\}$, with equality because both spaces have dimension two. Thus the flag gB has coordinates $([x], [y])$. □

Since $p \nmid p+1$, the group scheme μ_{p+1} is finite étale over $\bar{k}$, with $p+1$ points, all lying in $\mathbb{F}_{p^2}$. The function $\delta\colon Y(w) \to \mu_{p+1}$ therefore yields a decomposition into open and closed subschemes

$$Y(w) = \coprod_{\delta_0 \in \mu_{p+1}} Y_{\delta_0}(w), \qquad Y_{\delta_0}(w) := \{\delta = \delta_0\}. \tag{4.14}$$

Proposition 4.3 below shows that each $Y_{\delta_0}(w)$ is irreducible and determines its stabilizer in T^{wF}.

4.3. **Local equations and normalization of the cover.** Throughout this subsection we abbreviate $X = X(w)$ and $\overline{X} = \overline{X}(w)$, and we put $N_{\mathrm{un}} = p^2 - 1$.

4.3.1. *An affine chart of $\overline{X}$.* Let $\overline{X}_{0,2} = \overline{X} \cap \{x_0 \neq 0,\ y_2 \neq 0\}$, with affine coordinates $t = x_1/x_0$, $r = x_2/x_0$ and $s = y_1/y_2$. The incidence equation gives $y_0/y_2 = -(ts + r)$, so the points of $\overline{X}_{0,2}$ are

$$([x], [y]) = \big([1 : t : r],\ [-(ts + r) : s : 1]\big).$$

Substituting into the Hermitian equation for y gives $\mathcal{H}(-(ts+r), s, 1) = -\Phi$, where

$$\Phi = s^{p+1} + ts + t^p s^p + r + r^p.$$

Hence

$$\overline{X}_{0,2} = \operatorname{Spec} \bar{k}[t, r, s]/(\Phi). \tag{4.15}$$

Since $\partial\Phi/\partial r = 1$, the projection $\overline{X}_{0,2} \to \mathbb{A}^2_{t,s}$ is étale. Put

$$h := \mathcal{H}(1, t, r) = r + r^p - t^{p+1},$$

so that $\mathcal{H}(x) = x_0^{p+1} h$ on $\overline{X}_{0,2}$. By (4.11),

$$X \cap \overline{X}_{0,2} = \operatorname{Spec} R, \qquad R := \bar{k}[t, r, s, h^{-1}]/(\Phi).$$

4.3.2. *The boundary parameters u and v.* Define functions on $\overline{X}_{0,2}$ by

$$u := t^p + s, \qquad v := -(t + s^p).$$

Using $\Phi = 0$ to eliminate $r + r^p$, we obtain

$$h = -(s^{p+1} + ts + t^p s^p + t^{p+1}) = -(t + s^p)(t^p + s) = uv. \tag{4.16}$$

Comparing with (4.9) and (4.10), one checks that

$$B_u \cap \overline{X}_{0,2} = V(u), \qquad B_v \cap \overline{X}_{0,2} = V(v).$$

For instance, a point of $\overline{X}_{0,2}$ lies on B_v if and only if $[1 : t : r] = [1 : -s^p : -(ts+r)^p]$; the second coordinates give $v = 0$, and the third coordinates then agree because $\Phi = 0$. The argument for B_u is the same.

Let $o = ([1 : 0 : 0], [0 : 0 : 1])$. It is the point $t = r = s = 0$ of $\overline{X}_{0,2}$, and it lies in $X(1) = B_u \cap B_v$. Since the projection to $\mathbb{A}^2_{t,s}$ is étale, $\widehat{\mathcal{O}}_{\overline{X},o} = \bar{k}[[t, s]]$. The linear parts of u and v are s and $-t$, so (u, v) is again a regular system of parameters:

$$\widehat{\mathcal{O}}_{\overline{X},o} = \bar{k}[[u, v]]. \tag{4.17}$$

Thus, near o, the boundary $V(h) = B_u \cup B_v$ consists of the two smooth branches $V(u) = B_u$ and $V(v) = B_v$, which meet transversally. In particular, u is a uniformizer of the discrete valuation ring $\mathcal{O}_{\overline{X},B_u}$ and v is a unit in it, and symmetrically for B_v.

4.3.3. *The covering over the chart.* Fix $\delta_0 \in \mu_{p+1}$. Put $x' = (1, t, r)^t$ and $y' = (-(ts + r), s, 1)^t$. By Proposition 4.1 and Lemma 4.2, the points of $Y_{\delta_0}(w)$ over $X \cap \overline{X}_{0,2}$ are the triples $(x_0 x',\ y_2 y',\ \delta_0)$ with $x_0, y_2 \in \mathbb{G}_m$ satisfying (4.13). A direct computation, using $\Phi = 0$ for the first two coordinates, gives

$$x' \times J y'^{(p)} = v\, y'.$$

Hence (4.13) becomes

$$x_0^{p+1} h = 1, \qquad x_0 y_2^{p-1} v = \delta_0^p.$$

On X the function $h = uv$ is a unit, hence so is v. Solving the second equation for x_0, substituting into the first, and using $\delta_0^{p(p+1)} = 1$, we obtain $y_2^{N_{\mathrm{un}}} v^{p+1} = h = uv$, that is,

$$y_2^{N_{\mathrm{un}}} = u\, v^{-p}. \tag{4.18}$$

Conversely, every solution of (4.18) determines $x_0 = \delta_0^p y_2^{1-p} v^{-1}$. Therefore

$$Y_{\delta_0}(w) \times_X (X \cap \overline{X}_{0,2}) \cong \operatorname{Spec} R[y_2]/\big(v^p y_2^{N_{\mathrm{un}}} - u\big),$$

an étale Kummer covering of degree N_{un}.

4.3.4. *Components and their stabilizer.*

Proposition 4.3 (Components and their stabilizer)**.** *Each subscheme $Y_{\delta_0}(w)$ in (4.14) is irreducible. Consequently, $Y(w)$ has exactly $p+1$ irreducible components. Right multiplication by $t = \operatorname{diag}(\zeta, a, a^{-p}) \in T^{wF}$ acts on the coordinates by*

$$(x, y, \delta) \longmapsto (\zeta x,\ \zeta a\, y,\ \zeta a^{1-p}\delta). \tag{4.19}$$

Hence the stabilizer of every component is the cyclic group

$$\begin{aligned} H &= \ker\big(\det\colon T^{wF} \longrightarrow \mu_{p+1}\big) \\ &= \Big\{\operatorname{diag}(a^{p-1}, a, a^{-p}) : a^{p^2-1} = 1\Big\} \cong \mu_{N_{\mathrm{un}}}, \end{aligned} \tag{4.20}$$

and $T^{wF}/H \cong \mu_{p+1}$ acts simply transitively on the set of components.

Proof. Since $Y(w) \to X$ is a T^{wF}-torsor, the morphism $Y_{\delta_0}(w) \to X$ is finite étale. As X is normal and irreducible, $Y_{\delta_0}(w)$ is normal and each of its irreducible components dominates X; hence $Y_{\delta_0}(w)$ is irreducible if its generic fibre is a field. By (4.18), the generic fibre is $K[T]/(T^{N_{\mathrm{un}}} - uv^{-p})$, where $K = \bar{k}(X)$. Since u is a uniformizer of $\mathcal{O}_{\overline{X},B_u}$ and v is a unit there, this polynomial is Eisenstein, hence irreducible. This proves that $Y_{\delta_0}(w)$ is irreducible. The same argument shows that the covering is totally ramified along B_u; it is also totally ramified along B_v, because $\operatorname{ord}_{B_v}(uv^{-p}) = -p$ is prime to N_{un}.

Right multiplication by t multiplies the first two columns of g by ζ and a, respectively. Hence $x = v_1$ is multiplied by ζ, $y = v_1 \times v_2$ by ζa, and $\delta = \det(g)$ by $\det(t) = \zeta a^{1-p}$; this proves (4.19). In particular, t maps $Y_{\delta_0}(w)$ onto $Y_{\det(t)\delta_0}(w)$. The homomorphism $\det\colon T^{wF} \to \mu_{p+1}$ is surjective (take $a = 1$), and its kernel is obtained by setting $\zeta = a^{p-1}$. This proves (4.20) and the last assertion. □

4.3.5. *The normalization at a boundary point.* Let $\overline{Y}(w)$ be the normalization of $\overline{X}$ in $Y(w)$. By (4.14),

$$\overline{Y}(w) = \coprod_{\delta_0 \in \mu_{p+1}} \overline{Y}_{\delta_0}(w), \tag{4.21}$$

where $\overline{Y}_{\delta_0}(w)$ is the normalization of $\overline{X}$ in $Y_{\delta_0}(w)$. We fix δ_0 from now on and write $\overline{Y}^{\circ}(w) := \overline{Y}_{\delta_0}(w)$.

In $\widehat{\mathcal{O}}_{\overline{X},o} = \bar{k}[[u, v]]$ the function v is not a unit, so (4.18) does not exhibit y_2 as integral over $\widehat{\mathcal{O}}_{\overline{X},o}$. The function $z := vy_2$, however, satisfies

$$z^{N_{\mathrm{un}}} = u\, v^{N_{\mathrm{un}}-p} = u\, v^{p^2-p-1}.$$

The polynomial $T^{N_{\mathrm{un}}} - uv^{p^2-p-1}$ is Eisenstein with respect to the prime u of $\bar{k}[[u,v]]$, hence irreducible. Since $\overline{X}$ is excellent, normalization commutes with completion. It follows that $\overline{Y}_{\delta_0}(w)$ has a unique point o' over o, and that

$$\widehat{\mathcal{O}}_{\overline{Y}_{\delta_0}(w),o'} \cong \big(A_{\mathrm{Kum}}\big)^{\mathrm{norm}}, \qquad A_{\mathrm{Kum}} := \bar{k}[[u,v,z]]\big/\big(z^{p^2-1} - u\,v^{p^2-p-1}\big). \tag{4.22}$$

Let $\alpha \in \mathbb{F}_{p^2}^{\times}$ be a generator, so that α has order N_{un}, and put

$$\tau = \mathrm{diag}(\alpha^{p-1}, \alpha, \alpha^{-p}) \in H.$$

Then τ generates H. By (4.19), τ fixes δ and multiplies y by α^p. Hence τ fixes u and v, which are pulled back from $\overline{X}$, multiplies y_2 by α^p, and acts on the Kummer coordinate by

$$\tau(z) = \bar{\xi} z, \qquad \bar{\xi} := \alpha^p.$$

Since $\gcd(p, N_{\mathrm{un}}) = 1$, $\bar{\xi}$ is again a primitive N_{un}-th root of unity. This is the action of the component stabilizer used in all toric and Lefschetz computations below.

4.4. **The Bonnafé–Rouquier slice in the non-split case.** We next evaluate the slice construction of Section 2.8 for the unitary Frobenius. On the cocharacter lattice $Y(T) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3$ we have

$$F(e_1) = -pe_3, \qquad F(e_2) = -pe_2, \qquad F(e_3) = -pe_1,$$

and $w = s_1 s_2$ acts by $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$. Hence $wF = -ps_2$ on $Y(T)$. With $N_{\mathrm{un}} = p^2 - 1$ as before, the solutions of (2.27) are

$$\lambda_1 = (p-1, 1, -p), \qquad \lambda_2 = (p-1, -p, 1), \qquad m_1 = m_2 = N_{\mathrm{un}}. \tag{4.23}$$

Indeed,

$$\lambda_1 - wF(\lambda_1) = \lambda_1 + ps_2(\lambda_1) = N_{\mathrm{un}}(e_1 - e_2) = N_{\mathrm{un}}\,\alpha_1^{\vee},$$
$$\lambda_2 - wF(\lambda_2) = \lambda_2 + ps_2(\lambda_2) = N_{\mathrm{un}}(e_1 - e_3) = N_{\mathrm{un}}\,s_1(\alpha_2^{\vee}).$$

As in the split case, both cocharacters are primitive. For $(z_1, z_2) \in \mathbb{G}_m^2$,

$$\lambda_1(z_1)\lambda_2(z_2) = \mathrm{diag}\big((z_1 z_2)^{p-1},\ z_1 z_2^{-p},\ z_1^{-p} z_2\big),$$

and (2.30) becomes

$$K_{12} = \big\{(\zeta, \zeta^p) : \zeta^{N_{\mathrm{un}}} = 1\big\} \cong \mu_{N_{\mathrm{un}}}. \tag{4.24}$$

Thus $\zeta \in \mu_{N_{\mathrm{un}}}$ acts on the two-dimensional slice through $\mathrm{diag}(\zeta, \zeta^p)$. Consequently, at every point lying over the deepest boundary stratum $X(1)$, each irreducible component of the normal unitary compactification has a cyclic quotient singularity of type

$$\frac{1}{p^2-1}(1, p). \tag{4.25}$$

This agrees with the direct computation in Section 4.5, where the normalization of the local Kummer model is identified with the affine toric surface of the cone (4.26), whose Cox normal form has parameters (p^2-1, p). Since $(p^2-1)/p = [[p, p]]$, both descriptions lead to the same minimal resolution, a chain of two $(-p)$-curves.

4.5. **Toric normalization, resolution, and fixed locus.** We use the conventions for normalizations and toric resolutions collected in Section 2.7, including its divisor and character-coordinate formulas.

4.5.1. *The toric normalization.* Since normalization commutes with completion for excellent rings, (4.22) shows that it suffices to normalize the affine surface

$$Y_{\mathrm{Kum}} = \operatorname{Spec} R_{\mathrm{Kum}}, \qquad R_{\mathrm{Kum}} = \bar{k}[u,v,z]\big/\big(z^{p^2-1} - uv^{p^2-p-1}\big),$$

and to complete at the origin. The polynomial $z^{p^2-1} - uv^{p^2-p-1}$ has degree one in u and coprime coefficients, so it is irreducible and R_{Kum} is a domain. On the dense torus $\{vz \neq 0\}$ the relation gives $u = z^{p^2-1}v^{-(p^2-p-1)}$; hence

$$R_{\mathrm{Kum}} \cong \bar{k}\big[v,\ z,\ z^{p^2-1}v^{-(p^2-p-1)}\big] \subset \bar{k}[v^{\pm 1}, z^{\pm 1}].$$

Let $M \cong \mathbb{Z}^2$ be the character lattice of this torus, with $\chi^{(m_1,m_2)} = v^{m_1}z^{m_2}$, and let N be the dual lattice. Then

$$v = \chi^{(1,0)}, \qquad z = \chi^{(0,1)}, \qquad u = \chi^{m_u}, \quad m_u := \big(-(p^2-p-1),\, p^2-1\big),$$

and $R_{\mathrm{Kum}} = \bar{k}[S]$ for the semigroup

$$S = \mathbb{N}(1,0) + \mathbb{N}(0,1) + \mathbb{N}m_u \subset M.$$

Since

$$(0,1) = \frac{p^2-p-1}{p^2-1}(1,0) + \frac{1}{p^2-1}\, m_u,$$

the vector $(0,1)$ lies in the interior of $\mathrm{Cone}\big((1,0), m_u\big)$, which is therefore the cone spanned by S. Its dual cone is

$$\sigma = \big\{(n_1,n_2) \in N_{\mathbb{R}} : n_1 \geq 0,\ -(p^2-p-1)n_1 + (p^2-1)n_2 \geq 0\big\} = \mathrm{Cone}(u_0, u_3),$$

where $u_0 = (0,1)$ and $u_3 = (p^2-1, p^2-p-1)$. By (2.20), the normalization of Y_{Kum} is the affine toric surface

$$Y_{\mathrm{tor}} := Y_{\mathrm{Kum}}^{\mathrm{norm}} = \operatorname{Spec} \bar{k}[\sigma^\vee \cap M].$$

4.5.2. *The minimal toric resolution.* The vectors $e_1 = (1,1)$ and $e_2 = (0,1)$ form a basis of N, and $u_3 = (p^2-1)e_1 - pe_2$. Hence

$$\sigma = \mathrm{Cone}\big(e_2,\ (p^2-1)e_1 - pe_2\big) \tag{4.26}$$

is in Cox normal form with $d = p^2-1$ and $a = p$, where $\gcd(d,a) = 1$. Thus Y_{tor} is a cyclic quotient singularity of type $\frac{1}{p^2-1}(1,p)$, in accordance with (4.25). By (2.23), the relevant Hirzebruch–Jung continued fraction is

$$\frac{p^2-1}{p} = p - \frac{1}{p} = [[p,p]],$$

so $r = 2$ and $b_1 = b_2 = p$. The recursion $P_i = b_iP_{i-1} - P_{i-2}$, $Q_i = b_iQ_{i-1} - Q_{i-2}$ with $(P_0,Q_0) = (1,0)$ and $(P_1,Q_1) = (b_1,1)$ gives

$$(P_0,P_1,P_2) = (1,\ p,\ p^2-1), \qquad (Q_0,Q_1,Q_2) = (0,\ 1,\ p),$$

and the rays $u_i = P_{i-1}e_1 - Q_{i-1}e_2$ for $1 \leq i \leq 3$, together with $u_0 = e_2$, are

$$u_0 = (0,1), \qquad u_1 = (1,1), \qquad u_2 = (p,p-1), \qquad u_3 = (p^2-1, p^2-p-1).$$

They satisfy the wall relations (2.24),

$$u_0 + u_2 = pu_1, \qquad u_1 + u_3 = pu_2,$$

they form the Hilbert basis of $\sigma \cap N$, and $\det(u_{i-1}, u_i) = -1$ for $i = 1,2,3$. By (2.21), the fan Σ with maximal cones

$$\sigma_1 = \mathrm{Cone}(u_0,u_1), \qquad \sigma_2 = \mathrm{Cone}(u_1,u_2), \qquad \sigma_3 = \mathrm{Cone}(u_2,u_3)$$

is smooth, and $\widetilde{Y}_{\mathrm{tor}} := X_\Sigma \to Y_{\mathrm{tor}}$ is the minimal resolution. Its exceptional curves are the divisors corresponding to the rays u_1 and u_2, and both have self-intersection $-p$.

4.5.3. *Pullbacks of the boundary parameters.* For $0 \le i \le 3$, let D_i be the torus-invariant divisor of $\widetilde{Y}_{\mathrm{tor}}$ corresponding to u_i. By (2.26), $\mathrm{div}(\chi^m) = \sum_i \langle m, u_i\rangle D_i$. The pairings of the exponents of v, u and z with the rays are

	u_0	u_1	u_2	u_3
$v = \chi^{(1,0)}$	0	1	p	p^2-1
$u = \chi^{m_u}$	p^2-1	p	1	0
$z = \chi^{(0,1)}$	1	1	$p-1$	p^2-p-1

and therefore

$$\begin{aligned} \mathrm{div}(v) &= D_1 + pD_2 + (p^2-1)D_3, \\ \mathrm{div}(u) &= (p^2-1)D_0 + pD_1 + D_2, \\ \mathrm{div}(z) &= D_0 + D_1 + (p-1)D_2 + (p^2-p-1)D_3. \end{aligned} \tag{4.27}$$

In particular, the strict transforms of the reduced curves $\{u=0\}$ and $\{v=0\}$ are

$$\widetilde{\{u=0\}} = D_0, \qquad \widetilde{\{v=0\}} = D_3,$$

while D_1 and D_2 are the exceptional curves.

4.5.4. *The global resolution.* Let $\rho_{\delta_0} : \widetilde{\overline{Y}}_{\delta_0}(w) \longrightarrow \overline{Y}_{\delta_0}(w)$ be the minimal resolution of the chosen component. Away from $X(1)$, the branch locus $B_u \cup B_v$ is smooth and the covering is tamely ramified, so $\overline{Y}_{\delta_0}(w)$ is smooth there. The finite group $\mathrm{GL}_3(\bar{k})^F \cap \mathrm{SL}_3(\bar{k})$ acts on $Y(w)$ by left multiplication; this action commutes with that of T^{wF}, preserves δ, and is compatible with its action on $\overline{X}$, which is transitive on $X(1)$. Hence every point of $\overline{Y}_{\delta_0}(w)$ over $X(1)$ looks like o', and ρ_{δ_0} is, in completed local form, the toric resolution $\widetilde{Y}_{\mathrm{tor}} \to Y_{\mathrm{tor}}$ at each of these p^3+1 points. As in the split case, uniqueness of the minimal resolution lifts the action of H to $\widetilde{\overline{Y}}_{\delta_0}(w)$.

We fix the following notation. Write $I := p^3+1$ and $X(1) = \{o_1, \dots, o_I\}$. Let C_u and C_v be the strict transforms in $\widetilde{\overline{Y}}_{\delta_0}(w)$ of the prime divisors of $\overline{Y}_{\delta_0}(w)$ lying over B_u and B_v; these are unique by Proposition 4.3. Over o_j, let $E_{1,j}$ and $E_{2,j}$ be the exceptional curves corresponding to D_1 and D_2, and let $Q_j = E_{1,j} \cap E_{2,j}$. Near the fibre over o_j, the curves C_u and C_v correspond to D_0 and D_3, so that $\widetilde{\overline{Y}}_{\delta_0}(w)$ contains the chains

$$C_u - E_{1,j} - E_{2,j} - C_v \qquad (1 \le j \le I)$$

of transversally intersecting curves.

4.5.5. *Affine charts and the fixed locus.* By Section 4.3, $H = \langle \tau \rangle$ acts on the local model by

$$\tau(z) = \bar{\xi} z, \qquad \tau(u) = u, \qquad \tau(v) = v,$$

so that $\tau(\chi^{(m_1,m_2)}) = \bar{\xi}^{m_2}\chi^{(m_1,m_2)}$: the character $\chi^{(m_1,m_2)}$ has weight m_2 modulo p^2-1, in accordance with the character-coordinate rule of Section 2.7. Each chart U_{σ_i} is an affine plane whose coordinates are the characters dual to u_{i-1}, u_i. In

terms of u, v and z, these coordinates, the divisors they cut out, and their weights are as follows:

chart	coordinates	zero loci	weights
U_{σ_1}	$z/v,\ v$	$D_0,\ D_1$	$1,\ 0$
U_{σ_2}	$z^p v^{1-p},\ v/z$	$D_1,\ D_2$	$p,\ -1$
U_{σ_3}	$u,\ v^{p-1}z^{-p}$	$D_2,\ D_3$	$0,\ -p$

Since $\gcd(p, p^2-1) = 1$, the weights ± 1, p and $-p$ are nonzero modulo p^2-1. Hence the fixed locus of H is $D_0 \cap U_{\sigma_1}$ in the first chart, the origin $D_1 \cap D_2$ in the second, and $D_3 \cap U_{\sigma_3}$ in the third. In particular, D_1 and D_2 are not fixed pointwise, and $D_1 \cap D_2$ is the only fixed point on $D_1 \cup D_2$ that does not lie on $D_0 \cup D_3$.

Globally, H acts freely over X, and it fixes C_u and C_v pointwise, because the covering is totally ramified along B_u and B_v, so that the inertia groups of C_u and C_v are all of H. Together with the chart computation this gives

$$\big(\widetilde{\overline{Y}}_{\delta_0}(w)\big)^H = C_u \sqcup C_v \sqcup \{Q_1, \dots, Q_{p^3+1}\}. \tag{4.28}$$

4.6. **Boundary intersection calculations.** As before, B_u and B_v denote the boundary curves of the base $\overline{X} = \overline{X}(w)$, and C_u and C_v their strict transforms on the resolved component $\widetilde{\overline{Y}}_{\delta_0}(w)$.

4.6.1. *Boundary curves on the base compactification.* Let

$$A = \mathcal{O}_{\mathbb{P}^2_x}(1)|_{\overline{X}}, \qquad B = \mathcal{O}_{\mathbb{P}^2_y}(1)|_{\overline{X}}.$$

By (4.8), the surface $\overline{X}$ is a complete intersection of bidegrees $(1,1)$ and $(0, p+1)$ in $\mathbb{P}^2_x \times \mathbb{P}^2_y$. Computing in the Chow ring of $\mathbb{P}^2_x \times \mathbb{P}^2_y$, we obtain

$$A^2 = A \cdot B = p+1, \qquad B^2 = 0. \tag{4.29}$$

By Section 4.2.2, the first projection maps B_u isomorphically onto C_x and the second maps it to C_y with degree p, while the second projection maps B_v isomorphically onto C_y and the first maps it to C_x with degree p. Hence

$$A \cdot B_u = p+1, \quad B \cdot B_u = p(p+1), \qquad A \cdot B_v = p(p+1), \quad B \cdot B_v = p+1.$$

Since $\overline{X}$ is a $\mathbb{P}^1$-bundle over C_y, the numerical divisor class group $\mathrm{Num}(\overline{X})$ has rank two, and since the intersection matrix of A and B is non-singular, A and B form a basis of $\mathrm{Num}(\overline{X})_{\mathbb{Q}}$. Solving for the classes of the boundary curves yields

$$B_u \equiv pA - (p-1)B, \qquad B_v \equiv A + (p-1)B, \tag{4.30}$$

and consequently

$$\begin{aligned} B_u^2 &= p(p+1)(2-p), \\ B_v^2 &= (p+1)(2p-1), \\ B_u \cdot B_v &= p^3 + 1. \end{aligned} \tag{4.31}$$

The last equality agrees with the fact that B_u and B_v meet transversally in the p^3+1 points of $X(1)$.

4.6.2. *Strict transforms on the resolved cover.* Let

$$\widetilde{\pi}_{\delta_0}\colon \widetilde{\overline{Y}}_{\delta_0}(w) \xrightarrow{\rho_{\delta_0}} \overline{Y}_{\delta_0}(w) \longrightarrow \overline{X}$$

be the composite of the minimal resolution with the normal covering, which has degree $N_{\mathrm{un}} = p^2 - 1$. Since the covering is totally ramified along B_u and B_v (Proposition 4.3), the divisors (4.27) of the local equations u and v give the global pullback formulas

$$\widetilde{\pi}_{\delta_0}^{*} B_u = N_{\mathrm{un}} C_u + p \sum_{j=1}^{I} E_{1,j} + \sum_{j=1}^{I} E_{2,j}, \tag{4.32}$$

$$\widetilde{\pi}_{\delta_0}^{*} B_v = \sum_{j=1}^{I} E_{1,j} + p \sum_{j=1}^{I} E_{2,j} + N_{\mathrm{un}} C_v. \tag{4.33}$$

Here C_u meets each $E_{1,j}$ transversally in one point and is disjoint from the $E_{2,j}$ and from C_v; symmetrically for C_v.

Total ramification also implies that $C_u \to B_u$ and $C_v \to B_v$ have degree one. By the projection formula, $\widetilde{\pi}_{\delta_0}^{*} B_u \cdot C_u = B_u \cdot \widetilde{\pi}_{\delta_0 *} C_u = B_u^2$, while (4.32) gives $\widetilde{\pi}_{\delta_0}^{*} B_u \cdot C_u = N_{\mathrm{un}} C_u^2 + pI$. Arguing in the same way for B_v, we obtain

$$B_u^2 = N_{\mathrm{un}} C_u^2 + pI, \qquad B_v^2 = N_{\mathrm{un}} C_v^2 + pI.$$

Substituting (4.31) and $I = p^3 + 1$ yields

$$C_u^2 = -p(p+1), \qquad C_v^2 = -(p^2 - 1). \tag{4.34}$$

Finally, $C_u \to B_u$ and $C_v \to B_v$ are finite birational morphisms onto smooth curves, hence isomorphisms. Therefore

$$g(C_u) = g(C_v) = \frac{p(p-1)}{2}. \tag{4.35}$$

We conclude with the following figure:

4.7. **Brauer–Lefschetz calculation.** We employ the same methods discussed in Section 3.5.2. Put $N_{\mathrm{un}} = p^2 - 1$, and let τ generate $H = \ker(\det\colon T^{wF} \to \mu_{p+1}) \simeq \mu_{N_{\mathrm{un}}}$. On the local cyclic cover,

$$\tau(z) = \bar{\xi} z, \qquad \operatorname{ord}(\bar{\xi}) = N_{\mathrm{un}}, \qquad \widehat{\xi} = [\bar{\xi}] \in W(\mathbb{F}_{p^2})^{\times}.$$

The factors $1 - \widehat{\xi}$ and $1 - \widehat{\xi}^{p}$ are units. Set

$$c_0 := \frac{(p+1)(p-2)}{2}, \qquad c_u := \frac{(p+1)(3p-2)}{2}, \qquad c_v := \frac{(p+1)(3p-4)}{2}. \tag{4.36}$$

All three coefficients are integers.

4.7.1. *Fixed-locus data.* The table separates the fixed curves on the resolved cover from their images on the base. The last column gives cotangent eigenvalues at a point and the normal eigenvalue along a curve.

fixed stratum	number	genus	self-intersection	eigenvalue(s)
Q	p^3+1	–	–	$(\bar{\xi}^{-1}, \bar{\xi}^{p})$
C_u	1	$\frac{p(p-1)}{2}$	$-p(p+1)$	$\bar{\xi}^{-1}$
C_v	1	$\frac{p(p-1)}{2}$	$-(p^2-1)$	$\bar{\xi}^{p}$

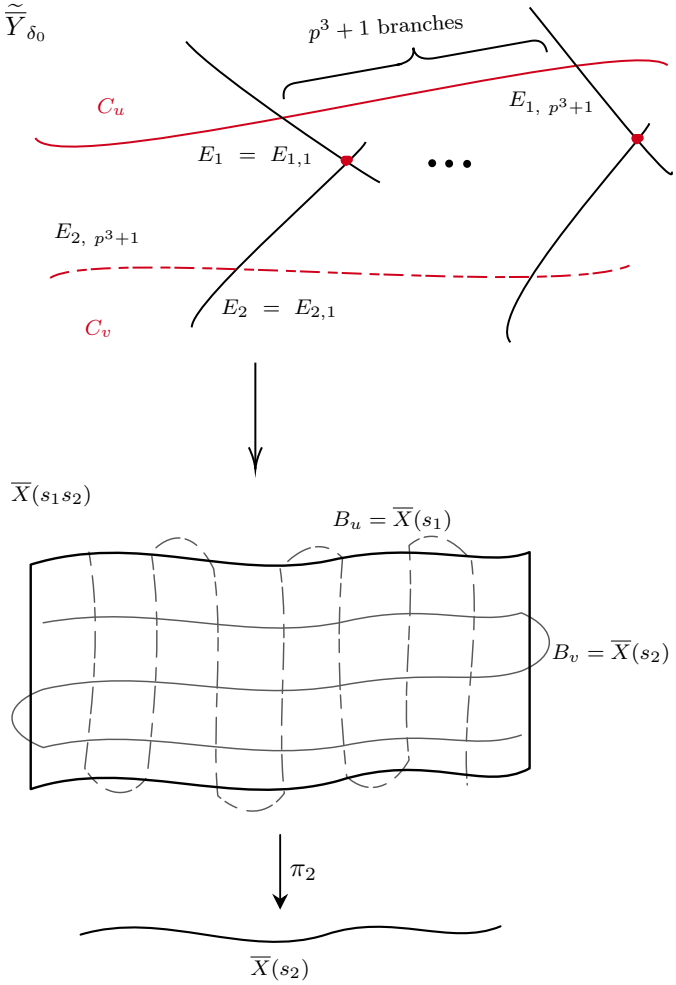


FIGURE 3. The surface $\overline{X}(s_1 s_2)$ ruled over $\overline{X}(s_2)$

4.7.2. *Isolated fixed points.* In the middle toric chart

$$U_{\sigma_2} = \operatorname{Spec} \bar{k}[y_1, y_2], \qquad y_1 = t_1 t_2^{-1}, \quad y_2 = t_1^{-p+1} t_2^p,$$

the lifted cotangent eigenvalues at the isolated fixed point Q are $\widehat{\xi}^{-1}$ and $\widehat{\xi}^p$. Hence

$$a_{\mathrm{Br}}(Q) = -\frac{\widehat{\xi}}{(1-\widehat{\xi})(1-\widehat{\xi}^p)}. \tag{4.37}$$

There are p^3+1 such points.

4.7.3. *Fixed curves.* By (4.34) and (4.35),

$$1 - g(C_u) = 1 - g(C_v) = -c_0.$$

The curve C_u. In $U_{\sigma_1} = \operatorname{Spec} \bar{k}[x_1, x_2]$, the curve C_u is given by $x_2 = 0$, and the normal eigenvalue is $\widehat{\xi}^{-1}$. Formula (2.18), together with $C_u^2 = -p(p+1)$, gives

$$\begin{aligned} b_{\mathrm{Br}}(C_u) &= -\frac{c_0}{1-\widehat{\xi}} + \frac{p(p+1)\widehat{\xi}}{(1-\widehat{\xi})^2} \\ &= \boxed{\frac{c_u\widehat{\xi} - c_0}{(1-\widehat{\xi})^2}}. \end{aligned} \tag{4.38}$$

The curve C_v. In $U_{\sigma_3} = \operatorname{Spec} \bar{k}[w_1, w_2]$, the curve C_v is given by $w_1 = 0$, and the normal eigenvalue is $\widehat{\xi}^{p}$. Since $C_v^2 = -(p^2-1)$,

$$\begin{aligned} b_{\mathrm{Br}}(C_v) &= \frac{c_0\widehat{\xi}^{p}}{1-\widehat{\xi}^{p}} + \frac{(p^2-1)\widehat{\xi}^{p}}{(1-\widehat{\xi}^{p})^2} \\ &= \boxed{\frac{\widehat{\xi}^{p}(c_v - c_0\widehat{\xi}^{p})}{(1-\widehat{\xi}^{p})^2}}. \end{aligned} \tag{4.39}$$

4.7.4. *Proof of the non-split main theorem.* The fixed locus of τ on $\widetilde{\overline{Y}}_{\delta_0}(w)$ is the disjoint union of C_u, C_v, and the p^3+1 isolated points. Adding the three types of contribution gives

$$\begin{aligned} L_{\mathrm{Br}}(\tau, \widetilde{\overline{Y}}_{\delta_0}(w)) &= \frac{c_u\widehat{\xi} - c_0}{(1-\widehat{\xi})^2} - \frac{(p^3+1)\widehat{\xi}}{(1-\widehat{\xi})(1-\widehat{\xi}^{p})} \\ &\quad + \frac{\widehat{\xi}^{p}(c_v - c_0\widehat{\xi}^{p})}{(1-\widehat{\xi}^{p})^2}. \end{aligned} \tag{4.40}$$

By (2.35), this is the value on $\overline{Y}_{\delta_0}(w)$, so it proves (1.4). Translations by T^{wF} identify the $p+1$ components and commute with τ. Therefore

$$L_{\mathrm{Br}}(\tau, \overline{Y}(w)) = (p+1) L_{\mathrm{Br}}(\tau, \overline{Y}_{\delta_0}(w)).$$

4.8. **Nonvanishing and Galois-module consequence.**

Proposition 4.4. *The non-split Brauer–Lefschetz numbers*

$$L_{\mathrm{Br}}(\tau, \overline{Y}_{\delta_0}(w)) \quad \textit{and} \quad L_{\mathrm{Br}}(\tau, \overline{Y}(w))$$

are nonzero for every prime p.

Proof. Let $R_p(T) \in \mathbb{Q}(T)$ denote the rational function on the right-hand side of (1.4). If ω is a primitive (p^2-1)-st root, then $\omega \neq 1$ and $\omega^p \neq 1$, because $p \nmid p^2-1$. Thus every denominator in $R_p(\omega)$ is nonzero. Lemma 2.1, with $M = p^2-1$ and $\widehat{\omega} = \widehat{\xi}$, reduces the claim to nonvanishing at $\omega = e^{2\pi i/(p^2-1)}$.

Put

$$A = \frac{\pi}{p^2-1}, \qquad x = e^{2iA}, \qquad K = p^3+1.$$

The self-intersection parts of (1.4) are real, because

$$\frac{x}{(1-x)^2} \qquad \text{and} \qquad \frac{x^p}{(1-x^p)^2}$$

are real. Applying the elementary cotangent identities to the remaining terms gives

$$\operatorname{Im} L_{\mathrm{Br}}(\tau, \overline{Y}_{\delta_0}(w)) = \frac{K + 2c_0}{4}\big(\cot(pA) - \cot A\big).$$

Here

$$K + 2c_0 = (p+1)(p^2-1) > 0,$$

whereas $0 < A < pA < \pi$. Since cot is strictly decreasing on $(0, \pi)$, the second factor is negative. Hence the component value is nonzero. The whole-space value is $p+1$ times the component value, so it is nonzero as well. □

Corollary 4.5 (Non-split Galois-module consequence)**.** *Put* $A_{\mathrm{un}} = H = \langle \tau \rangle$. *Then*

$$\chi_{A_{\mathrm{un}}}\big(\overline{Y}^{\circ}(w), \mathcal{O}_{\overline{Y}^{\circ}(w)}\big)$$

is not virtually free over $\bar{k}[A_{\mathrm{un}}]$. *Moreover,*

$$\chi_{T^{wF}}\big(\overline{Y}(w), \mathcal{O}_{\overline{Y}(w)}\big)$$

is not virtually free over $\bar{k}[T^{wF}]$.

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